

PARTIAL LOCAL MORAN

HANDLING LOCAL CONFOUNDING

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LOCAL CONFOUNDING

PRIOR ART IN ESDA

PARTIAL CONDITIONING

AFFORDABILITY OUTLIERS



Bristol's big
housing
conversation



**Bristol Living Rent
Commission**



BRISTOL
ONE CITY



HOW CAN WE FIND RENT AFFORDABILITY OUTLIERS?



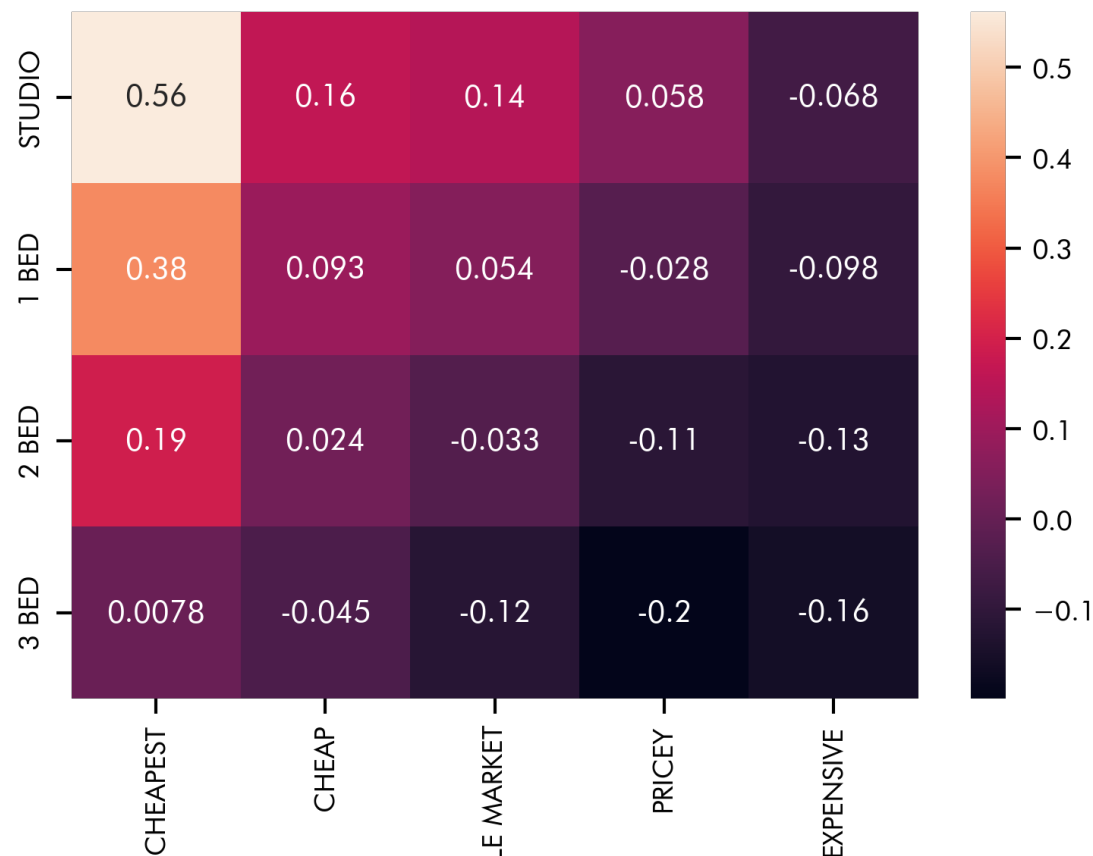
Bristol's big housing conversation



Bristol Living Rent Commission



BRISTOL
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HOW CAN WE FIND RENT AFFORDABILITY OUTLIERS?



Bristol's big housing conversation

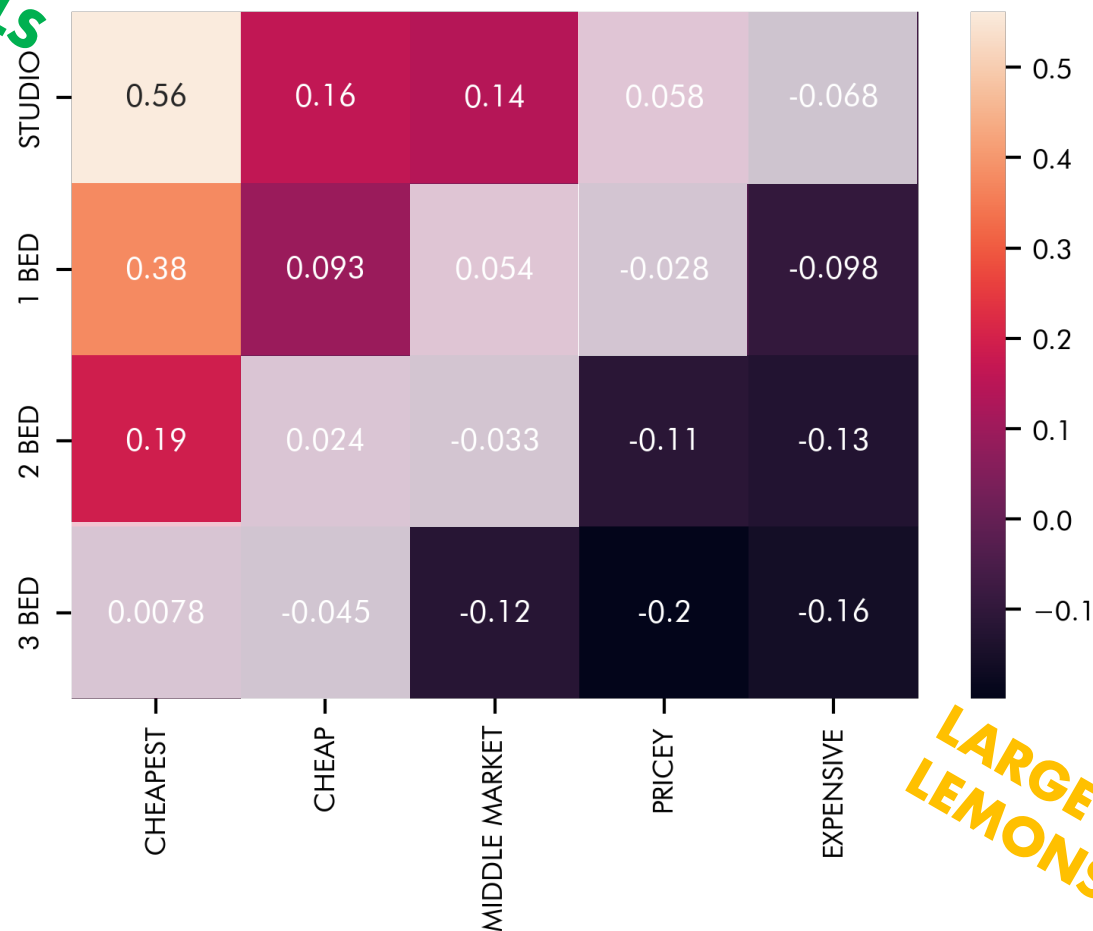
SMALL STEALS



Bristol Living Rent Commission



BRISTOL
ONE CITY



LARGE LEMONS

HOW CAN WE FIND RENT AFFORDABILITY OUTLIERS?

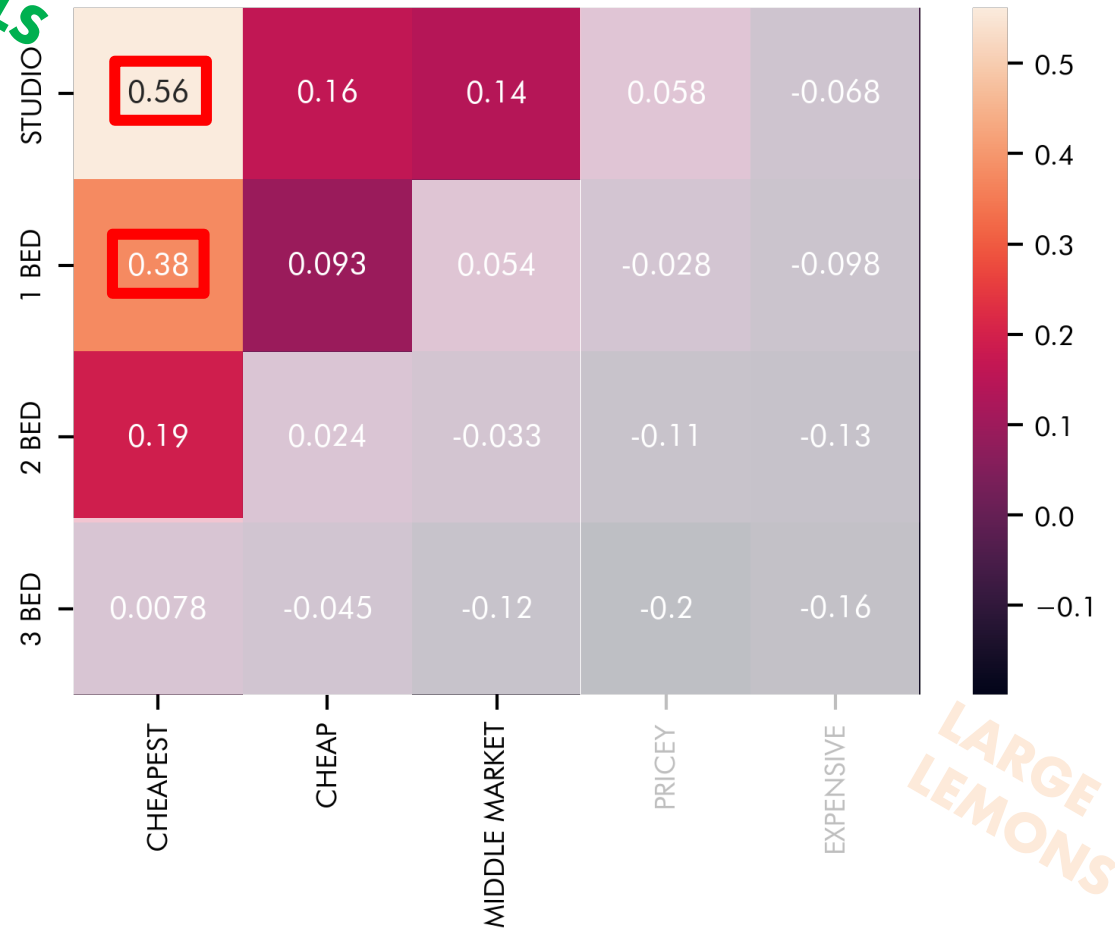


Bristol's big housing conversation



SMALL STEALS

DOUBLED RENT W/IN 2 YEARS



LARGE LEMONS

Bristol Living Rent Commission



BRISTOL ONE CITY



HOW CAN WE FIND RENT AFFORDABILITY OUTLIERS?



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housing
conversation



How the rent will be decided depends on the type of rent case...

The tribunal will look at the market or rents of similar properties in the local area.

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HOW CAN WE FIND RENT AFFORDABILITY OUTLIERS?



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How the rent will be decided depends on the type of rent case...

The tribunal will look at the market or rents of similar properties in the local area.

What is the “rent?”

No central database of what rents are charged, only rents that are advertised.

HOW CAN WE FIND RENT AFFORDABILITY OUTLIERS?



Bristol's big
housing
conversation



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Commission**



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How the rent will be decided depends on the type of rent case...

The tribunal will look at the market or rents of similar properties in the local area.

What is “similar?”

Depends on judgement of the assessor during inspection, structural factors, and local intangibles.

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Bristol's big
housing
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**Bristol Living Rent
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How the rent will be decided depends on the type of rent case...

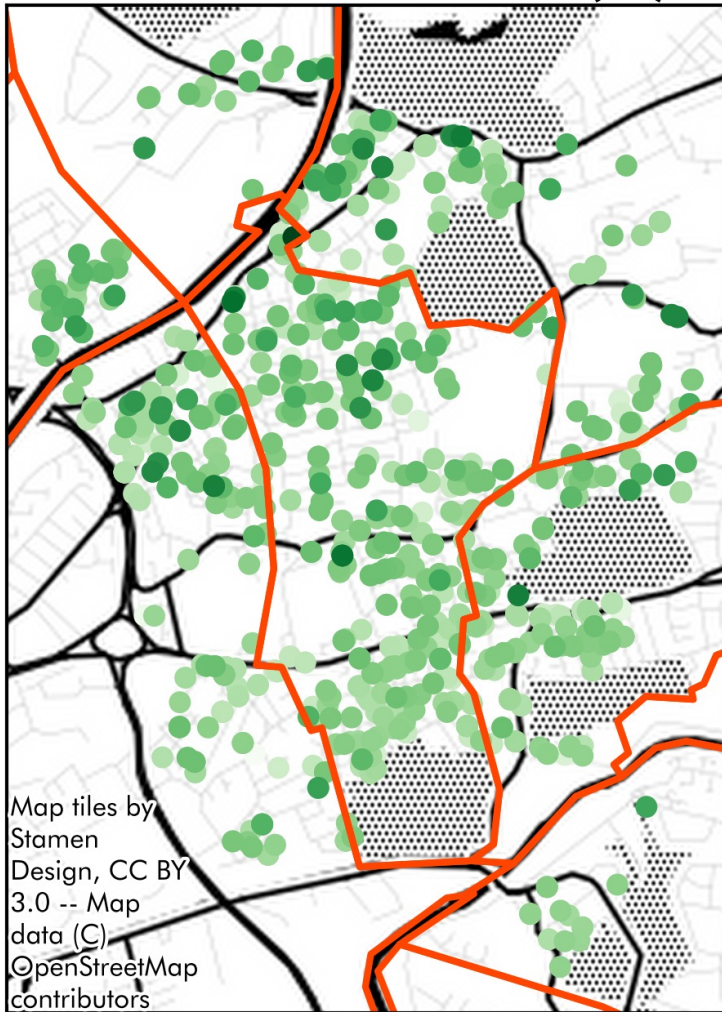
The tribunal will look at the market or rents of similar properties in the local area.

What is “local area?”

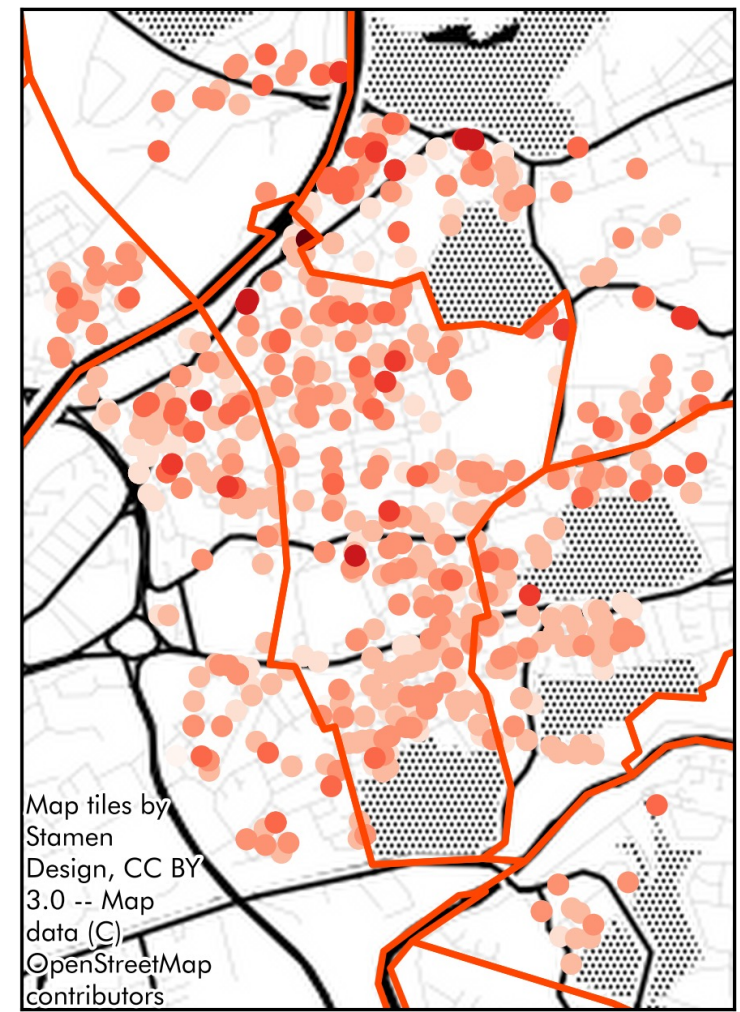
No specific guidance to decide upon
street < n’hood < town < labor market

HOW CAN WE FIND RENT AFFORDABILITY OUTLIERS?

WEEKLY RENT (£)

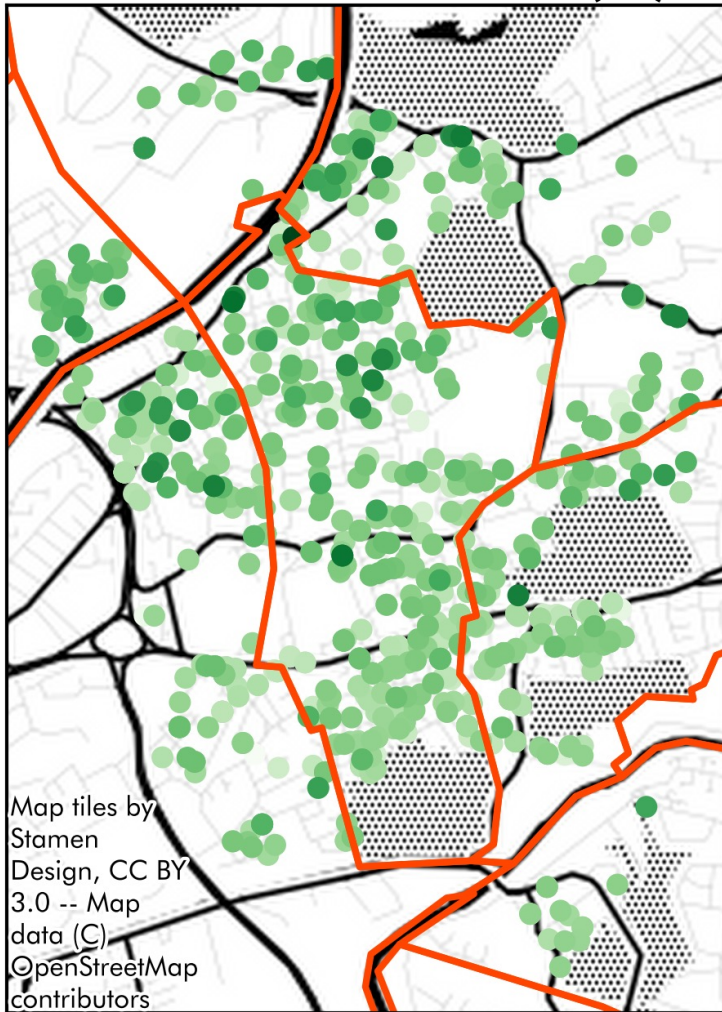


BEDROOMS



HOW CAN WE FIND RENT AFFORDABILITY OUTLIERS?

WEEKLY RENT (£)

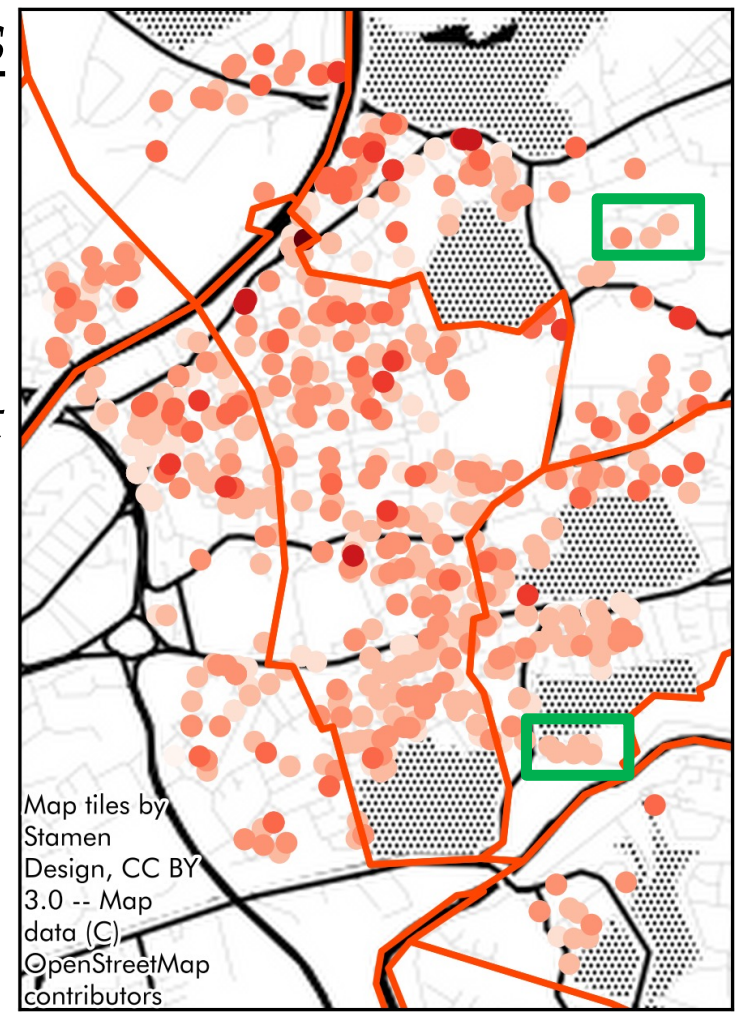


similar properties

Lettings that are substitutes for a reasonable tenant looking to house themselves or their family.

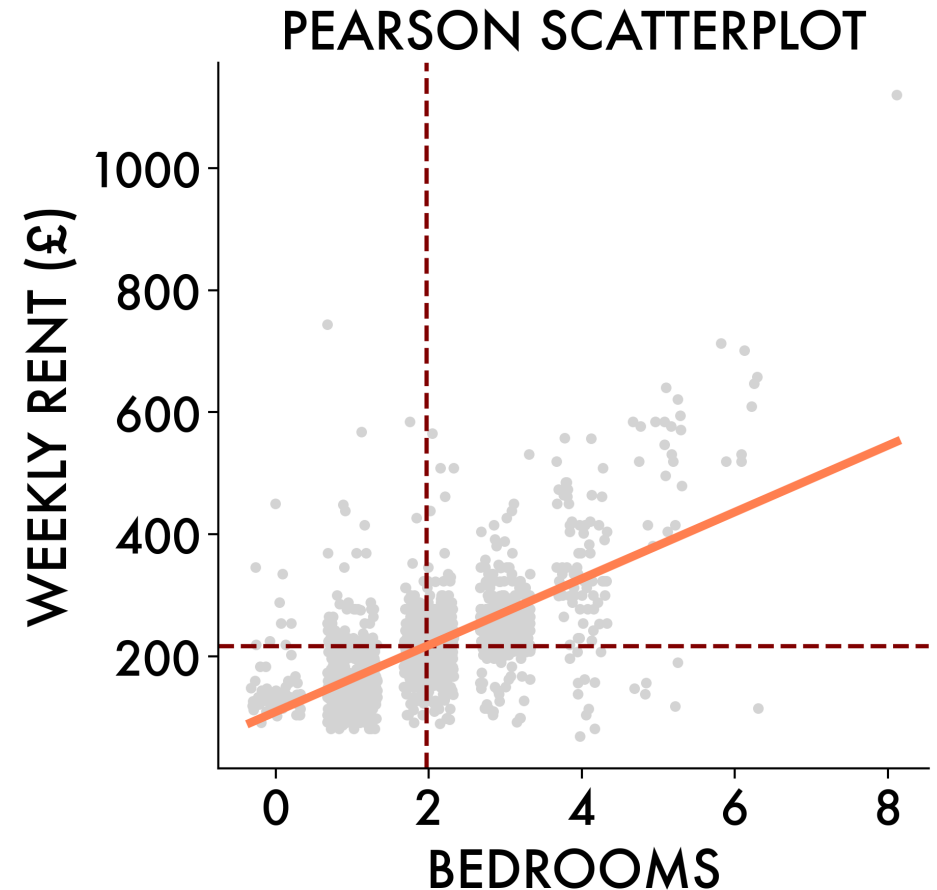
Same bedrooms, ca. baths, receptions within some distance

BEDROOMS



HOW CAN WE FIND RENT AFFORDABILITY OUTLIERS?

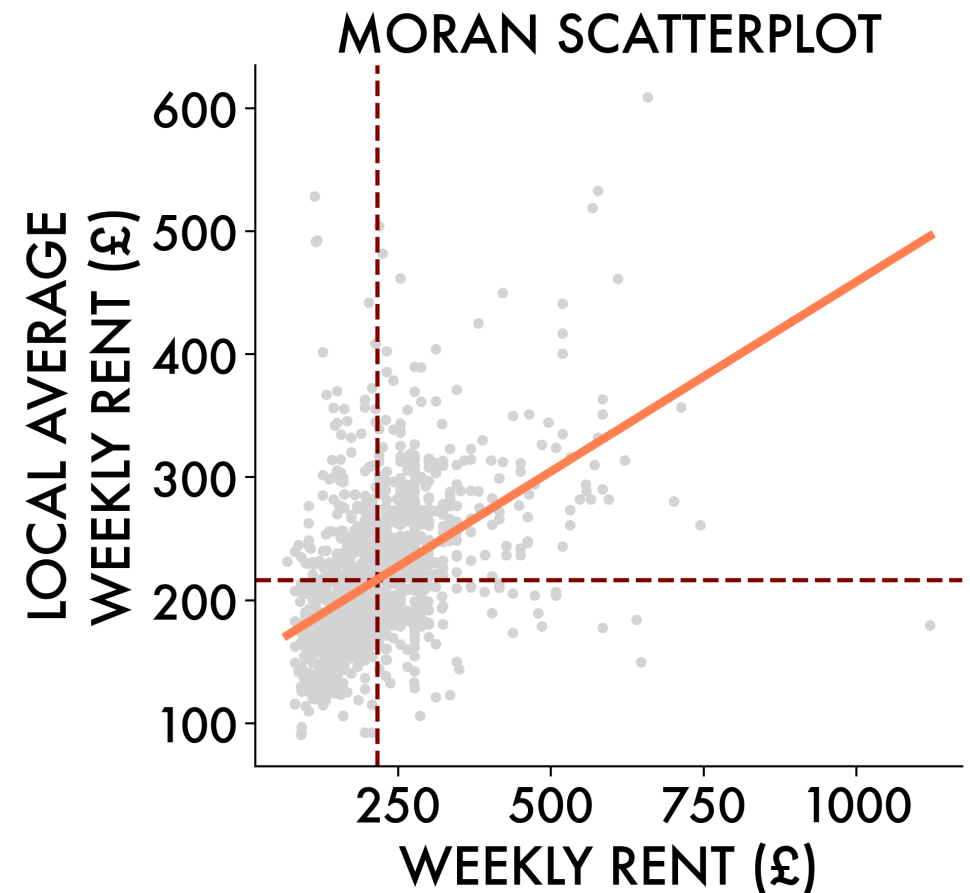
*Properties with fewer bedrooms
are generally lower rent.*



HOW CAN WE FIND RENT AFFORDABILITY OUTLIERS?

*Properties with fewer bedrooms
are generally lower rent.*

*Nearby properties tend to have
similar rent levels*

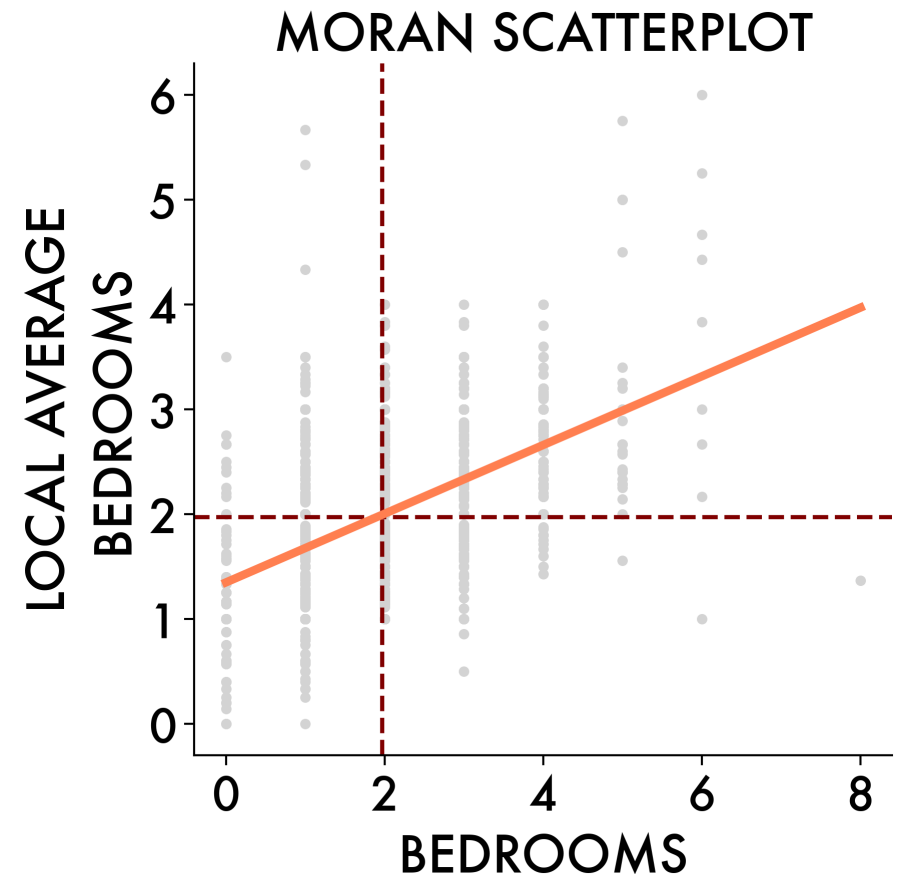


HOW CAN WE FIND RENT AFFORDABILITY OUTLIERS?

*Properties with fewer bedrooms
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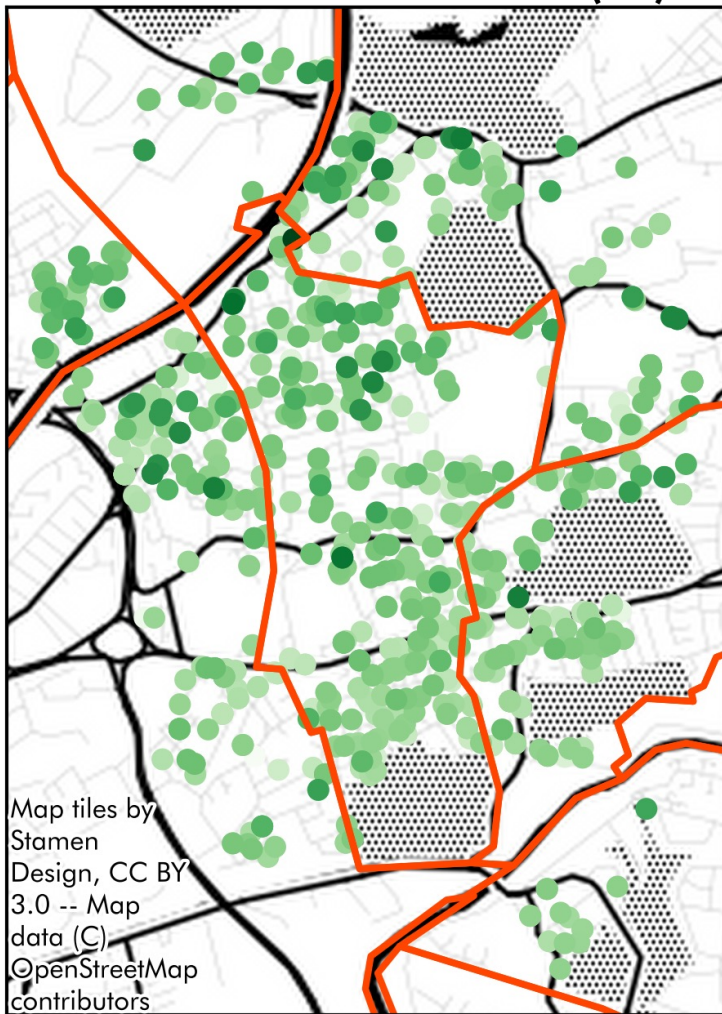
*Nearby properties tend to have
similar rent levels*

*Nearby properties also tend to be
of similar size as well*



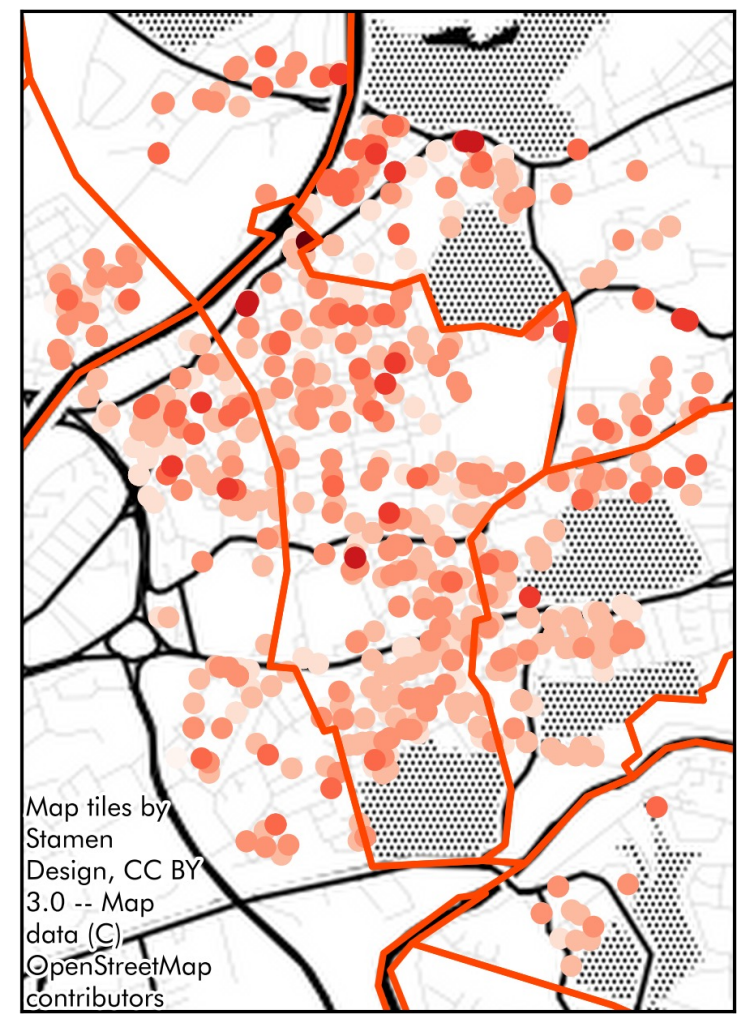
HOW CAN WE FIND RENT AFFORDABILITY OUTLIERS?

WEEKLY RENT (£)



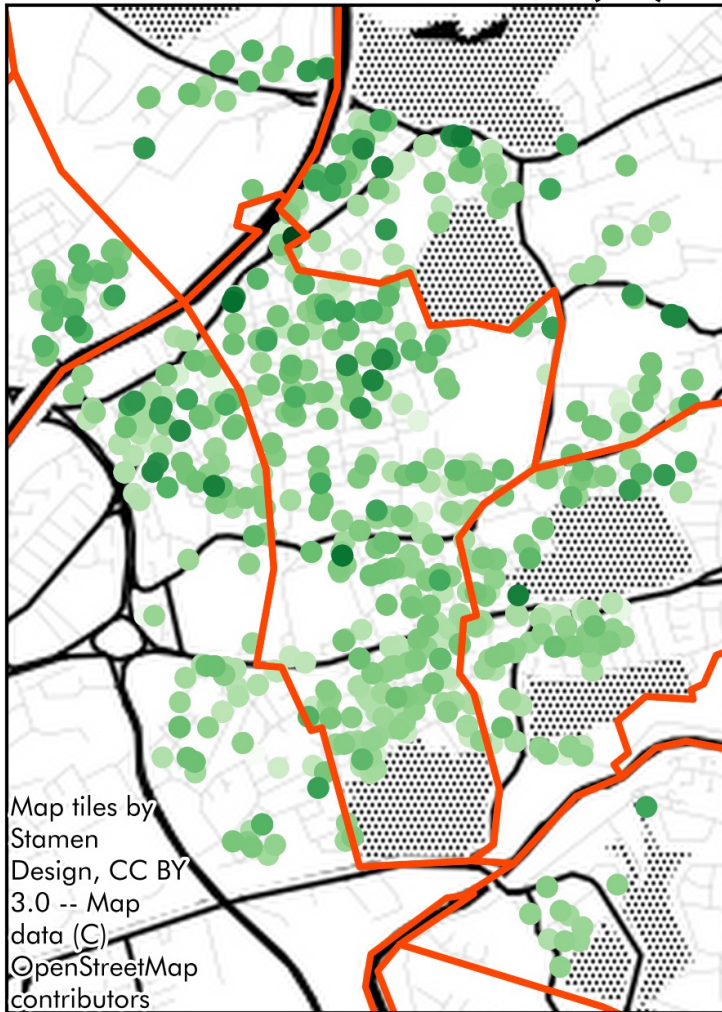
How can we find locally-unusual rents given the characteristics of properties?

BEDROOMS



HOW CAN WE FIND RENT AFFORDABILITY OUTLIERS?

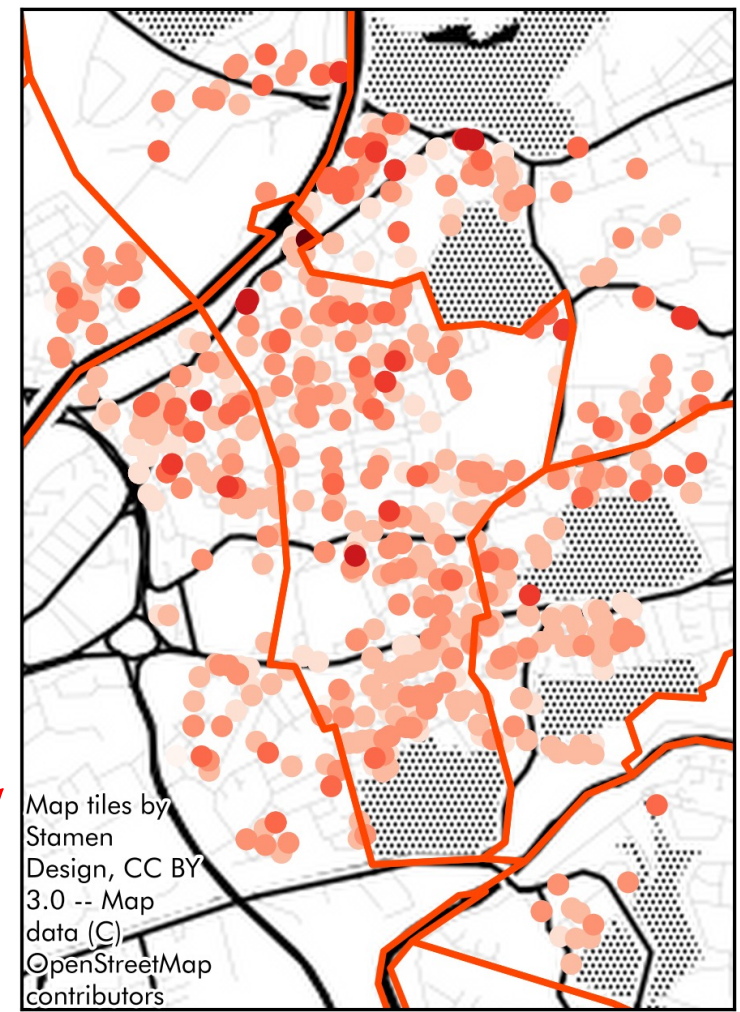
WEEKLY RENT (£)



*How can we find
locally-unusual
rents given the
characteristics of
properties?*

*How can we find
local affordability
outliers?*

BEDROOMS



HOW CAN WE FIND RENT AFFORDABILITY OUTLIERS?

LOCAL CONFOUNDING

size effects on rent vary

PRIOR ART IN ESDA

PARTIAL CONDITIONING

AFFORDABILITY OUTLIERS

$$\rho = \sum_i^N \frac{(x'_i - \bar{x}') (y'_i - \bar{y}')}{\sigma(x') \sigma(y')}$$

CLASSICAL CORRELATION & LOCAL PEARSON

$$x = \frac{x' - \bar{x}'}{\sigma(x')}$$

$$y = \frac{y' - \bar{y}'}{\sigma(y')}$$

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CLASSICAL CORRELATION & LOCAL PEARSON

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CLASSICAL CORRELATION & LOCAL PEARSON

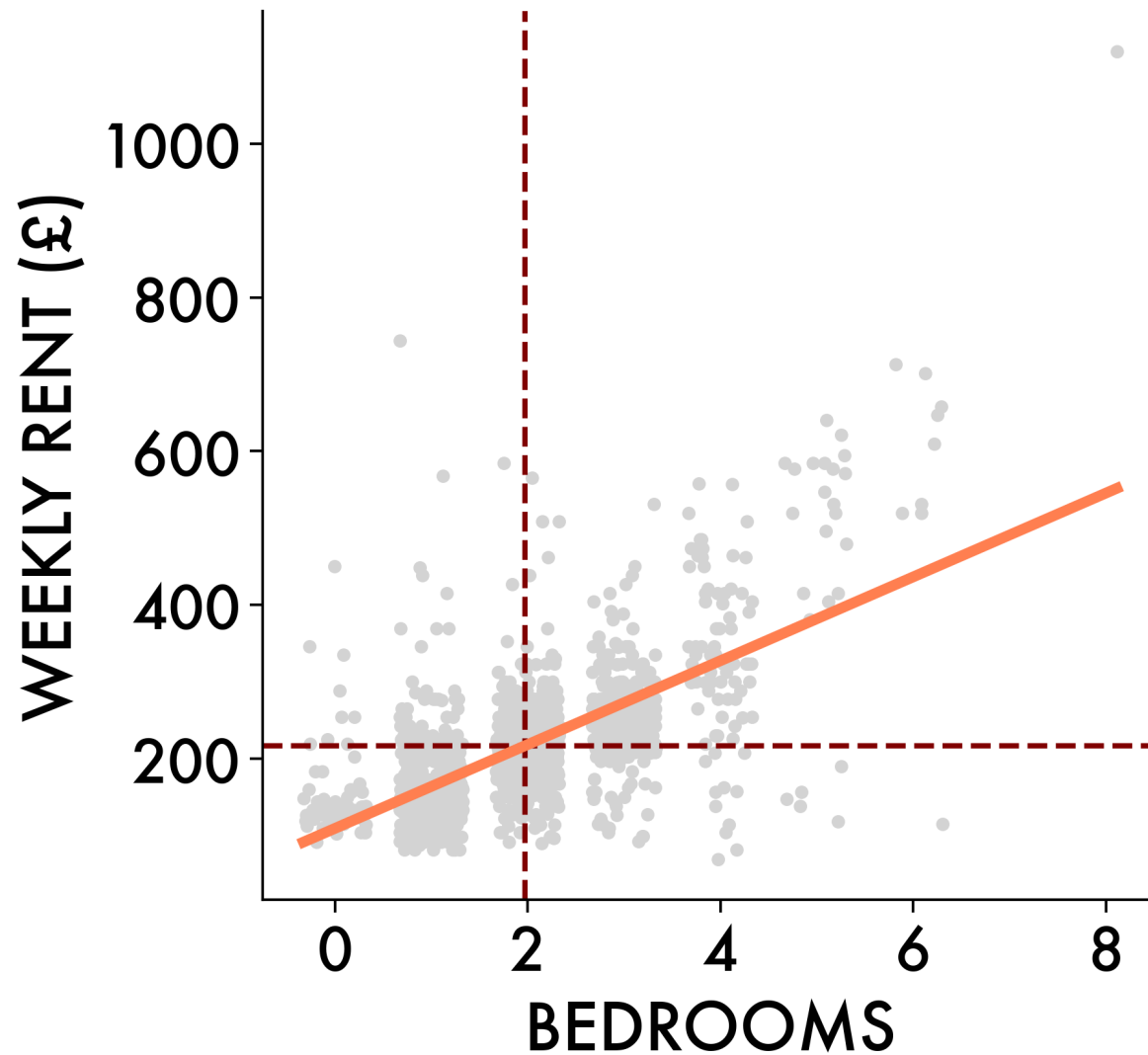
$$x = \frac{x' - \bar{x}'}{\sigma(x')}$$

$$y = \frac{y' - \bar{y}'}{\sigma(y')}$$

$$\rho_i = \frac{(x'_i - \bar{x}')(y'_i - \bar{y}')}{\sigma(x')\sigma(y')} = x_i y_i$$

CLASSICAL CORRELATION & LOCAL PEARSON

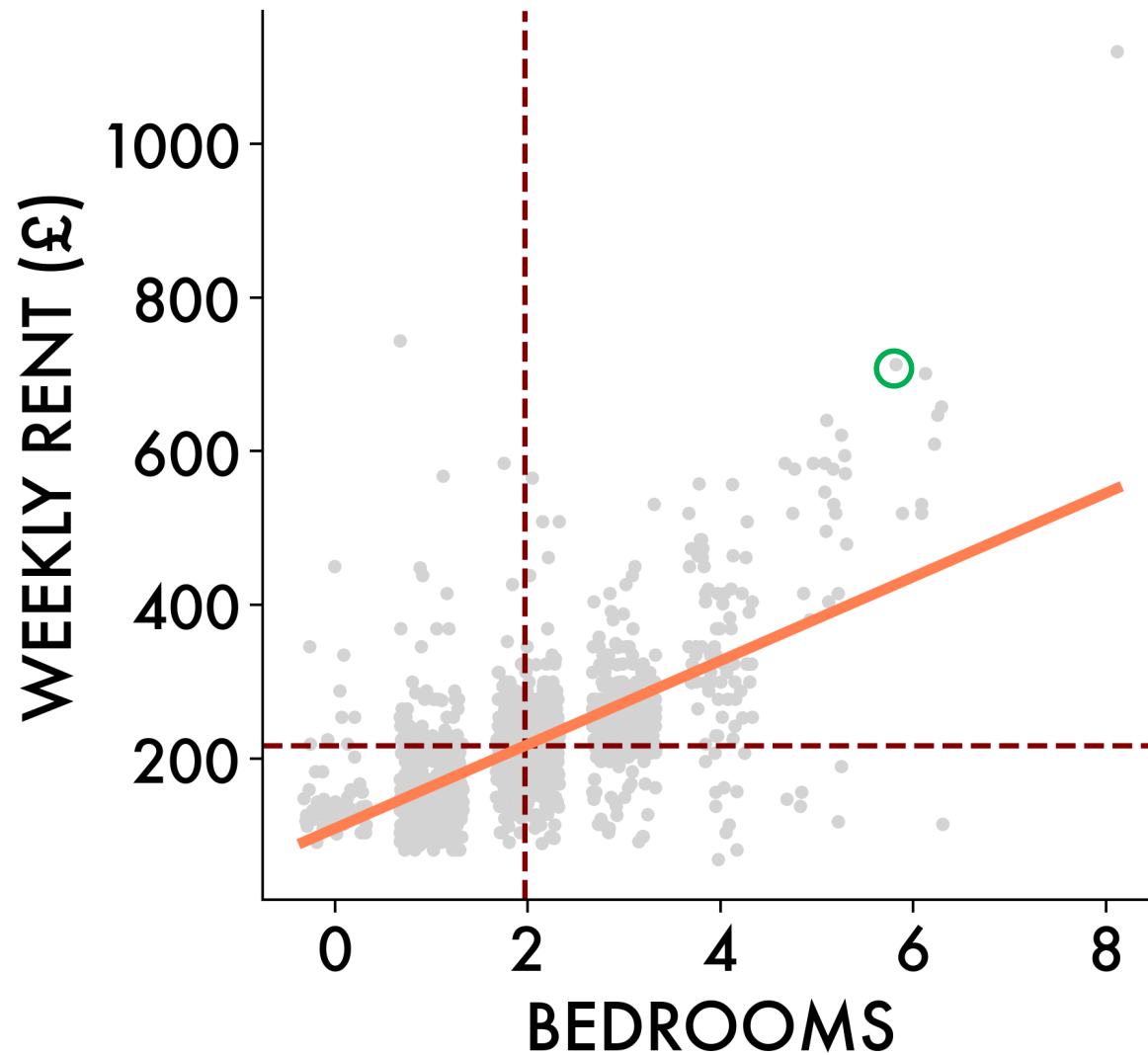
PEARSON SCATTERPLOT



$$x_i y_i$$

CLASSICAL CORRELATION & LOCAL PEARSON

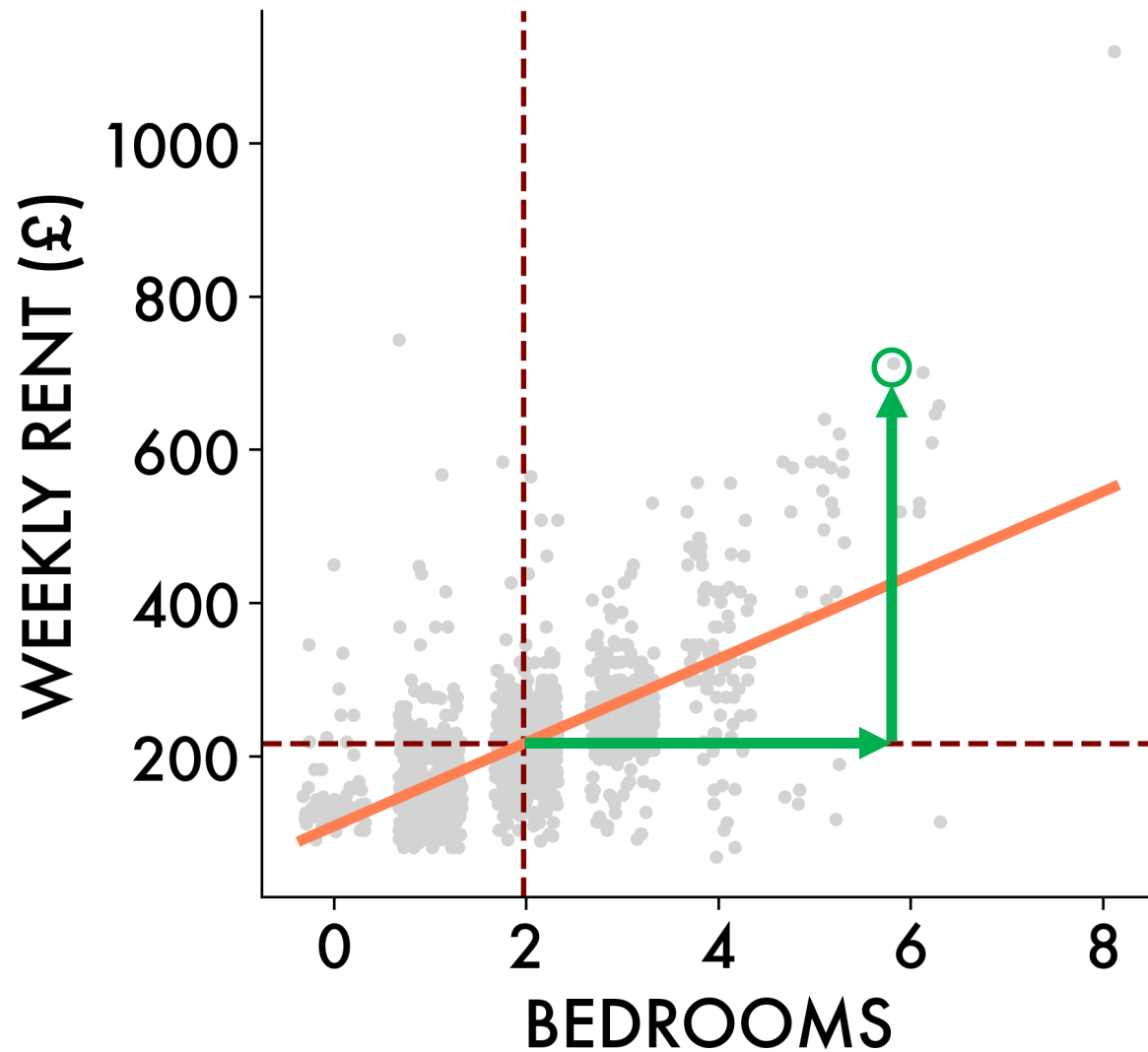
PEARSON SCATTERPLOT



$$x_i y_i$$

CLASSICAL CORRELATION & LOCAL PEARSON

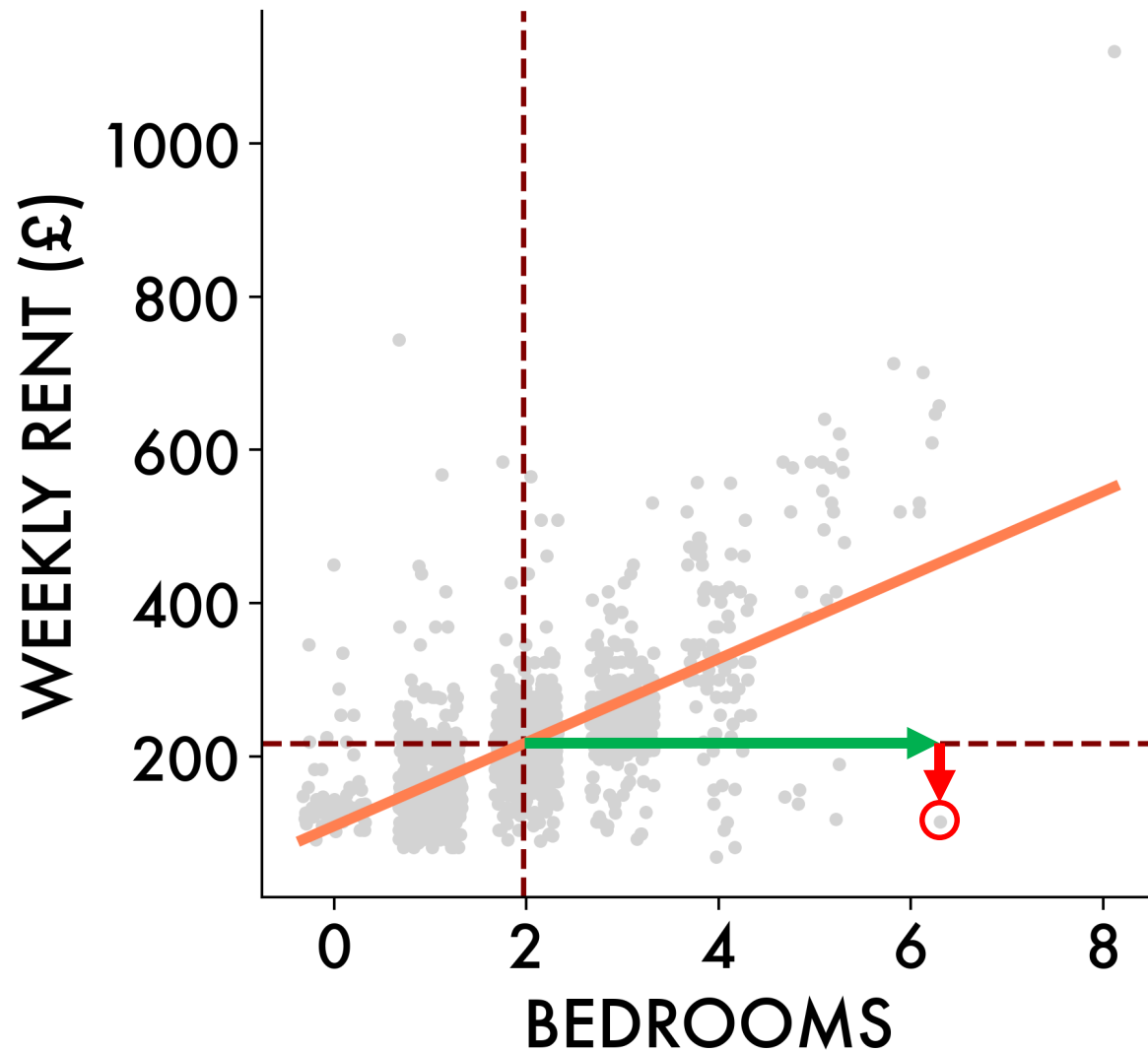
PEARSON SCATTERPLOT



$x_i y_i$

CLASSICAL CORRELATION & LOCAL PEARSON

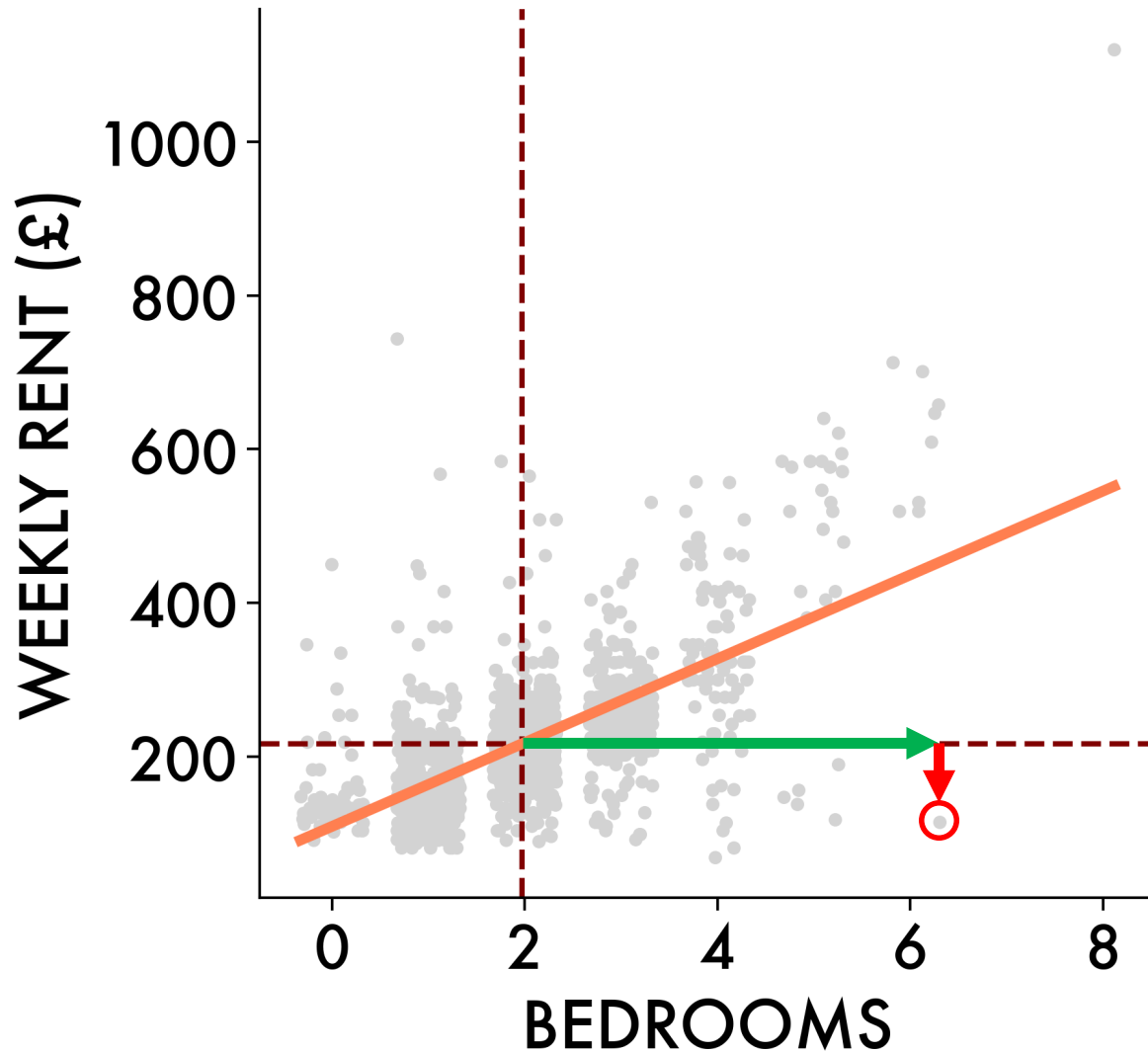
PEARSON SCATTERPLOT



$x_i y_i$

CLASSICAL CORRELATION & LOCAL PEARSON

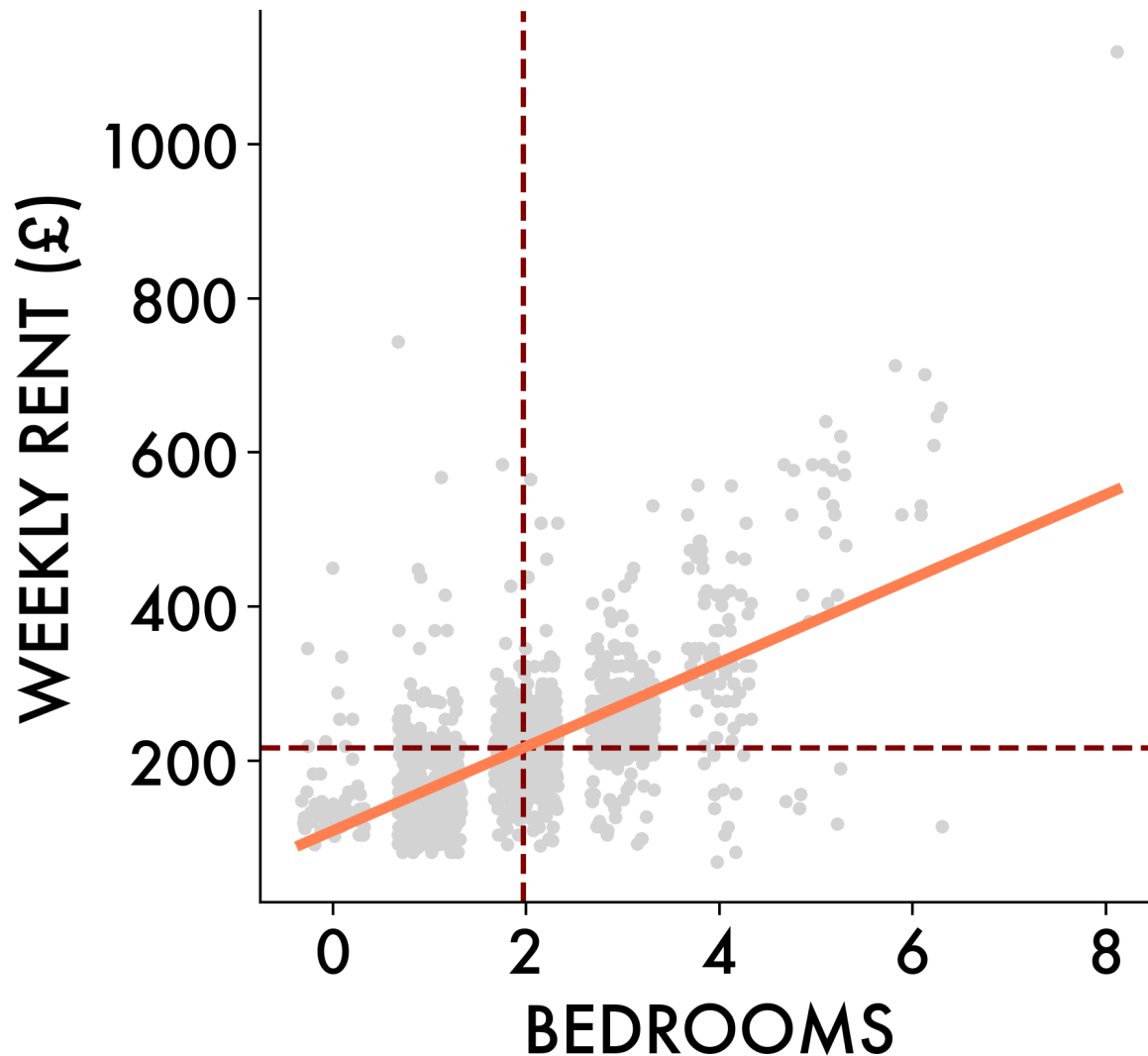
PEARSON SCATTERPLOT



$$\rho_i = x_i y_i$$

CLASSICAL CORRELATION & LOCAL PEARSON

PEARSON SCATTERPLOT



$$\rho = \begin{bmatrix} x_1 y_1 \\ x_2 y_2 \\ \vdots \\ x_N y_N \end{bmatrix} = x \circ y$$

$$\sum_i^N \rho_i = \rho$$

CLASSICAL CORRELATION & LOCAL PEARSON

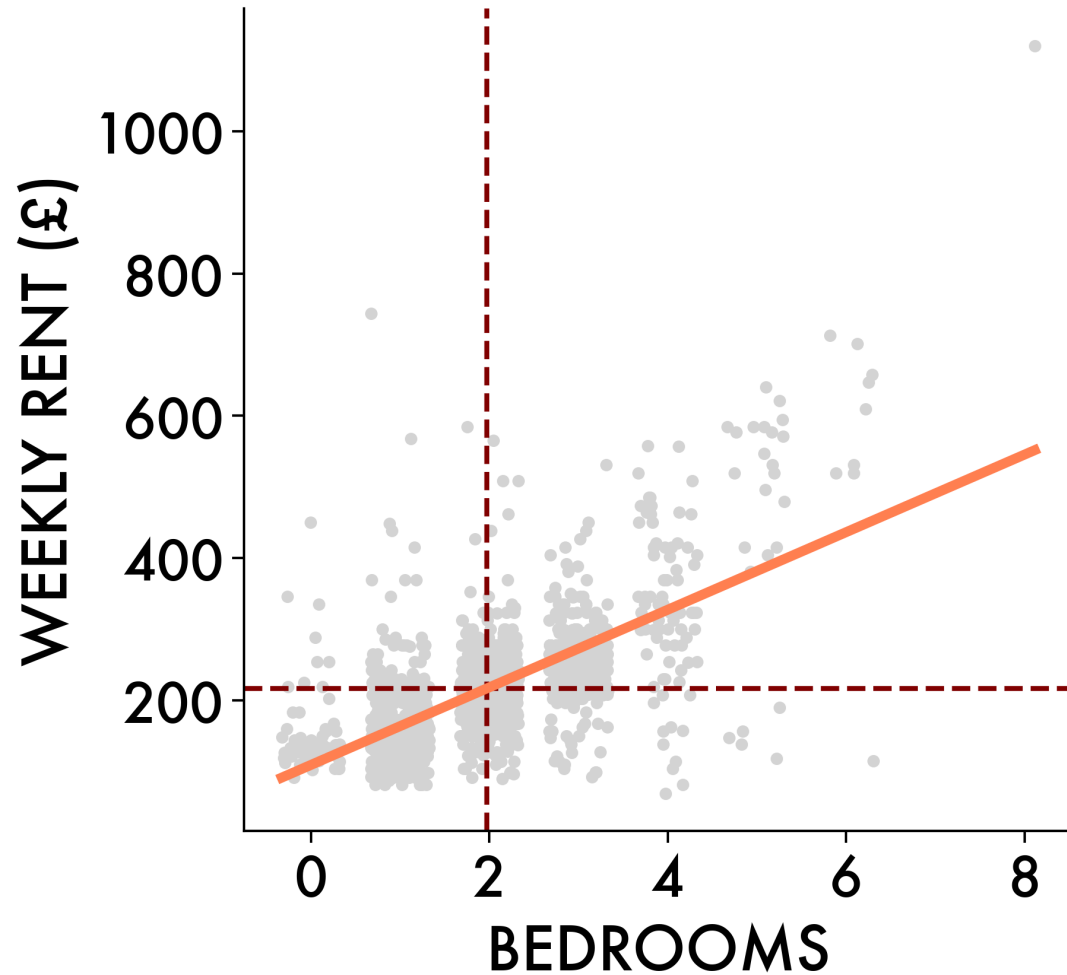
A sufficient condition to turn a global summary statistic into an Anselin (1995) LISA statistic is to swap in an elementwise product where an inner product is used.

$$\rho = \begin{bmatrix} x_1 y_1 \\ x_2 y_2 \\ \vdots \\ x_N y_N \end{bmatrix} = x \circ y$$

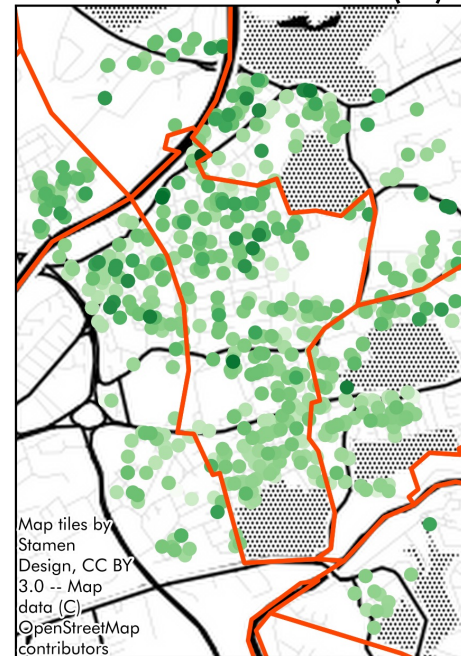
$$\rho = x \cdot y$$

CLASSICAL CORRELATION & LOCAL PEARSON

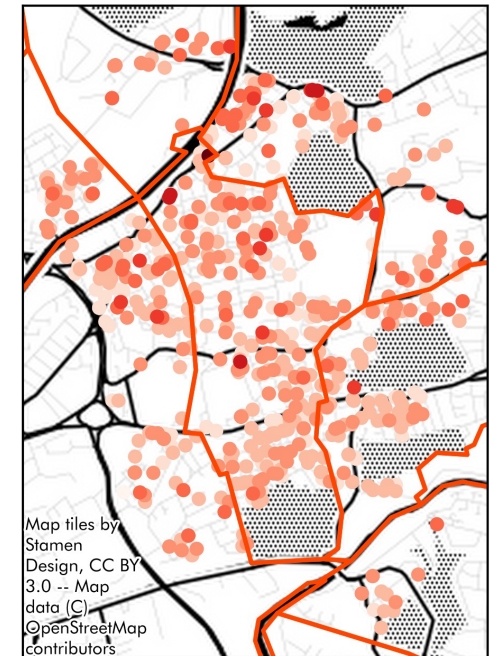
PEARSON SCATTERPLOT



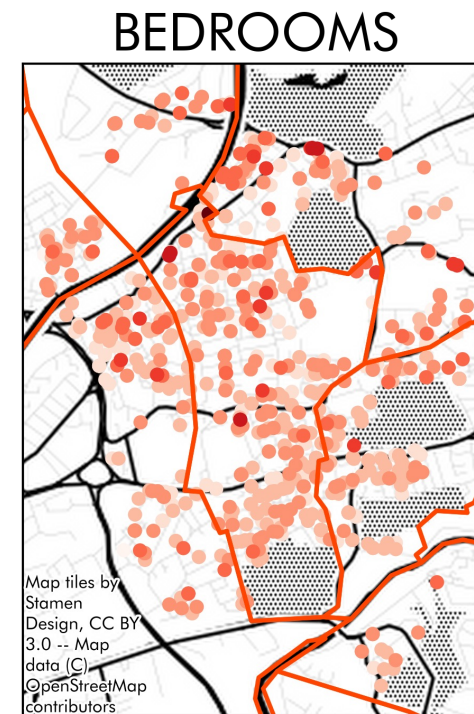
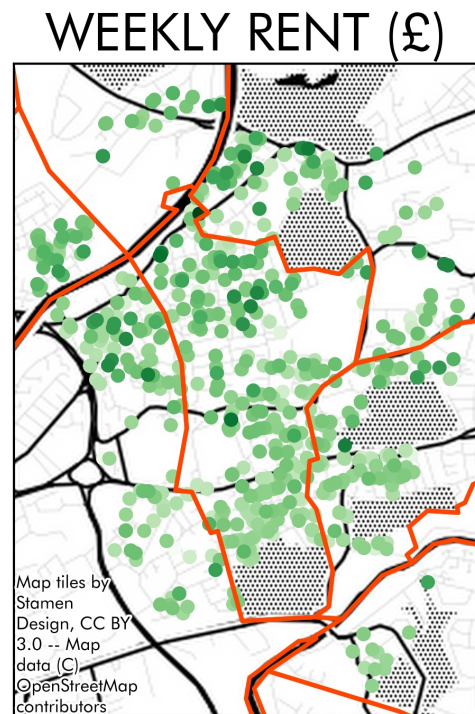
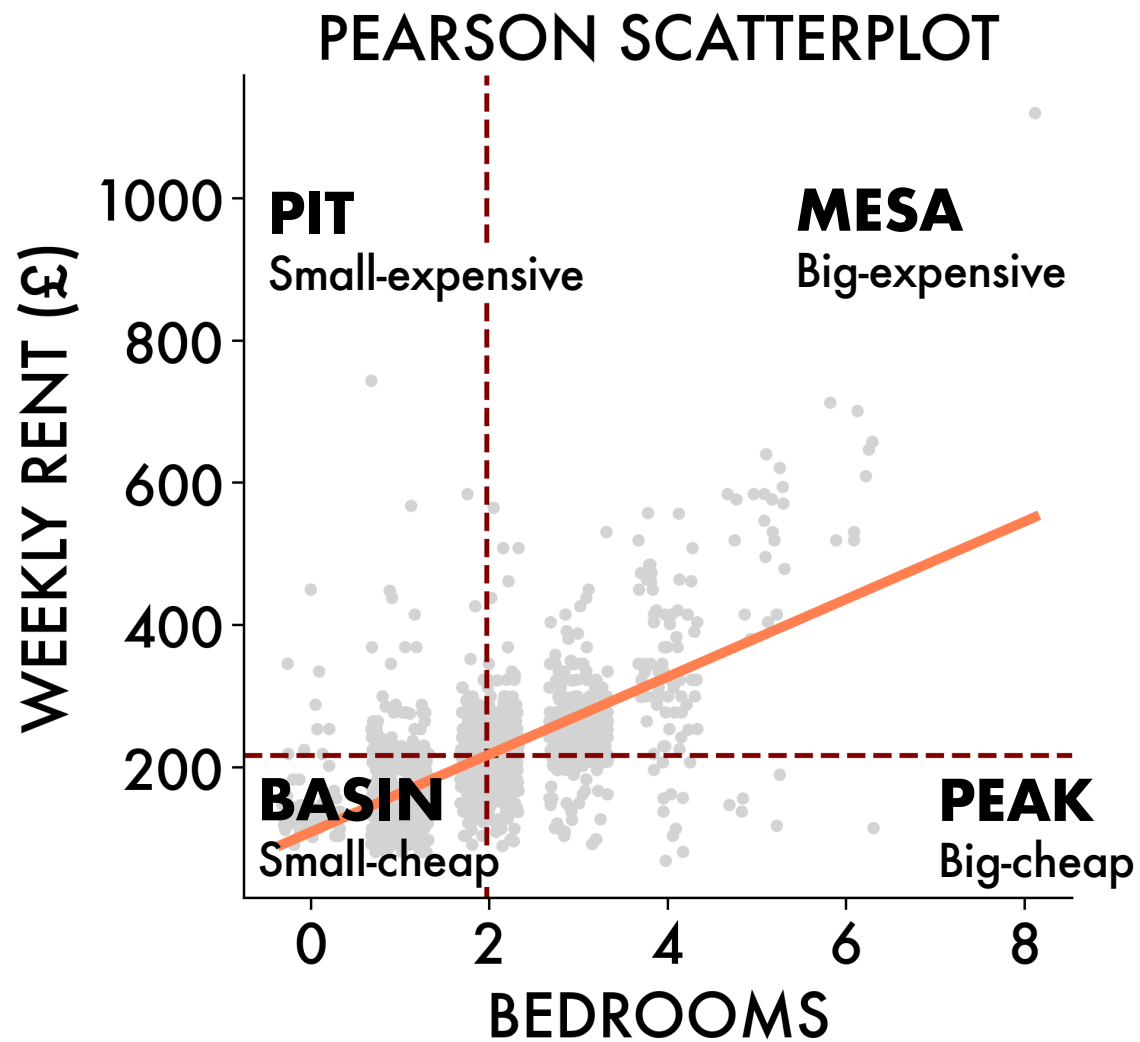
WEEKLY RENT (£)



BEDROOMS

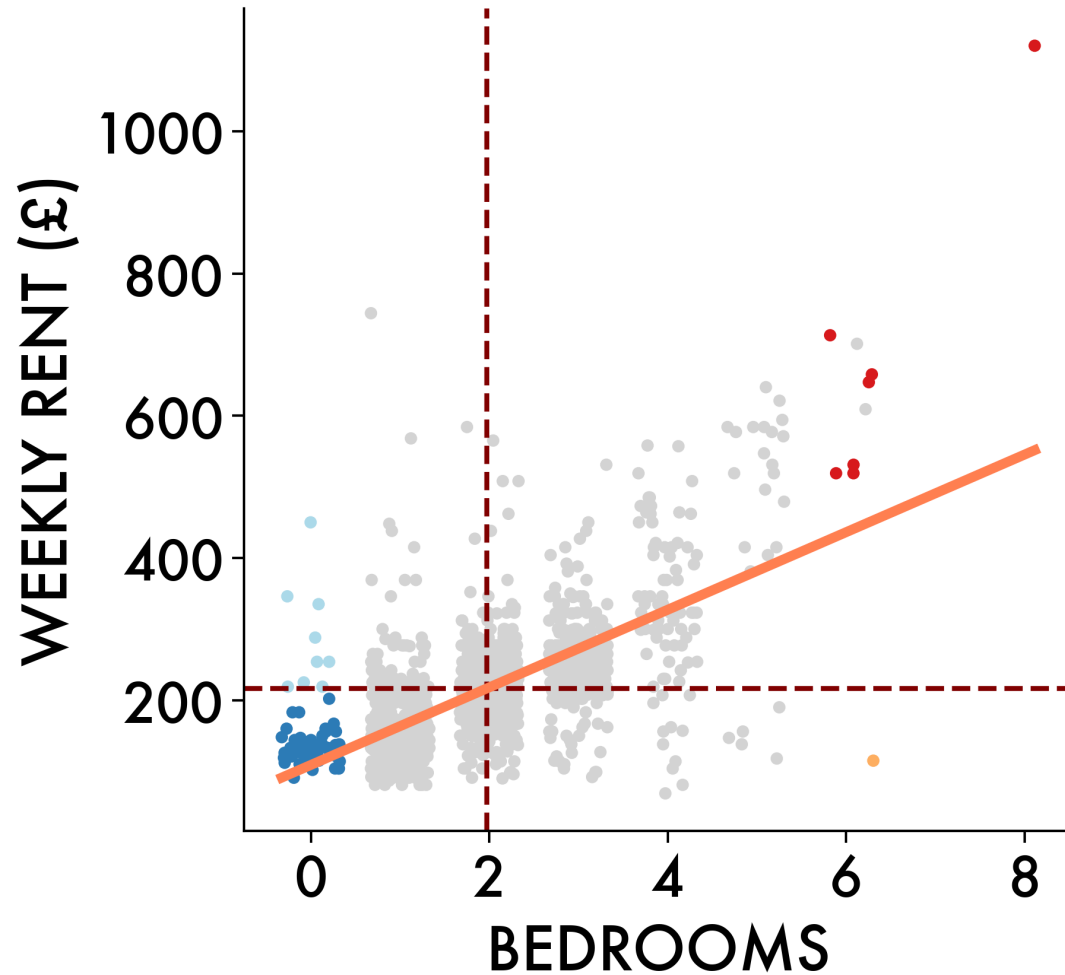


LOCAL PEARSON SCATTERPLOT

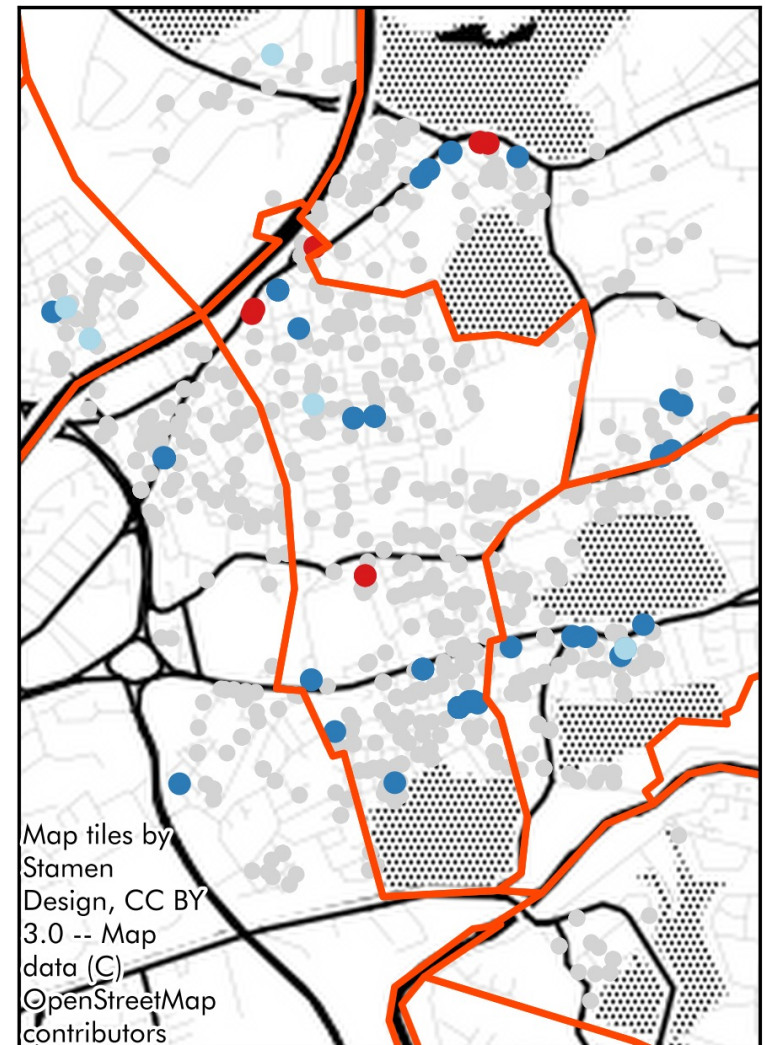


LOCAL PEARSON SCATTERPLOT

PEARSON SCATTERPLOT

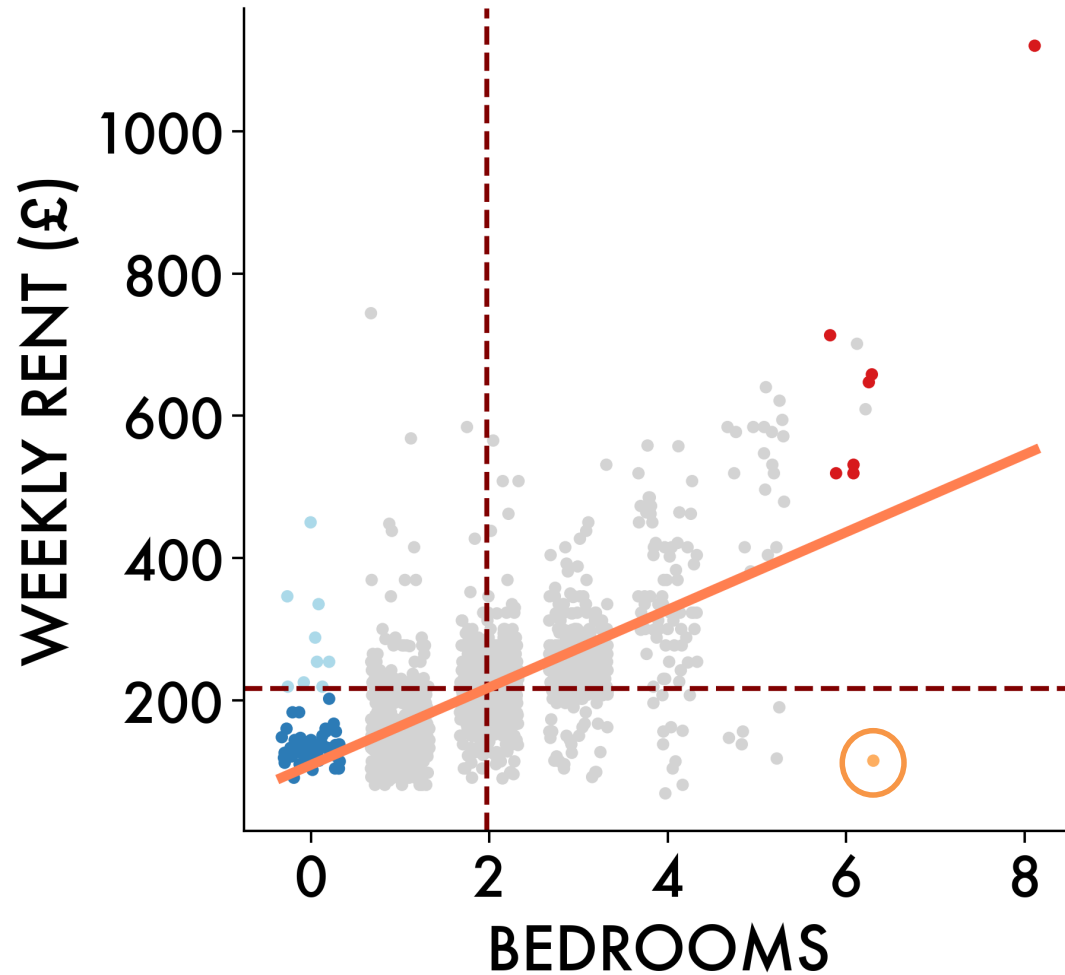


PEARSON CLASSES

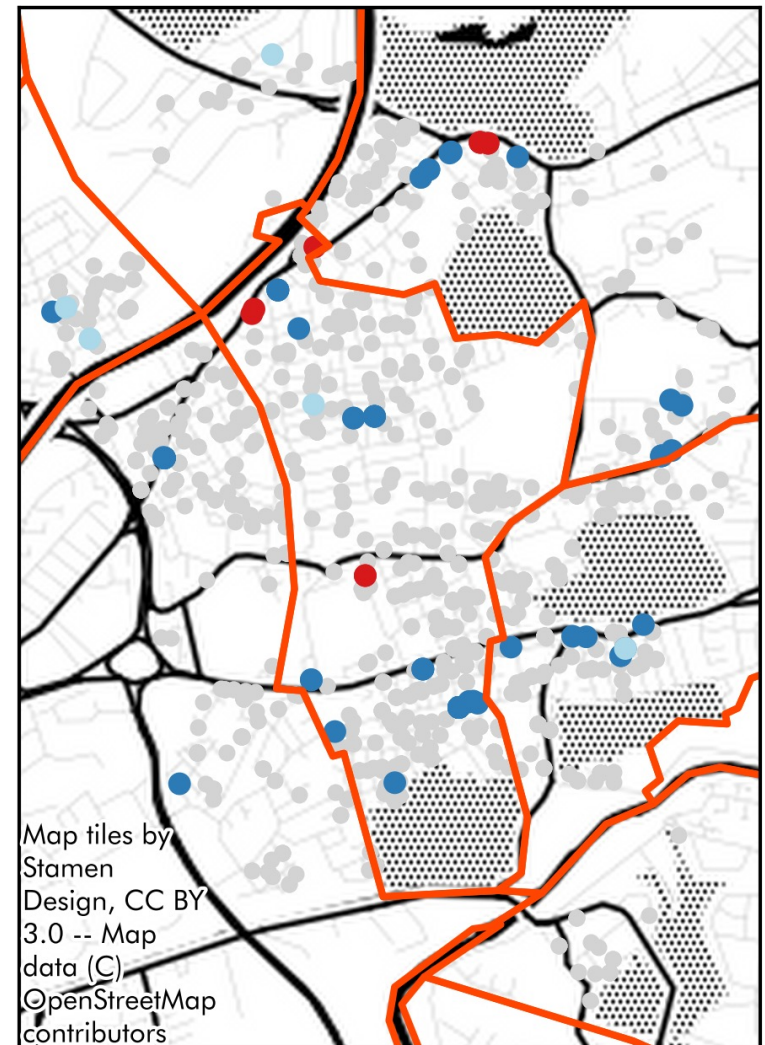


LOCAL PEARSON SCATTERPLOT

PEARSON SCATTERPLOT

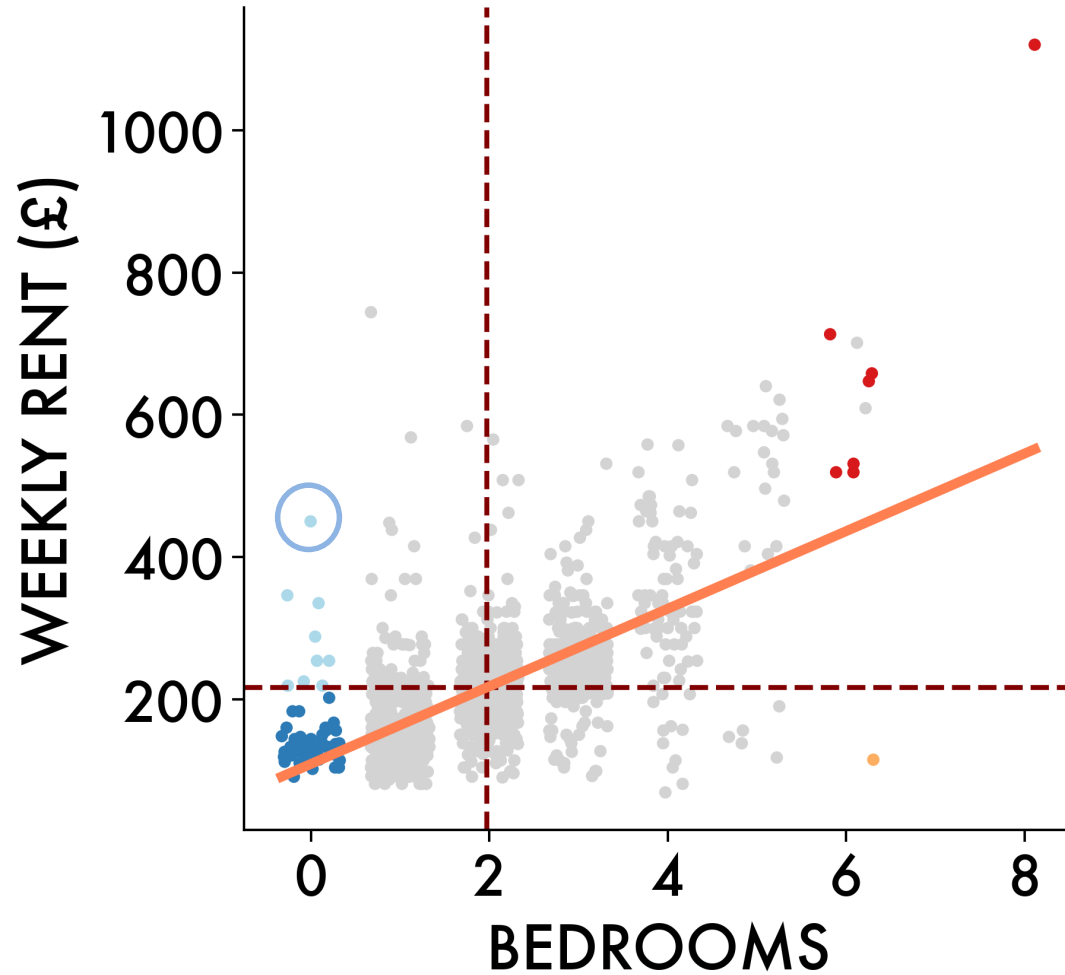


PEARSON CLASSES

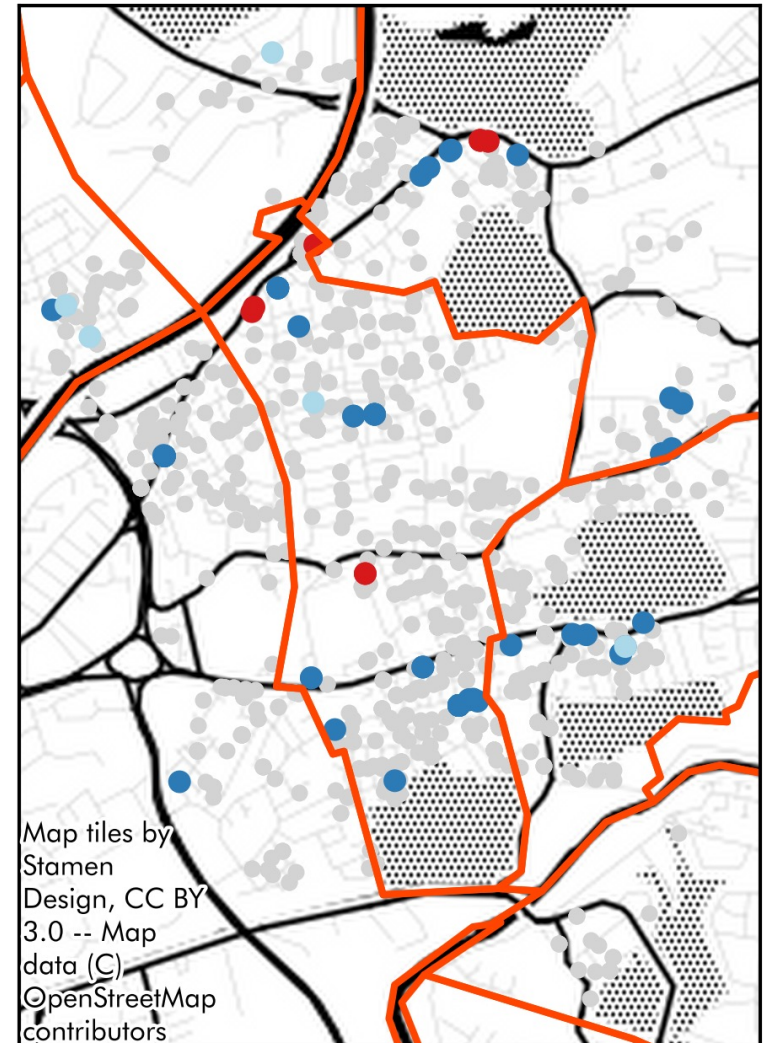


LOCAL PEARSON SCATTERPLOT

PEARSON SCATTERPLOT

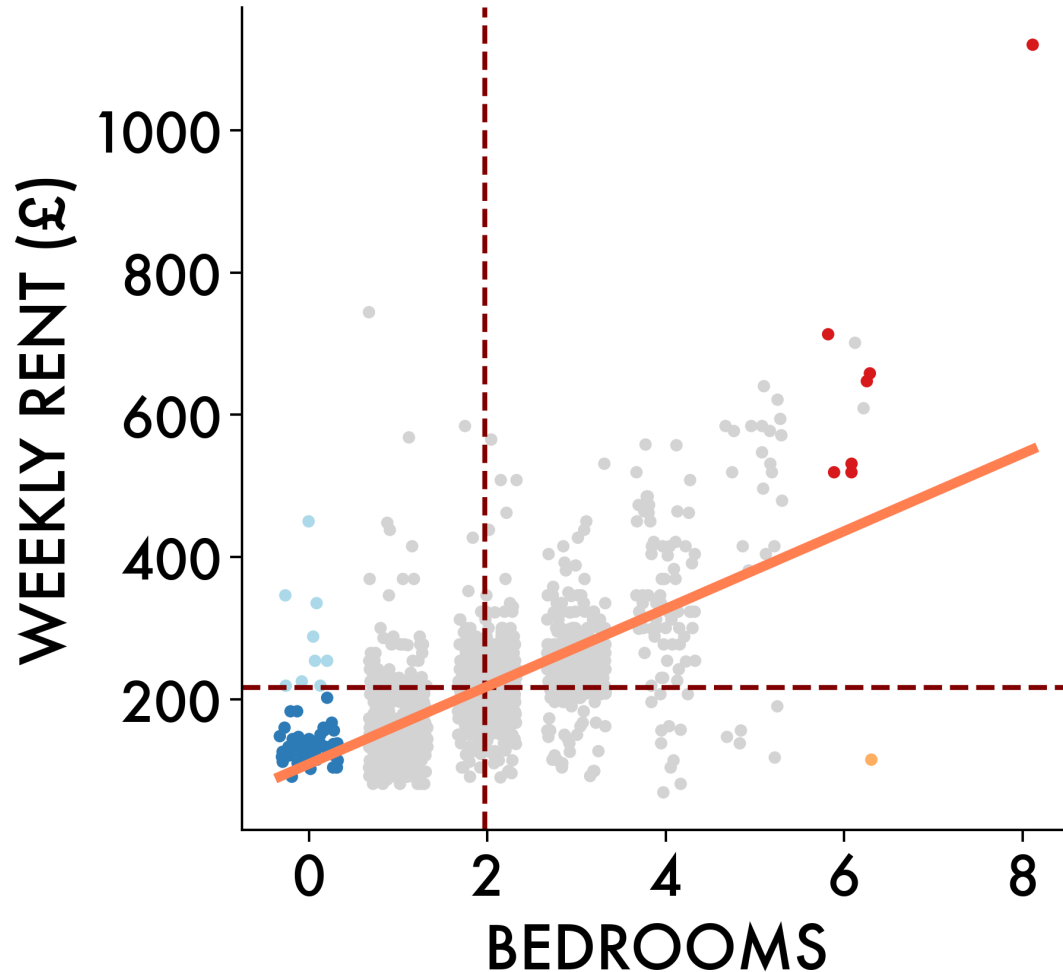


PEARSON CLASSES



LOCAL PEARSON SCATTERPLOT

PEARSON SCATTERPLOT



How can we find locally-unusual rents given the characteristics of properties nearby?

Local correlation only summarizes site by site

LOCAL PEARSON SCATTERPLOT

LOCAL CONFOUNDING

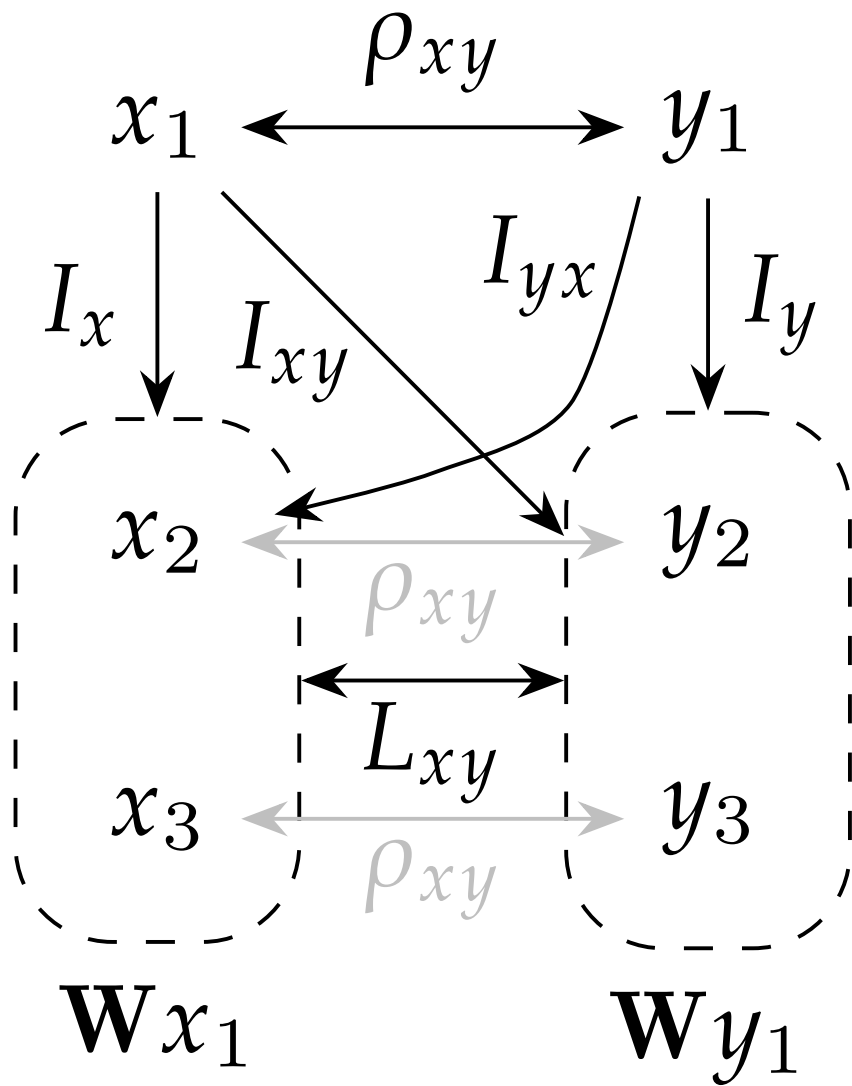
size effects on rent vary

PRIOR ART IN ESDA

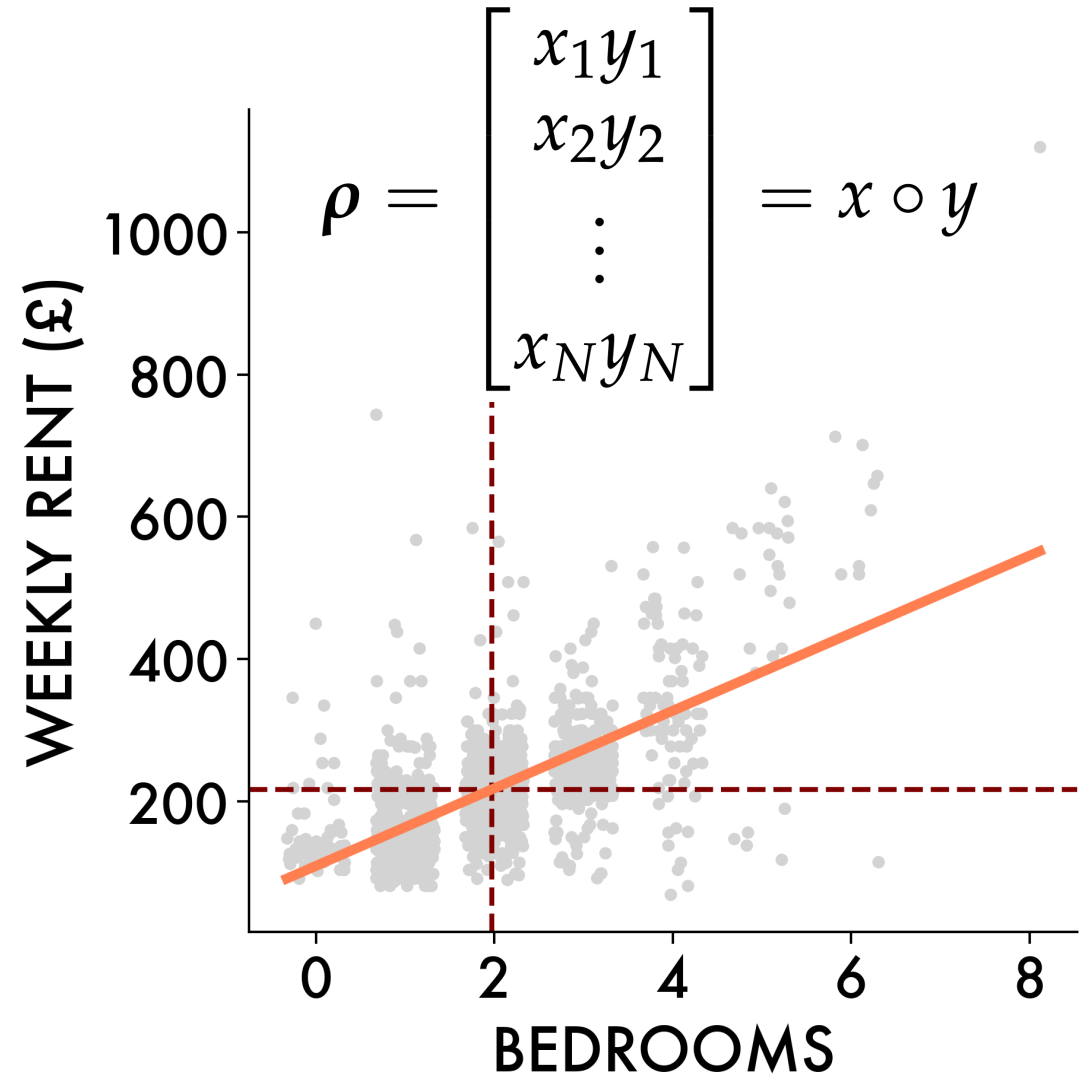
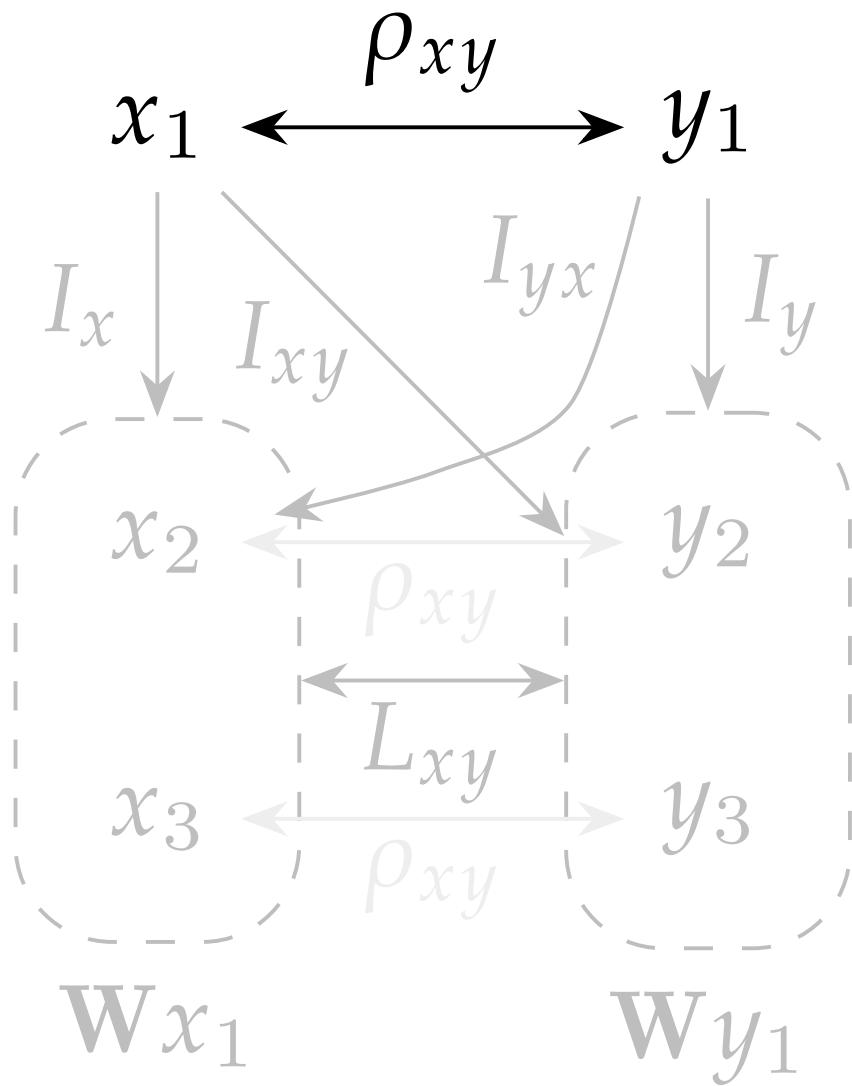
The Moran-Form Regression

PARTIAL CONDITIONING

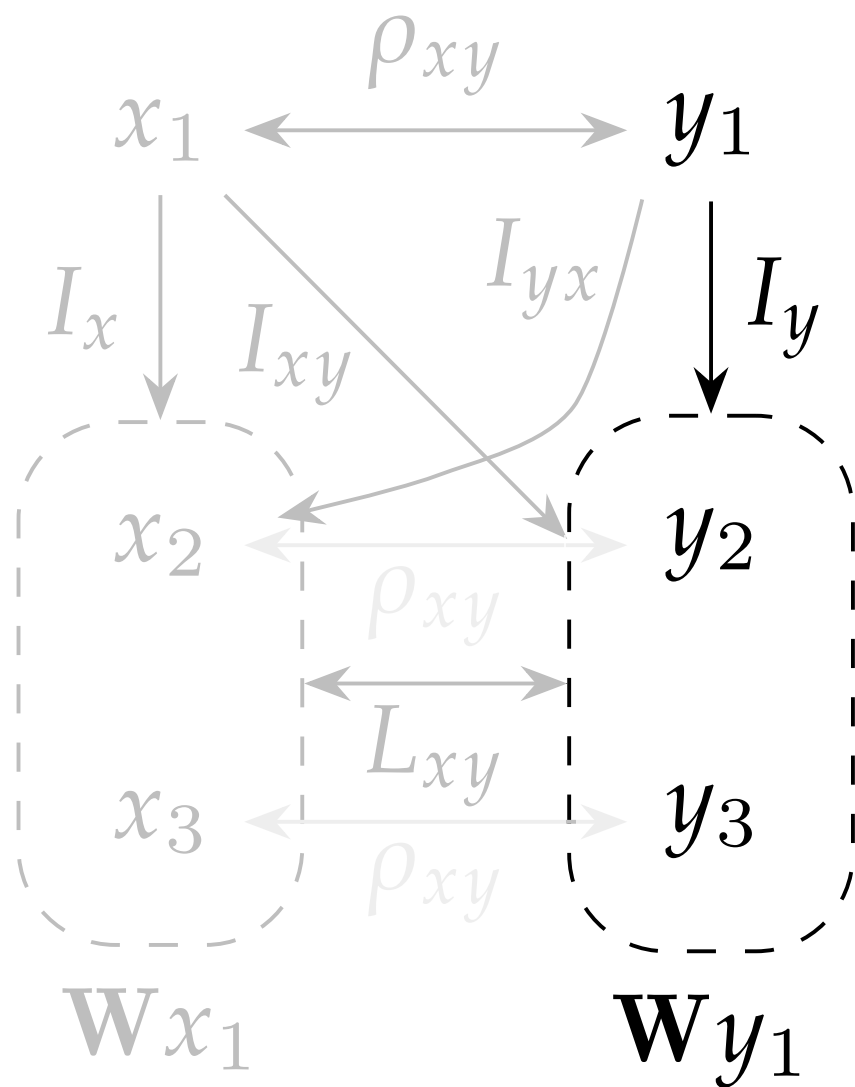
AFFORDABILITY OUTLIERS



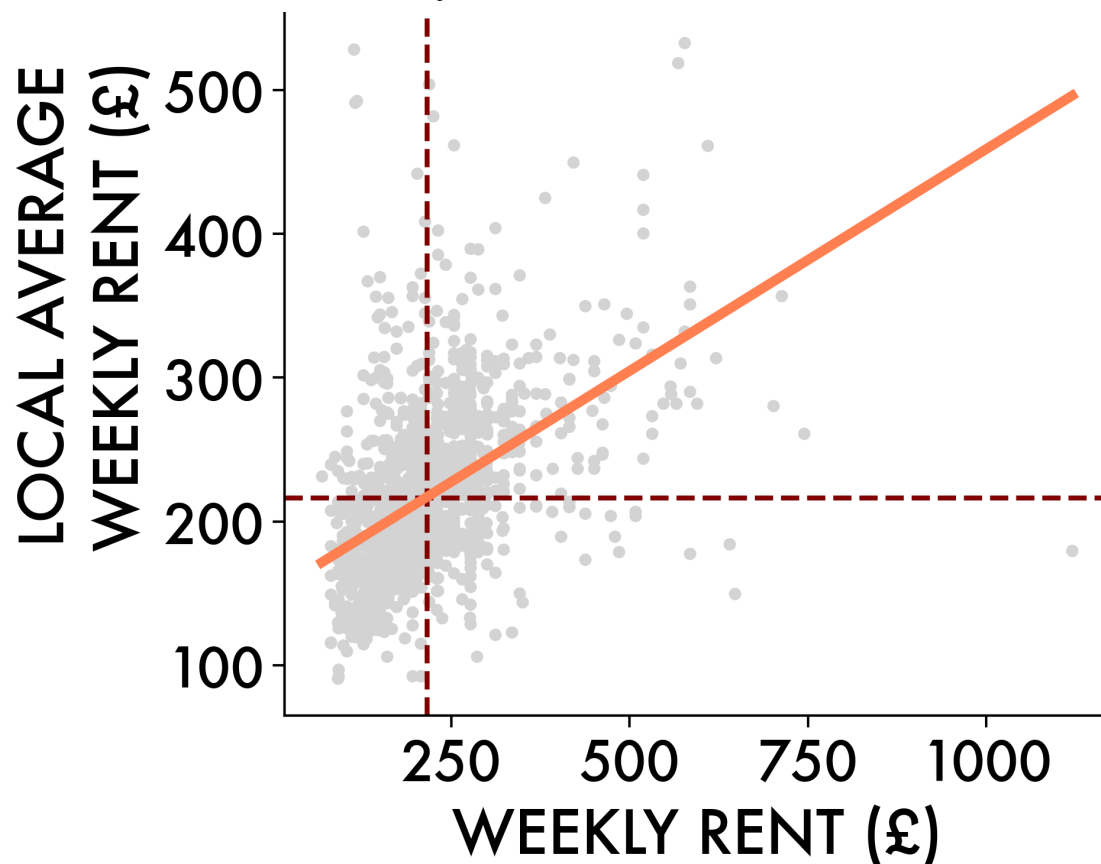
LOCAL COVARIANCE DAGS



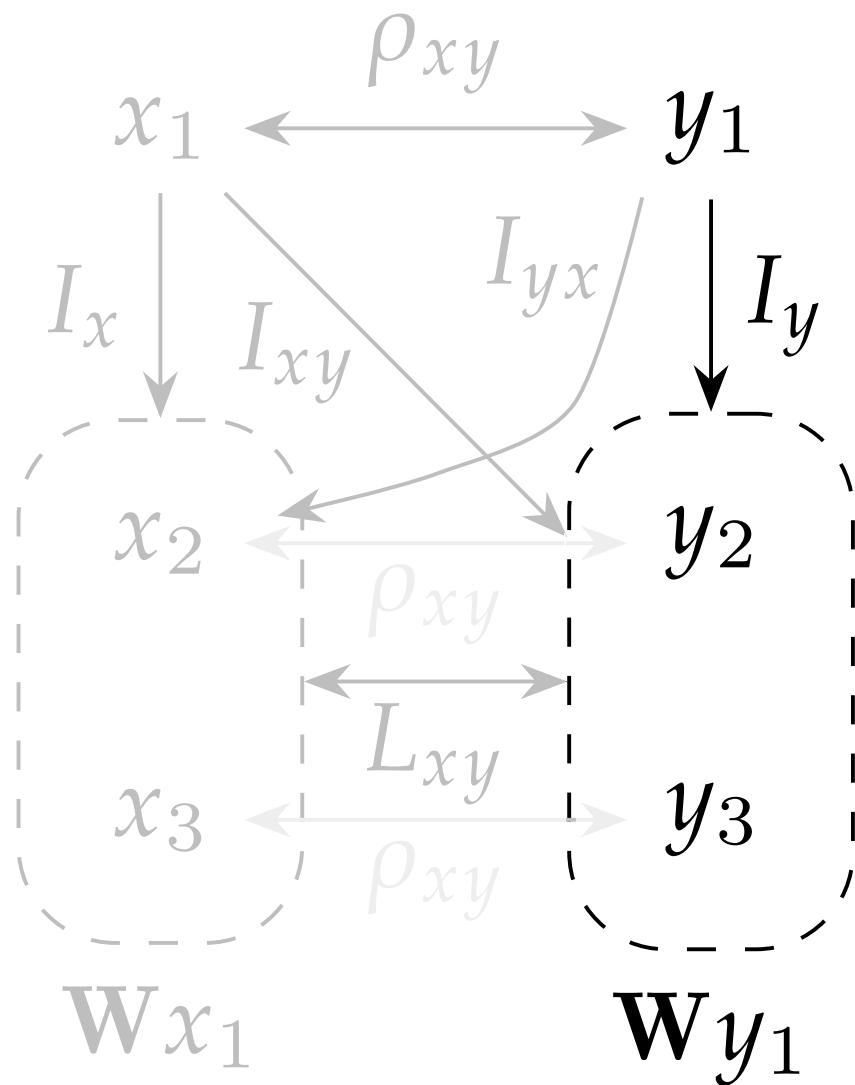
PEARSON CORRELATION



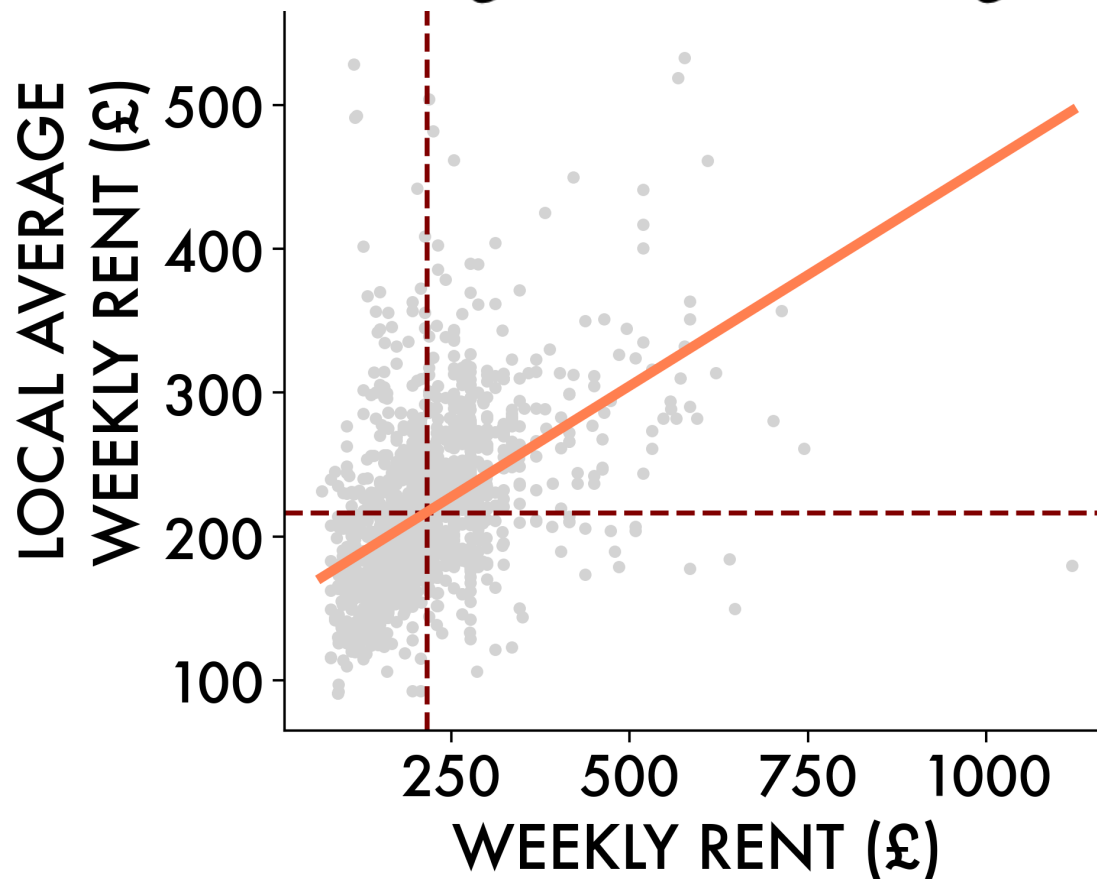
$$I = \sum_i^N \sum_j^N w_{ij} y_i y_j = y \cdot \mathbf{W} \cdot y$$



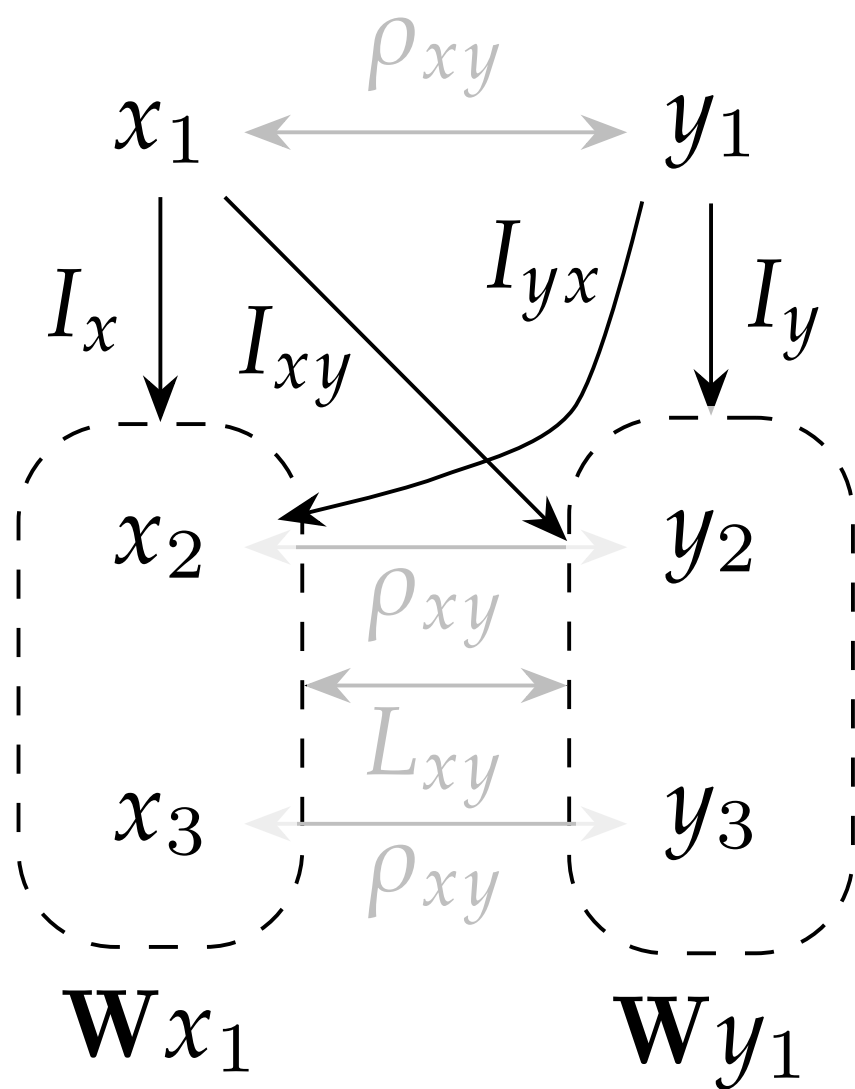
MORAN ASSOCIATION



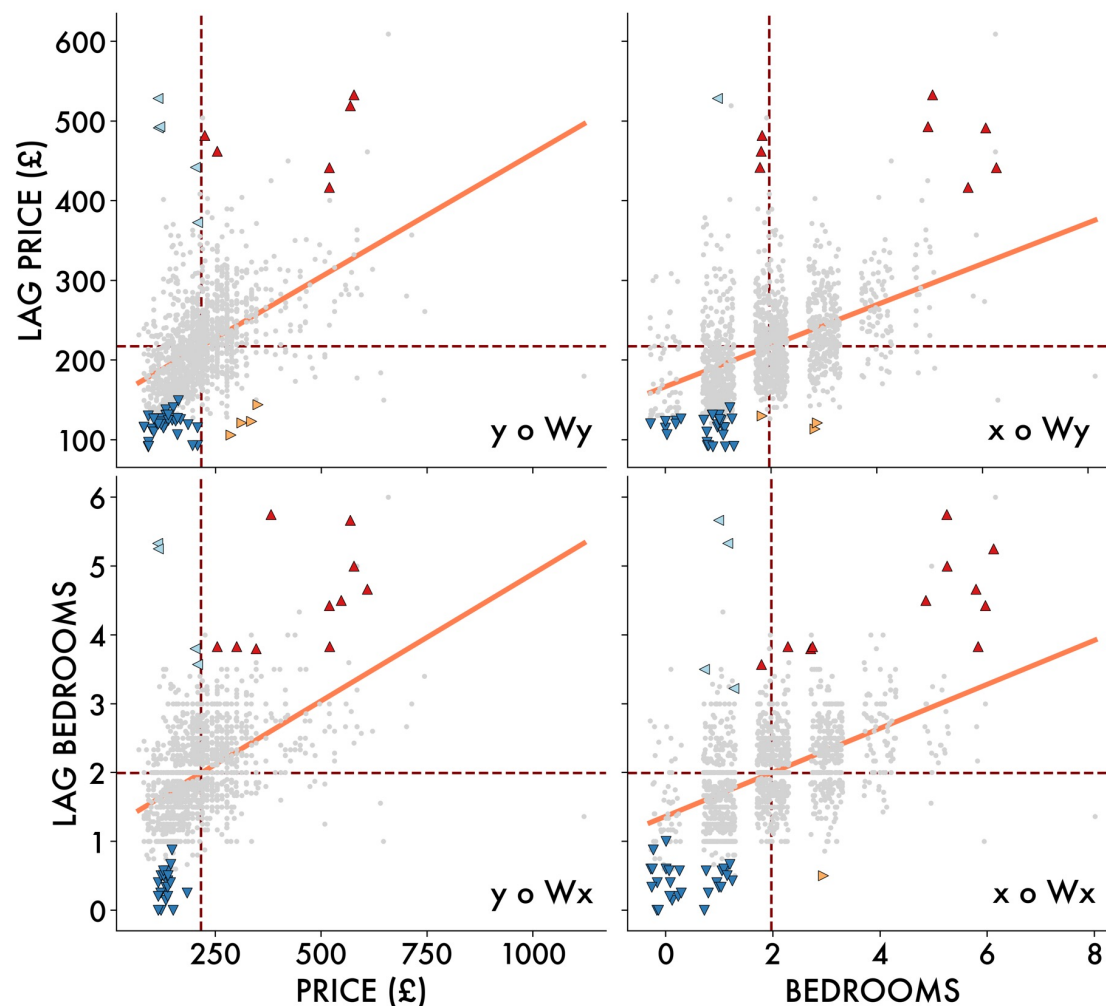
$$\mathbf{I} = y \circ \mathbf{W} \cdot y$$



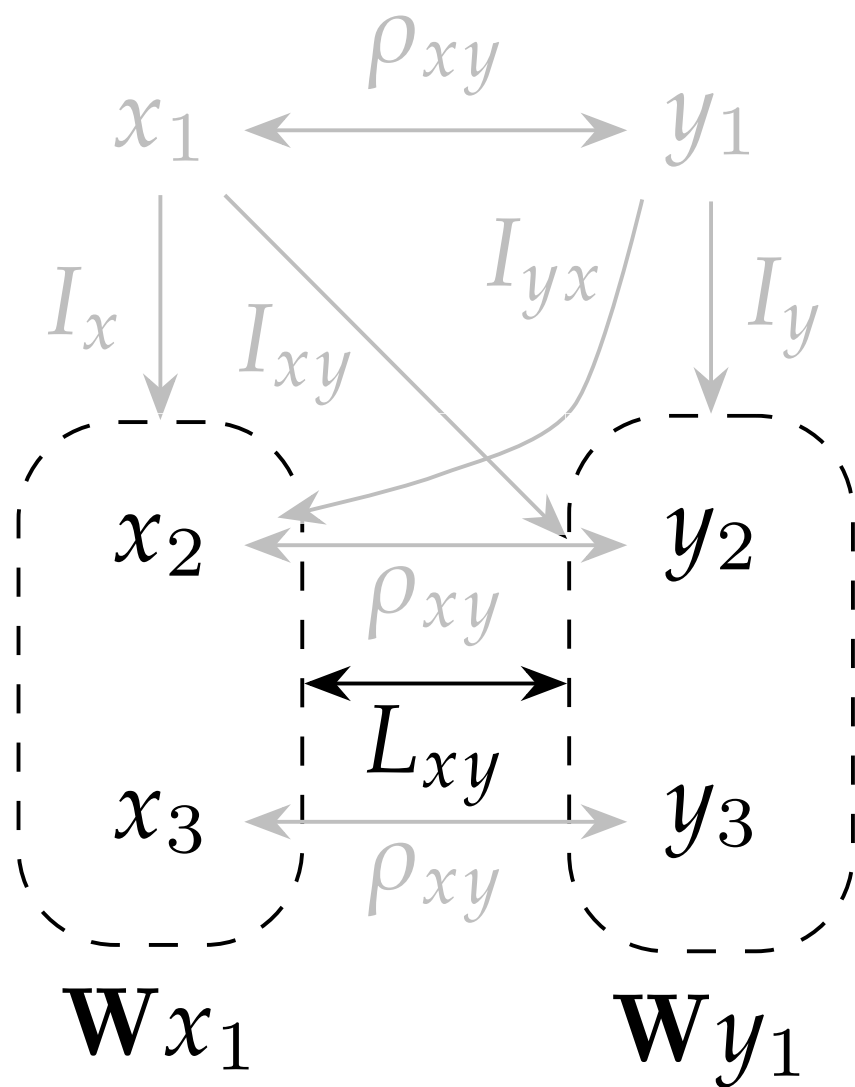
MORAN ASSOCIATION



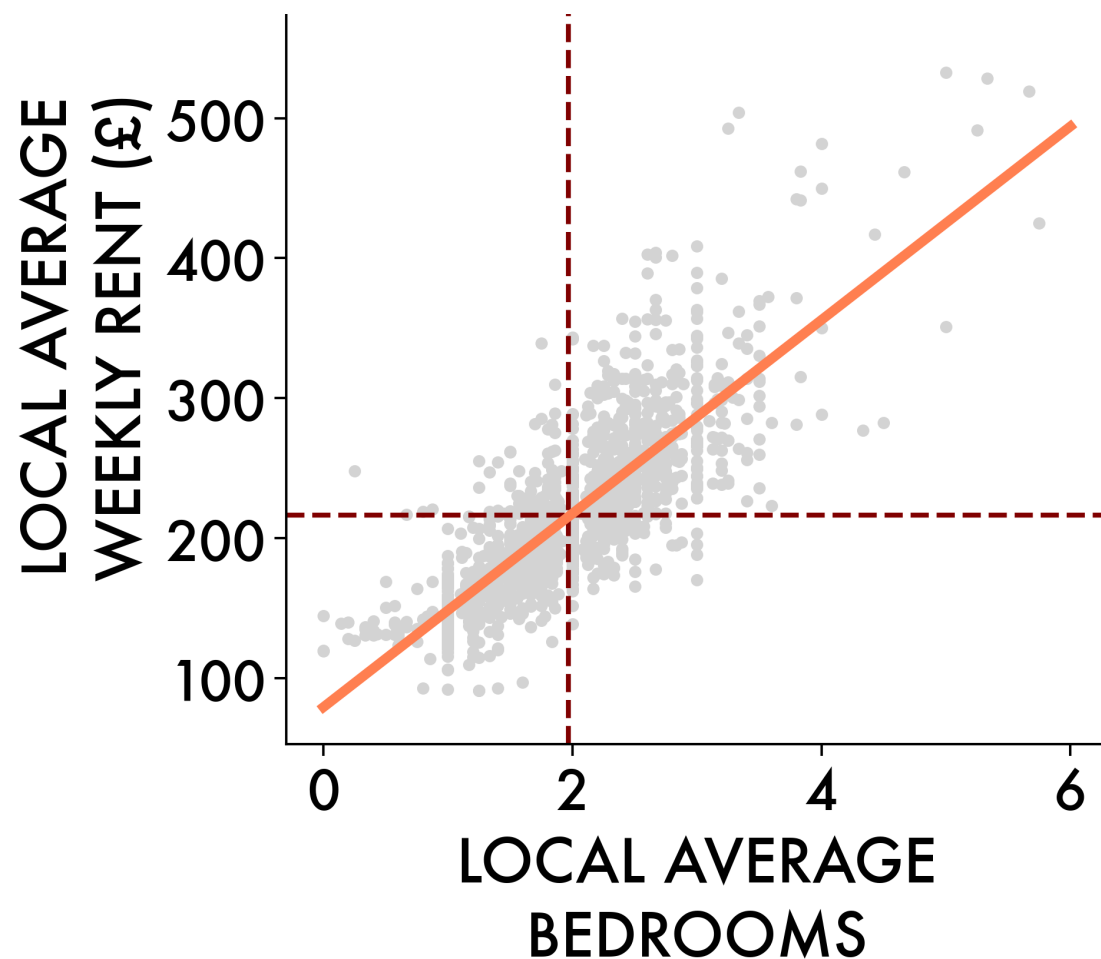
$$\mathbf{I}_{xy} = x \circ \mathbf{W} \cdot y$$



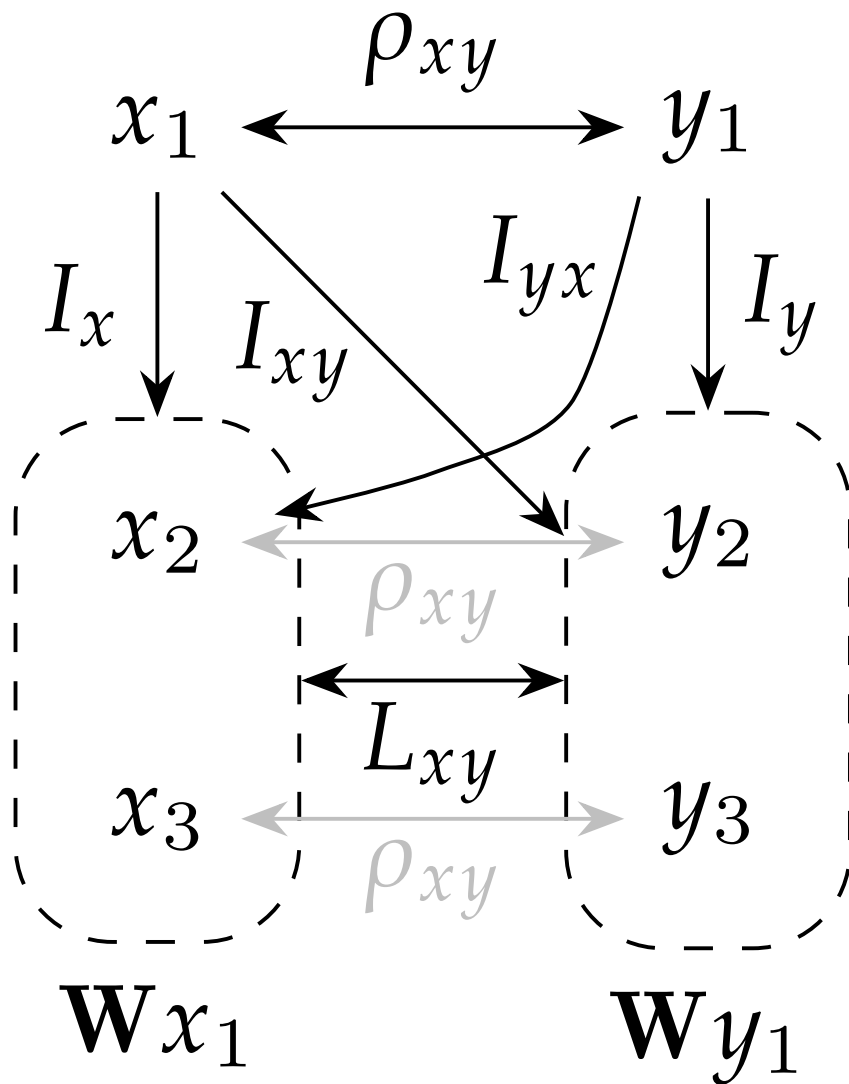
WARTENBERG MORAN MATRIX



$$\mathbf{L}_{xy} = \mathbf{W} \cdot x \circ \mathbf{W} \cdot y$$



LEE'S SPATIAL PEARSON



Is this house's rent unusual for its area, given its size?

$$\mathbf{I} = y \circ \mathbf{W} \cdot y$$

Is this house's rent unusual for its area?

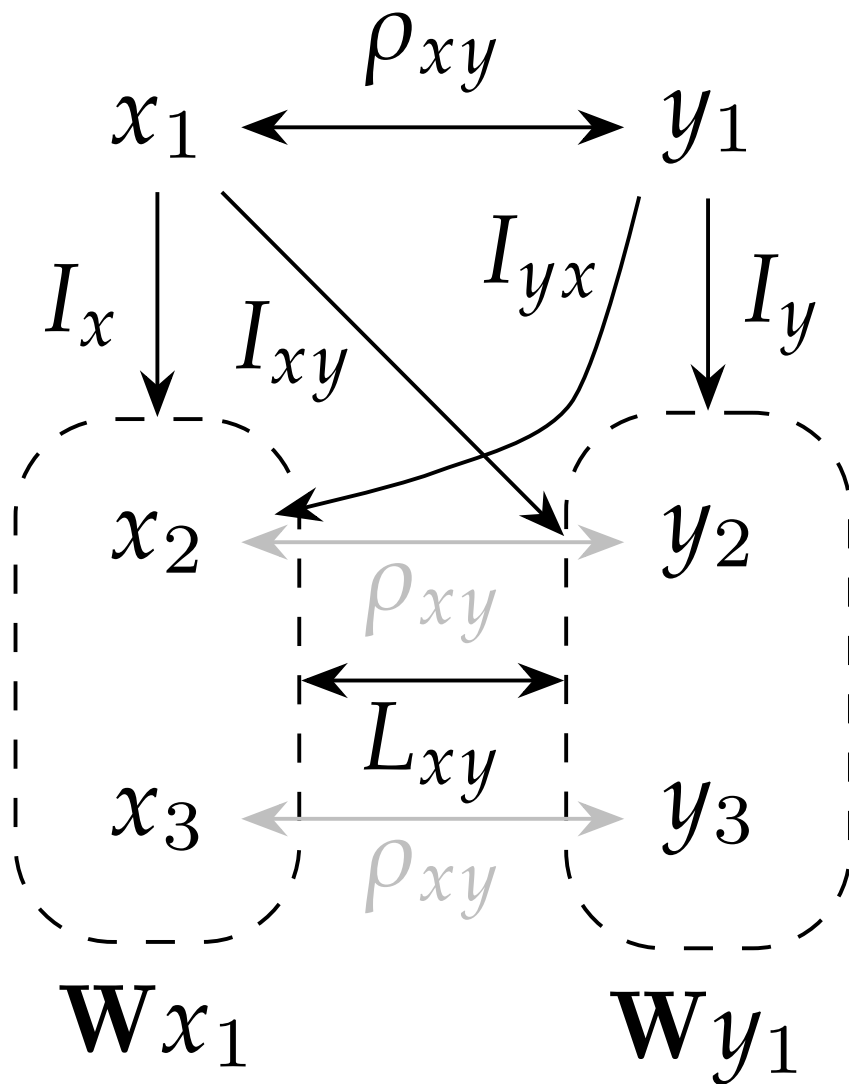
$$\mathbf{I}_{xy} = x \circ \mathbf{W} \cdot y$$

Is this lot size unusual for house rent in this area?

$$\mathbf{L}_{xy} = \mathbf{W} \cdot x \circ \mathbf{W} \cdot y$$

Are this area's rents unusual for this area's lot sizes?

DO ANY OF THESE WORK?



Is this house's rent unusual for its area, given its size?

$$\mathbf{I} = y \circ \mathbf{W} \cdot y$$

Is this house's rent unusual for its area?

$$\mathbf{I}_{xy} = x \circ \mathbf{W} \cdot y$$

Is this lot size unusual for house rent in this area?

$$\mathbf{L}_{xy} = \mathbf{W} \cdot x \circ \mathbf{W} \cdot y$$

Are this area's rents unusual for this area's lot sizes?

DO ANY OF THESE WORK? NOPE!

LOCAL CONFOUNDING

size effects on rent vary

PRIOR ART IN ESDA

The Moran-Form Regression

PARTIAL CONDITIONING

Three approaches to "control"

AFFORDABILITY OUTLIERS

$$y = (\mathbf{I} - \lambda \mathbf{W})^{-1} (x\beta + \epsilon)$$

THE SPATIAL LAG MODEL

$$y = (\mathbf{I} - \lambda \mathbf{W})^{-1} (x\beta + \epsilon)$$

Rents At Site Nearby Bedrooms

THE SPATIAL LAG MODEL

$$y = (\mathbf{I} - \lambda \mathbf{W})^{-1} (x\beta + \epsilon)$$

Rents At Site Nearby Bedrooms Premium?

THE SPATIAL LAG MODEL

$$y = (\mathbf{I} - \lambda \mathbf{W})^{-1} (x\beta + \epsilon)$$

Rents At Site Nearby Bedrooms Premium?

$$e = y - (\mathbf{I} - \hat{\lambda} \mathbf{W})^{-1} x \hat{\beta}$$

$$e = y - x \hat{\beta} - \hat{\lambda} \mathbf{W} x \hat{\beta} - \hat{\lambda}^2 \mathbf{W}^2 x \hat{\beta} + \dots$$

THE SPATIAL LAG MODEL

$$y = (\mathbf{I} - \lambda \mathbf{W})^{-1} (x\beta + \epsilon)$$

Rents At Site Nearby Bedrooms Premium?

$$e = y - (\mathbf{I} - \hat{\lambda} \mathbf{W})^{-1} x \hat{\beta}$$

$$e = y - x \hat{\beta} - \hat{\lambda} \mathbf{W} x \hat{\beta} - \hat{\lambda}^2 \mathbf{W}^2 x \hat{\beta} + \dots$$

Premium? Rents Site Size Nearby Sizes

THE SPATIAL LAG MODEL

$$y = (\mathbf{I} - \lambda \mathbf{W})^{-1} (x\beta + \epsilon)$$

Rents At Site Nearby Bedrooms Premium?

$$e = y - (\mathbf{I} - \hat{\lambda} \mathbf{W})^{-1} x \hat{\beta}$$

Still “filtered” by $(\mathbf{I} - \lambda \mathbf{W})^{-1}$

Can ID unusual prices

Not a *covariance*

THE SPATIAL LAG MODEL

$$y = x\beta + \epsilon$$

Rents

Bedrooms

Premium at site

AUXILIARY REGRESSION

$$y = x\beta + \epsilon$$

$$e = y - x\hat{\beta}$$

AUXILIARY REGRESSION

$$y = x\beta + \epsilon$$

$$e = y - x\hat{\beta}$$

$$\mathbf{I}_{x \rightarrow y} = e \circ \mathbf{W}e$$

Is this premium unusual given the premium people are paying for space in the area?

AUXILIARY REGRESSION

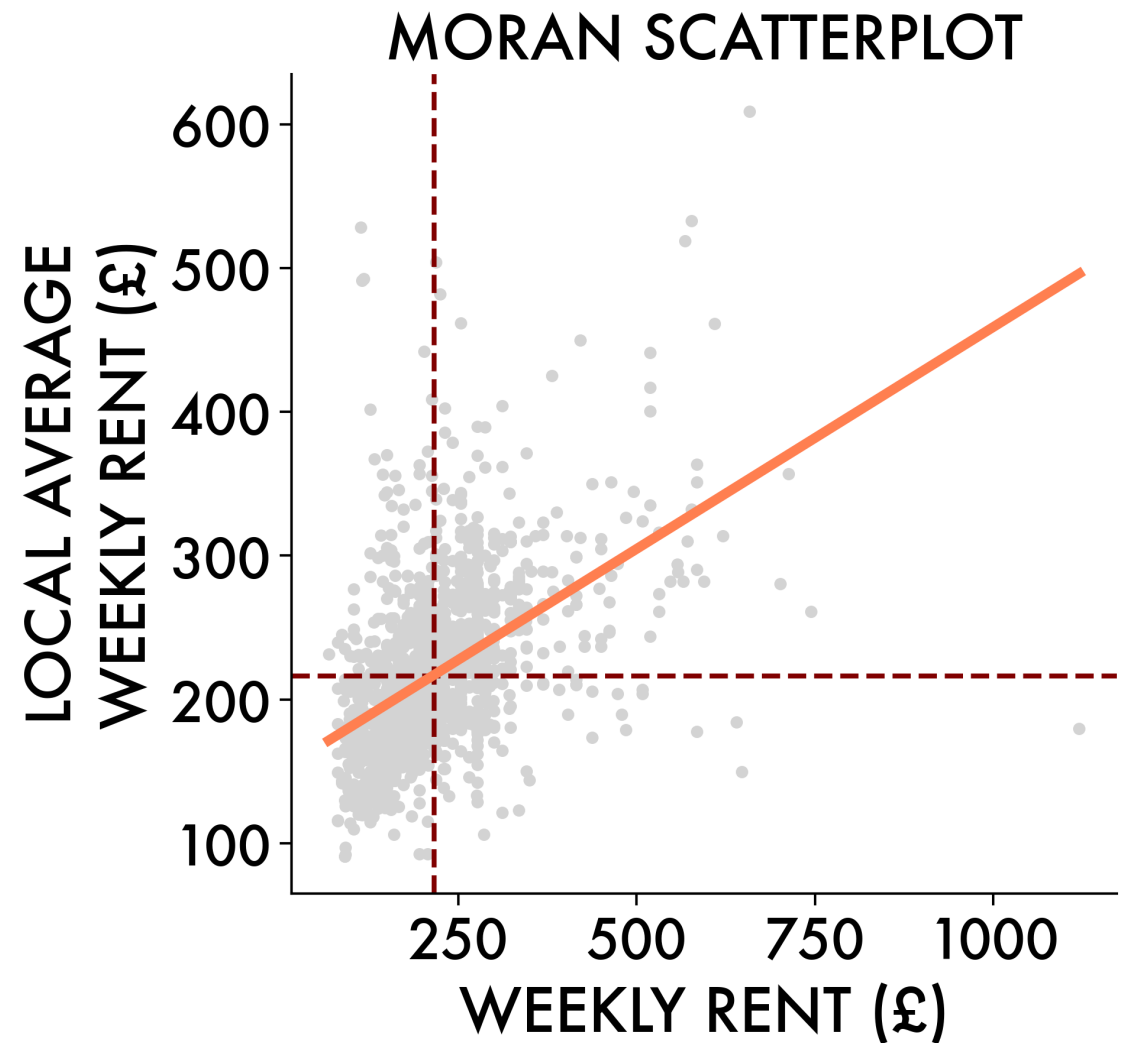
$$y = x\beta + \epsilon$$

$$e = y - x\hat{\beta}$$

$$\mathbf{I}_{x \rightarrow y} = e \circ \mathbf{W}e$$

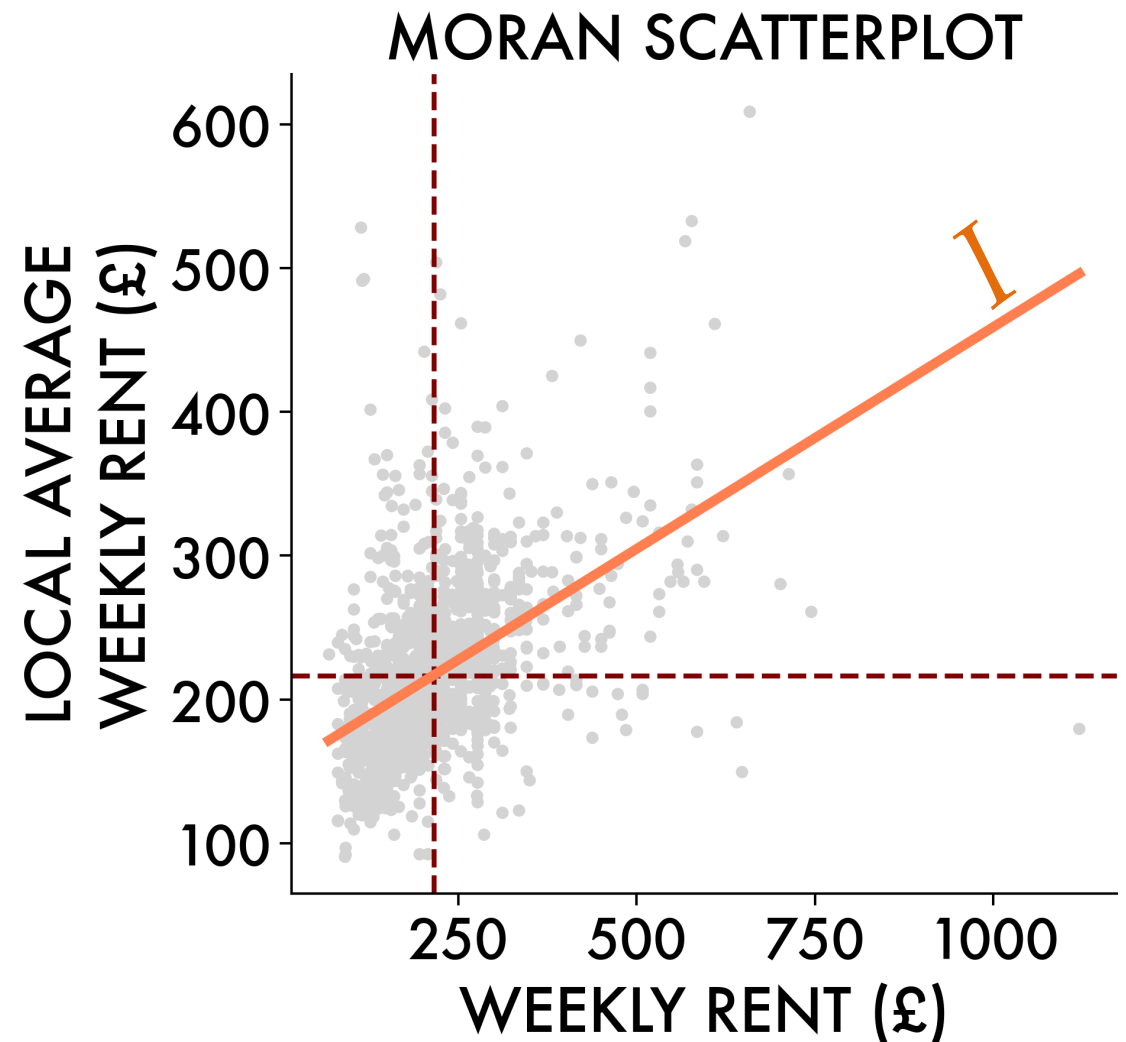
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AUXILIARY REGRESSION



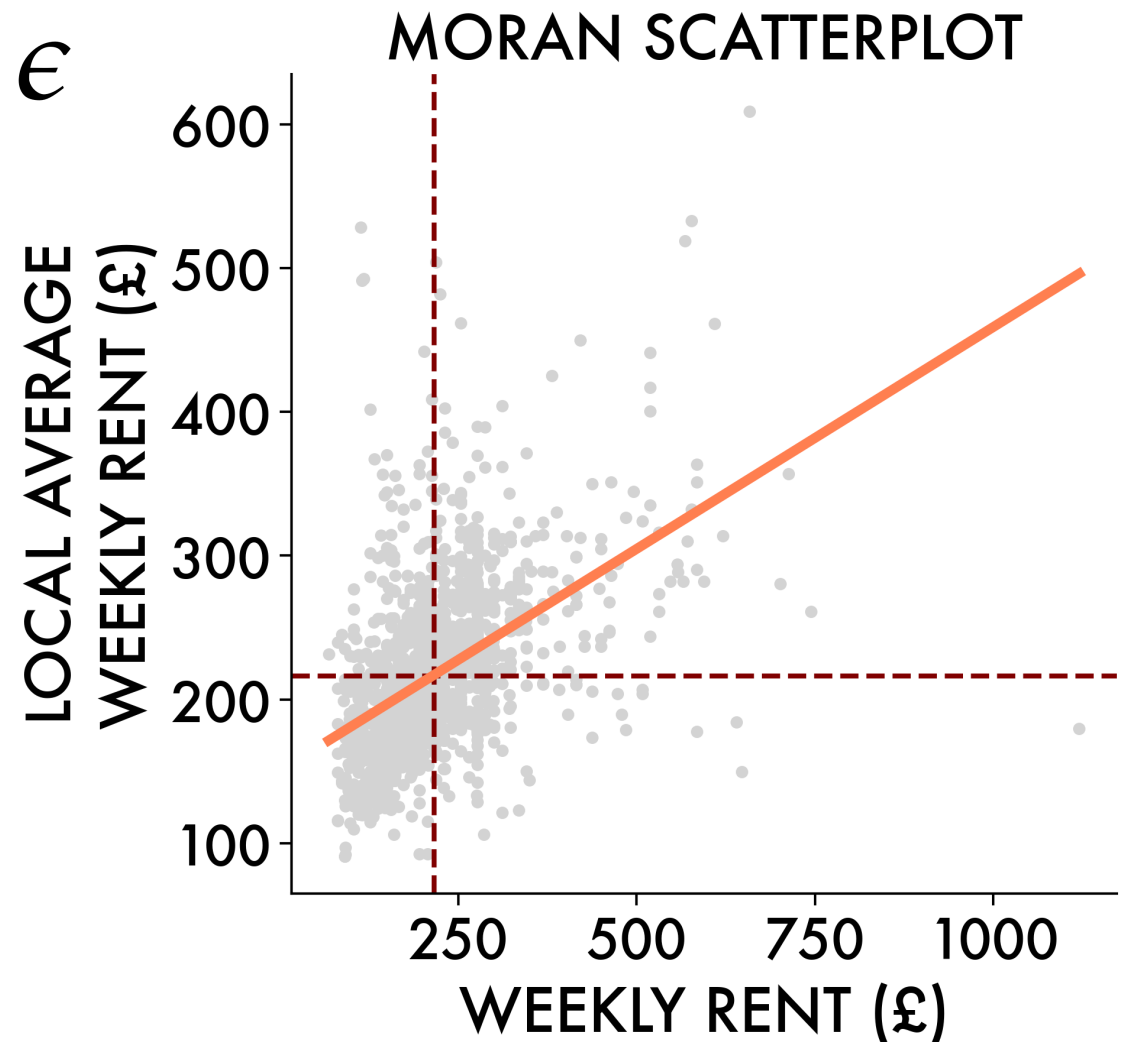
MORAN-FORM REGRESSIONS

$$Wy = Iy + \epsilon$$



MORAN-FORM REGRESSIONS

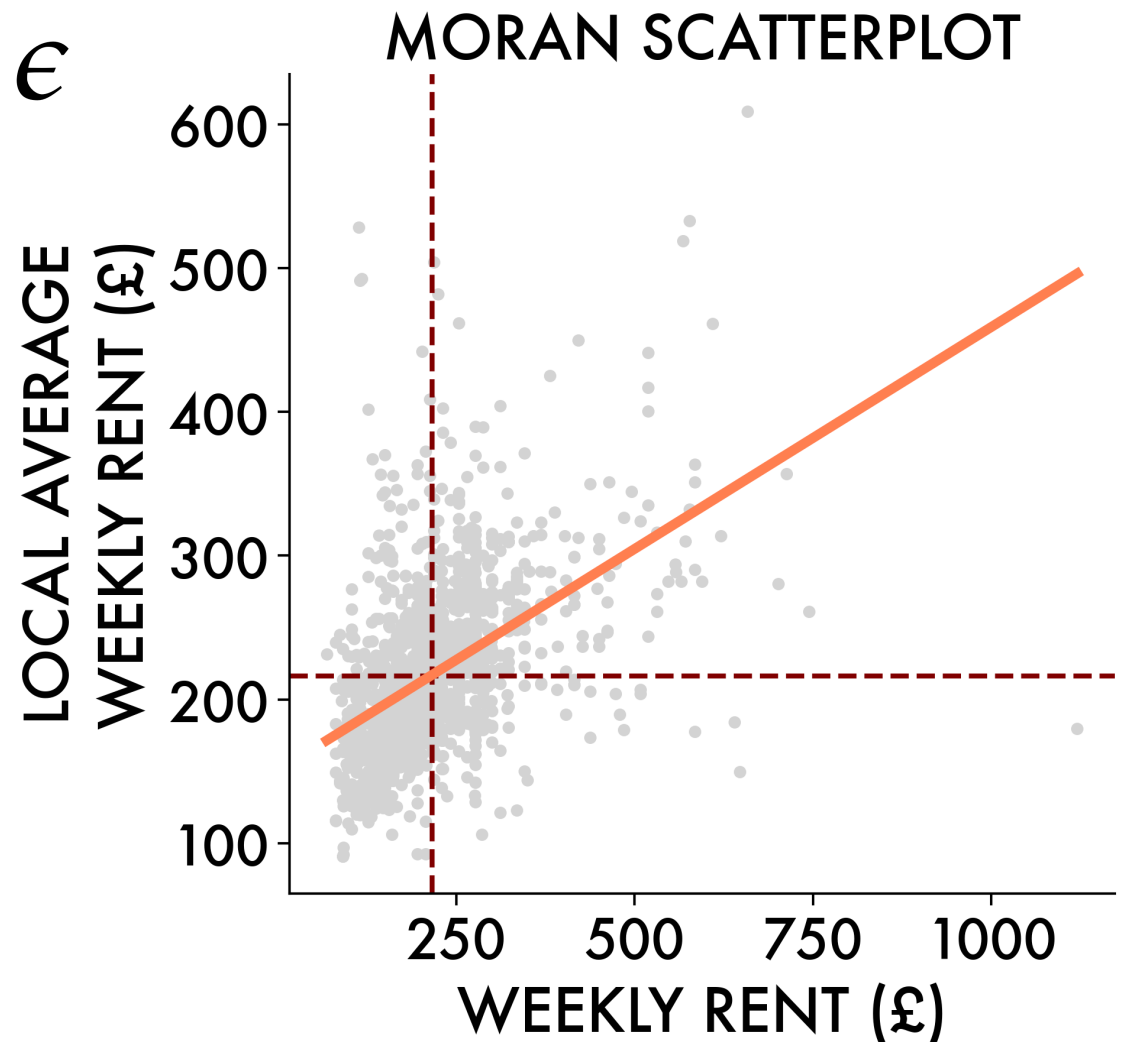
$$Wy = Iy + x\beta + \epsilon$$



MORAN-FORM REGRESSIONS

$$Wy = Iy + x\beta + \epsilon$$

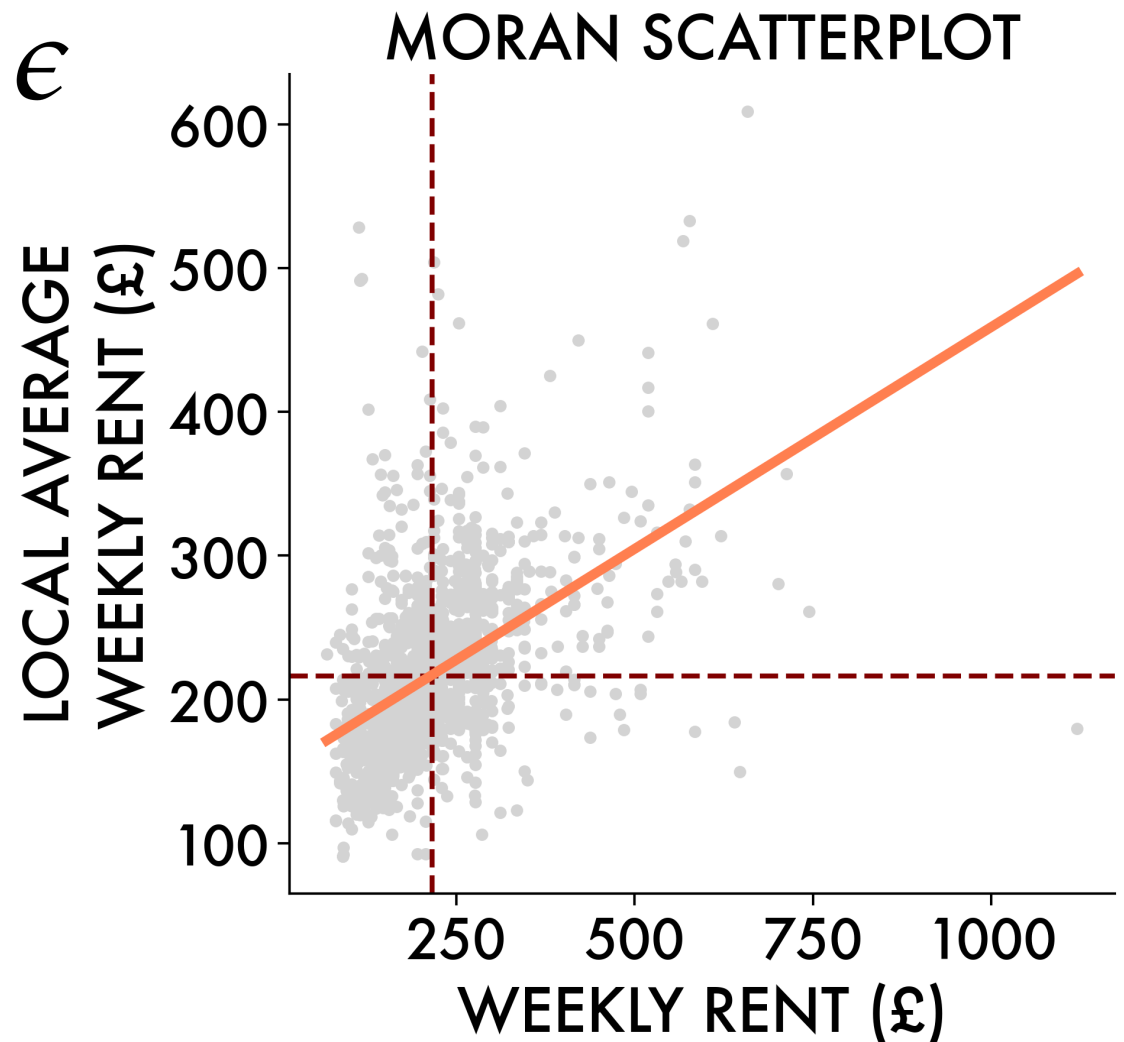
What is the relationship between a house's rent and nearby rents, holding the house's size constant?



MORAN-FORM REGRESSIONS

$$W_y = Iy + x\beta + \epsilon$$

What is the relationship between a house's rent and nearby rents, holding the house's size constant? ✓



MORAN-FORM REGRESSIONS

$$\mathbf{W}y = Iy + x\beta + \epsilon \quad \mathbf{W}\vec{y} = \mathbf{W} \begin{bmatrix} y & y & \dots \end{bmatrix}$$

$$\mathbf{D} = \begin{bmatrix} y & x \end{bmatrix}$$

$$\mathbf{I}_{y|x} = (\mathbf{D}'\mathbf{D})^{-1}\mathbf{D} \circ \mathbf{W}\vec{y}$$

A LOCAL “CETERIS PARIBUS” STATISTIC

$$\mathbf{W}y = Iy + x\beta + \epsilon \quad \mathbf{W}\vec{y} = \mathbf{W} \begin{bmatrix} y & y & \dots \end{bmatrix}$$

$$\mathbf{D} = \begin{bmatrix} y & x \end{bmatrix}$$

$$\mathbf{I}_{y|x} = (\mathbf{D}'\mathbf{D})^{-1}\mathbf{D} \circ \mathbf{W}\vec{y}$$

$$\mathbf{I}_{y|x} \propto \mathbf{I}_y - \rho\mathbf{I}_{xy}$$

A LOCAL “CETERIS PARIBUS” STATISTIC

$$\mathbf{I}_{y|x} \propto \mathbf{I}_y - \rho \mathbf{I}_{xy}$$

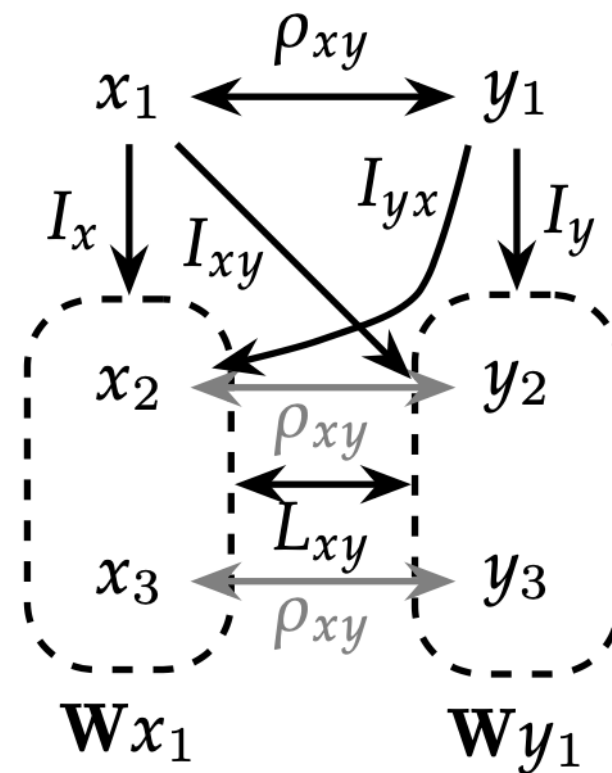
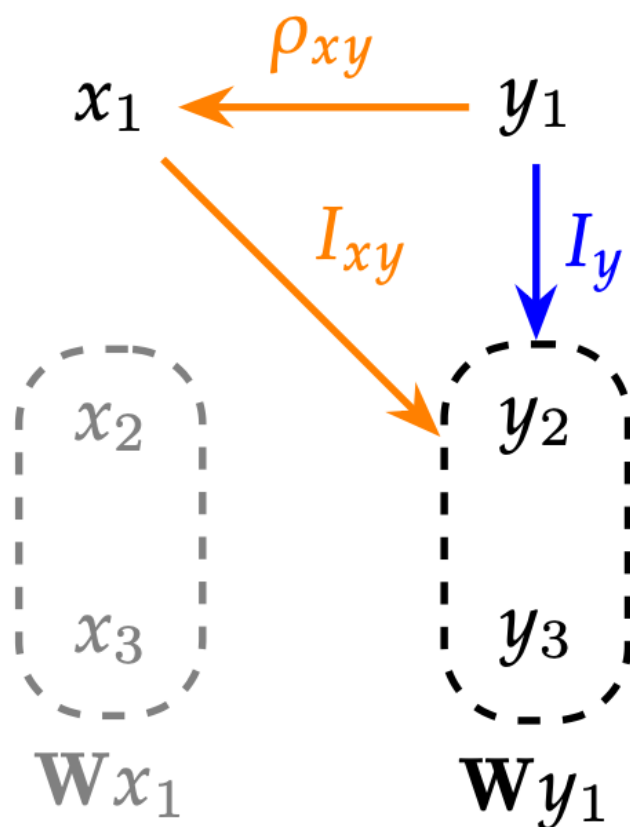
THE PARTIAL LOCAL MORAN

Spatial association within y

Adjusted by x 's association
with nearby y and y itself

$$\mathbf{I}_{y|x} \propto \mathbf{I}_y - \rho \mathbf{I}_{xy}$$

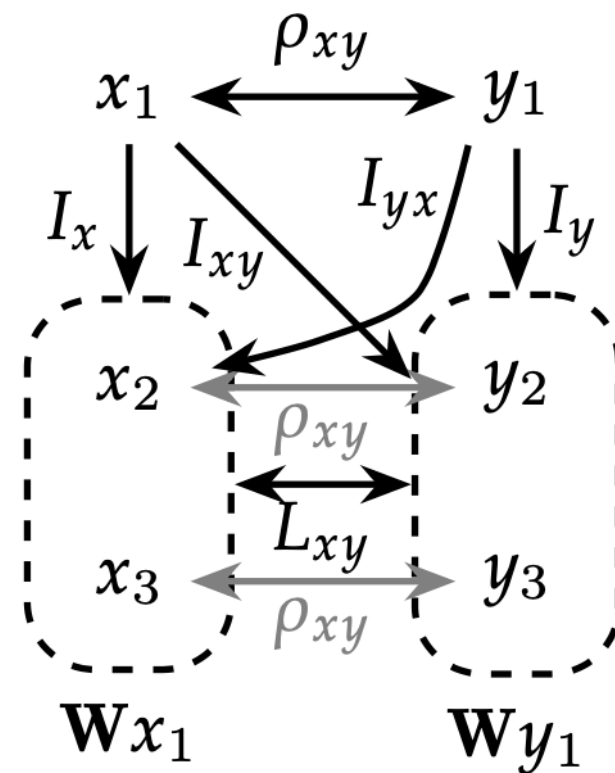
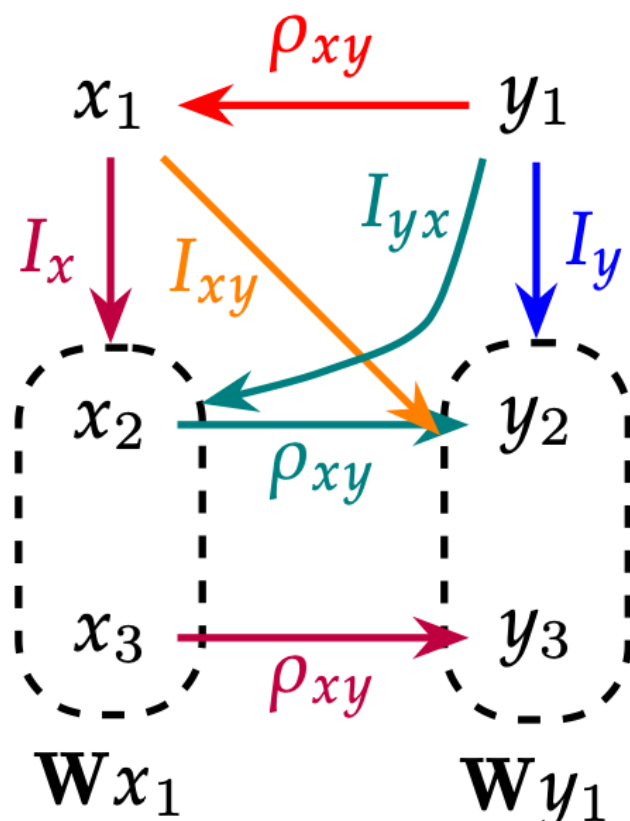
PARTIAL LOCAL MORAN



Classic Statistics

$$\mathbf{I}_{y|x} \propto \underline{\mathbf{I}_y} - \underline{\rho \mathbf{I}_{xy}}$$

PARTIAL LOCAL MORAN



$$I_{x \rightarrow y} \propto \underline{I_y} - \underline{\rho I_{xy}} - \underline{\rho I_{yx}} + \underline{\rho^2 I_x}$$

AUXILIARY REGRESSION MORAN IS FULL CONDITIONING

$$Wy = X\beta + Iy + e$$

HIGHER DIMENSIONAL PARTIAL CONDITIONING BY FWL

$$Wy = X\beta + Iy + e$$

Now have p features
needing adjustment

HIGHER DIMENSIONAL PARTIAL CONDITIONING BY FWL

$$Wy = X\beta + Iy + e$$

$$\mathbf{M} = \mathbf{I} - \mathbf{X}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$$

Residual Maker Matrix, so-called because $\mathbf{M}y = e$

HIGHER DIMENSIONAL PARTIAL CONDITIONING BY FWL

$$Wy = X\beta + Iy + e$$

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Residual Maker Matrix, so-called because $\mathbf{M}y = e$

$$\mathbf{M}Wy = I_{y|x}\mathbf{M}y + \epsilon$$

HIGHER DIMENSIONAL PARTIAL CONDITIONING BY FWL

$$\mathbf{M}W_y = I_{y|x} \mathbf{M}y + \epsilon$$

$$I_{y|x} \propto (\mathbf{M}y)' \mathbf{M}W_y$$

HIGHER DIMENSIONAL PARTIAL CONDITIONING BY FWL

$$\mathbf{M}\mathbf{W}y = I_{y|x}\mathbf{M}y + \epsilon$$

$$I_{y|x} \propto (\mathbf{M}y)' \mathbf{M}\mathbf{W}y$$

$$\propto y' \mathbf{W}y - (\mathbf{X}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'y)' \mathbf{W}y$$

$$I_y$$

HIGHER DIMENSIONAL PARTIAL CONDITIONING BY FWL

$$\mathbf{M}\mathbf{W}y = I_{y|x}\mathbf{M}y + \epsilon$$

$$I_{y|x} \propto (\mathbf{M}y)' \mathbf{M}\mathbf{W}y$$

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$$I_y \quad (\mathbf{X}'\beta)' \mathbf{W}y$$

HIGHER DIMENSIONAL PARTIAL CONDITIONING BY FWL

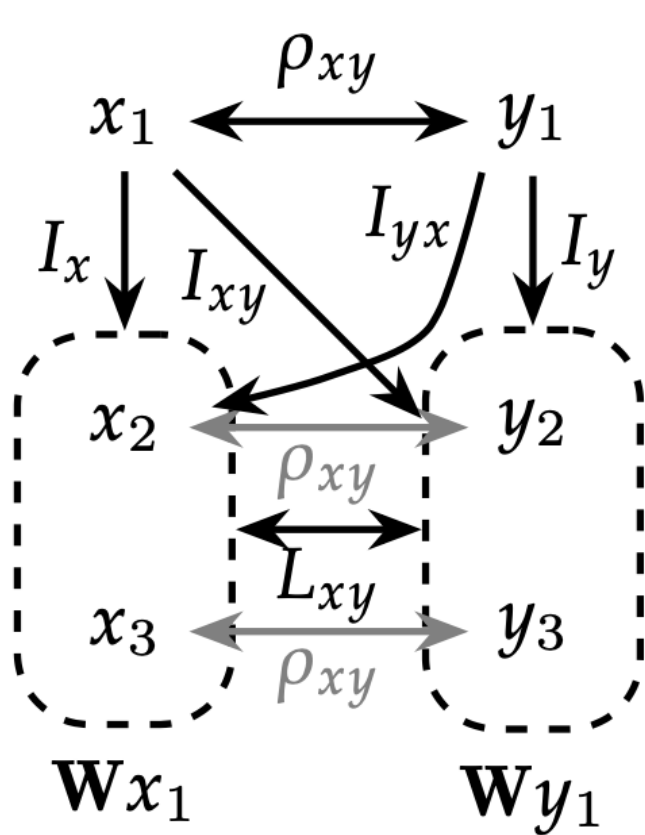
$$\mathbf{M}\mathbf{W}y = I_{y|x}\mathbf{M}y + \epsilon$$

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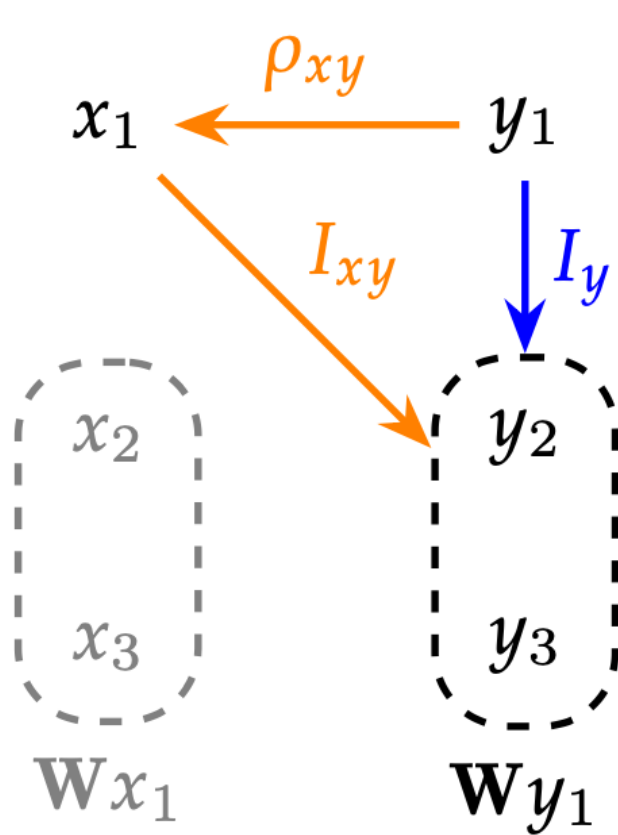
$$\propto y' \mathbf{W}y - (\mathbf{X}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'y)' \mathbf{W}y$$

$$I_y \quad (\mathbf{P}y)' \mathbf{W}y$$

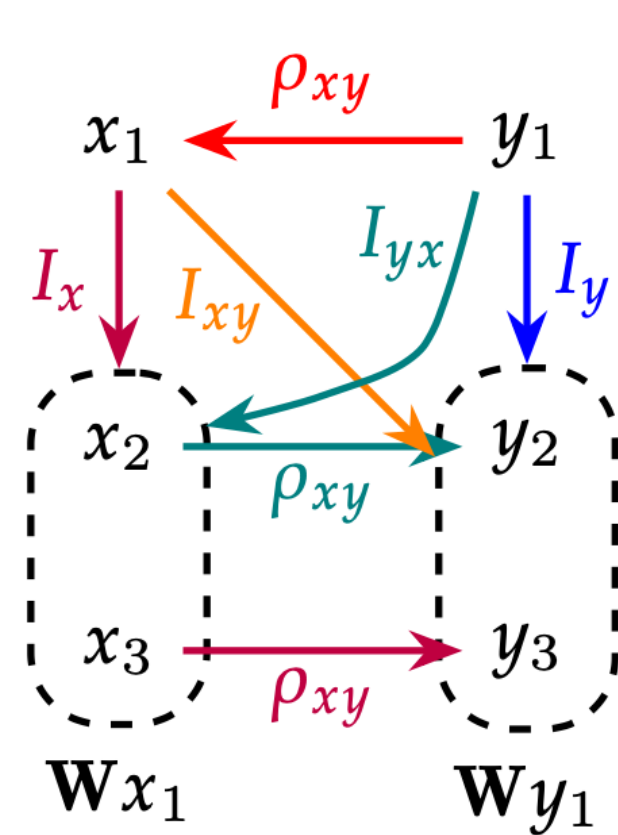
HIGHER DIMENSIONAL PARTIAL CONDITIONING BY FWL



Classic Statistics



Partial $I_{y|x}$



Full $I_{x \rightarrow y}$

THE POINT REMAINS: CORRECT Y FOR X (& MAYBE WX)

LOCAL CONFOUNDING

confounding is not stationary

PRIOR ART IN ESDA

The Moran-Form Regression

PARTIAL CONDITIONING

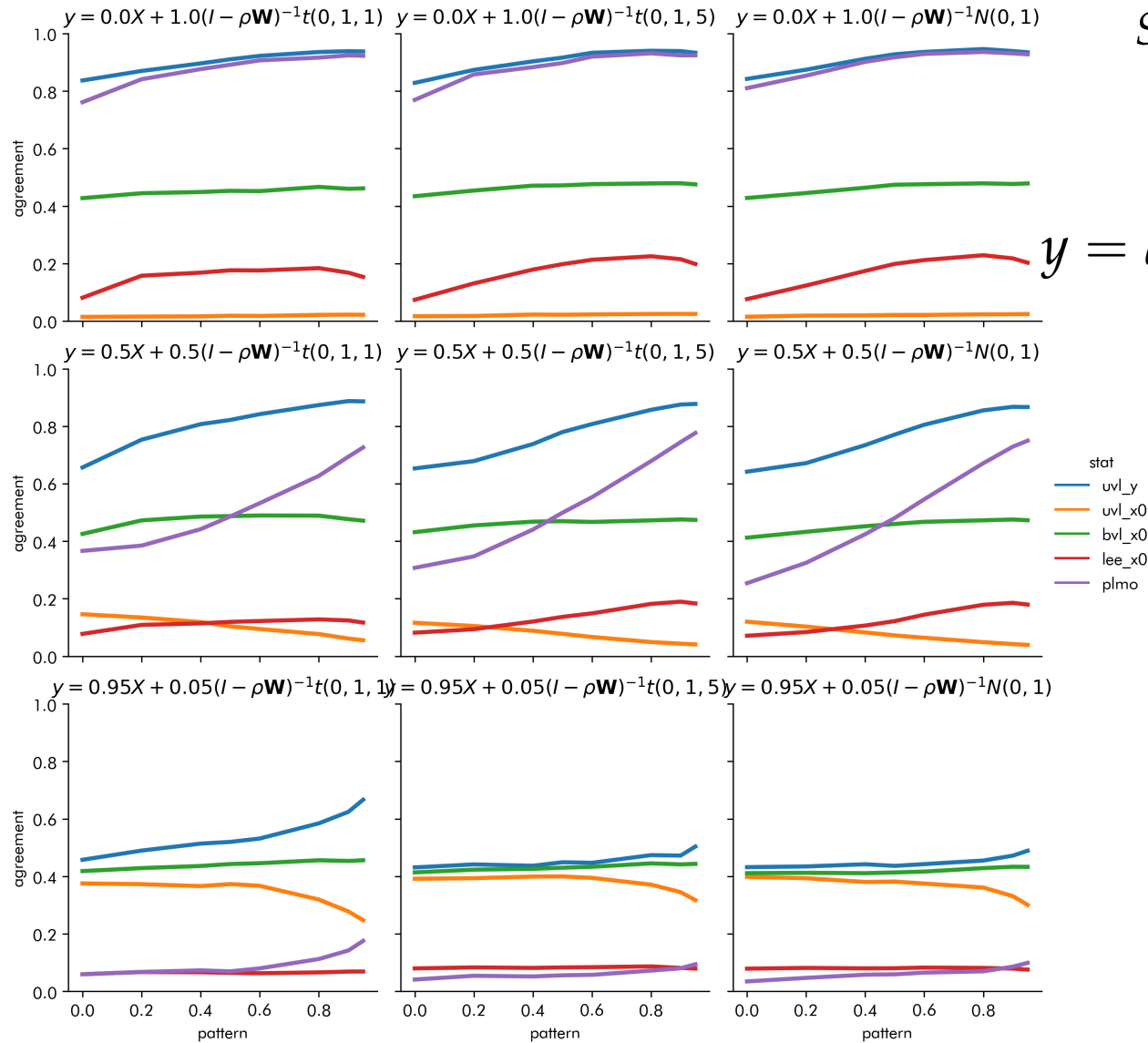
Two approaches to "control"

AFFORDABILITY OUTLIERS

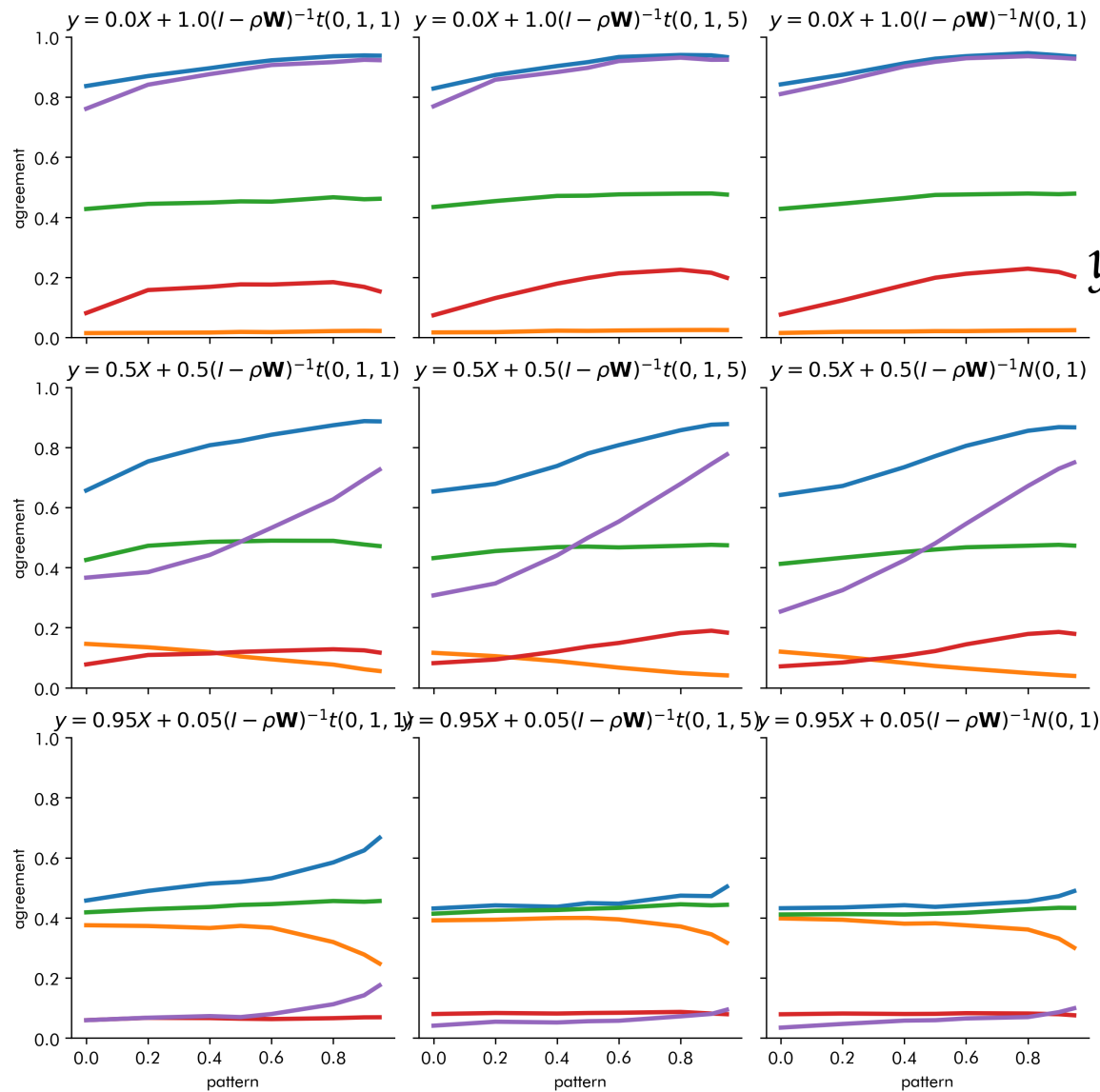
Where are the good deals?

*SAR t -distributed random effects process
(SARTRE)*

$$y = \alpha x + (1 - \alpha)(I - \lambda \mathbf{W})^{-1} t(\mu, \tau^2, \nu)$$



SURPRISE! SIMULATION RESULTS FIRST!



SAR t -distributed random effects process (SARTRE)

$$y = \alpha x + (1 - \alpha)(I - \lambda \mathbf{W})^{-1} t(\mu, \tau^2, \nu)$$

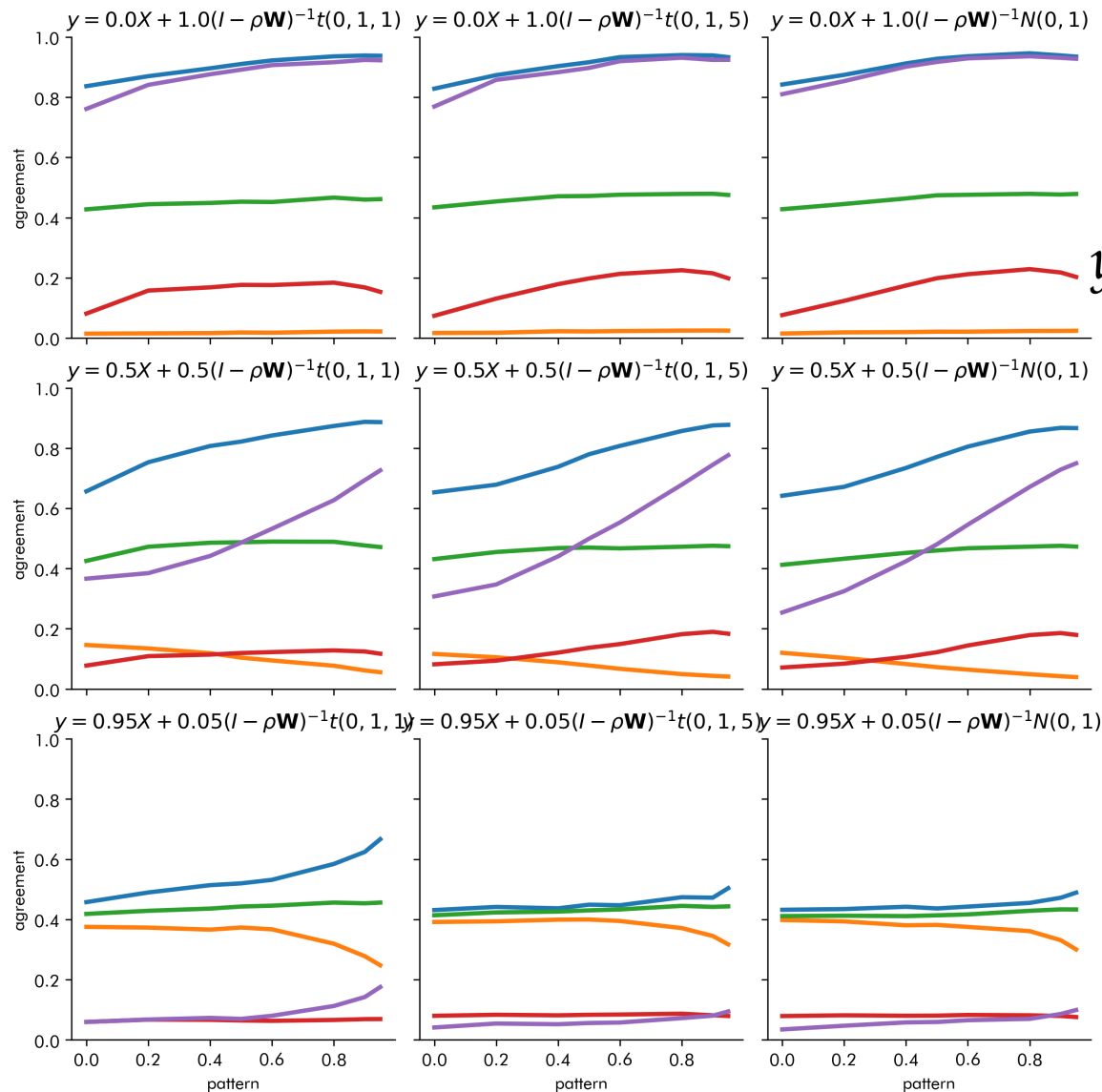
confounder

$\alpha \in [0, 1]$

SAR patterning

random effect

SURPRISE! SIMULATION RESULTS FIRST!



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$$y = \alpha x + (1 - \alpha)(I - \lambda\mathbf{W})^{-1}t(\mu, \tau^2, \nu)$$

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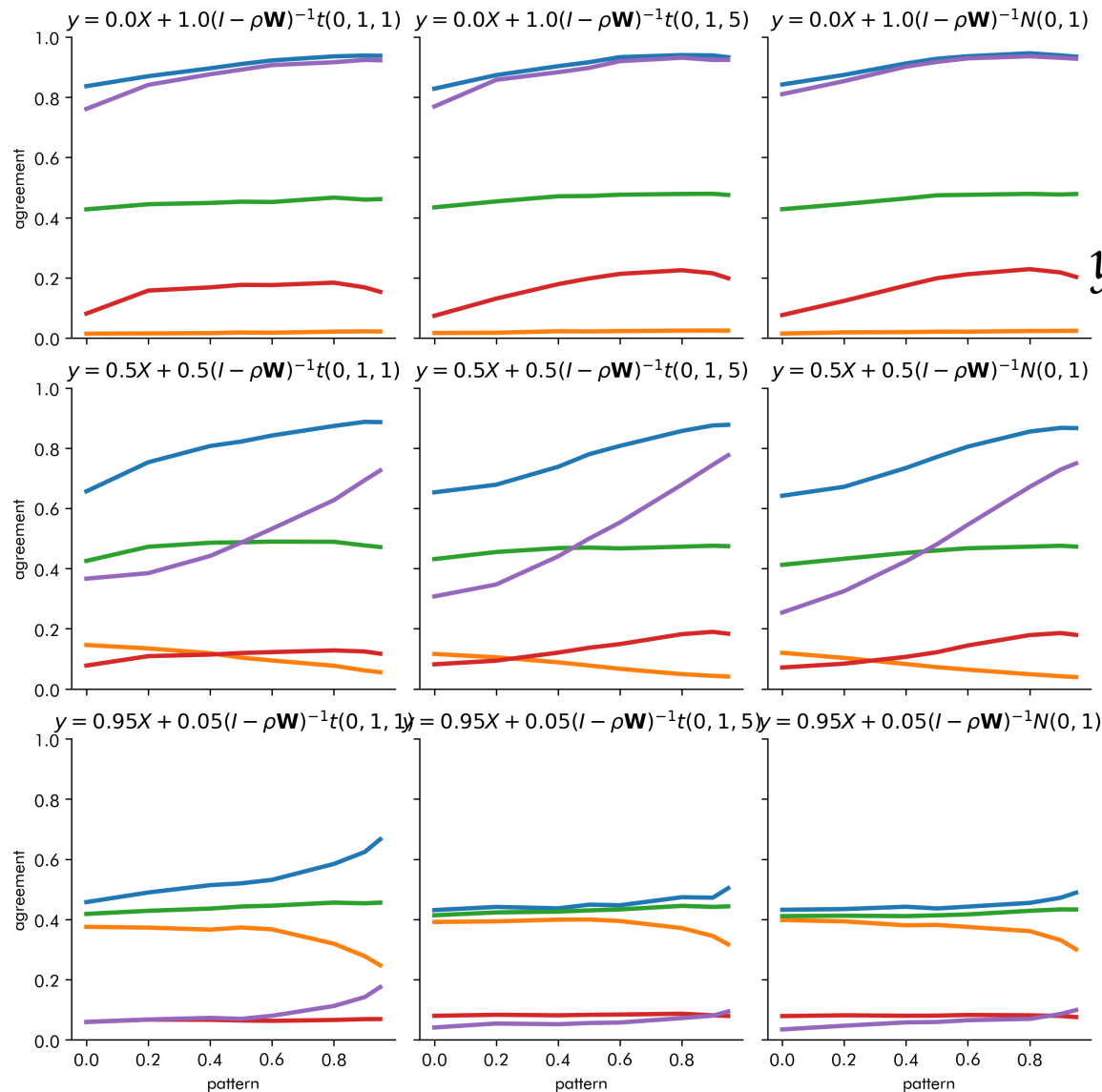
$\alpha \in [0, 1]$

SAR patterning

random effect

Blend a spatially-unpatterned "confounding effect" with a spatially-structured part with controlled tail thickness.

SURPRISE! SIMULATION RESULTS FIRST!

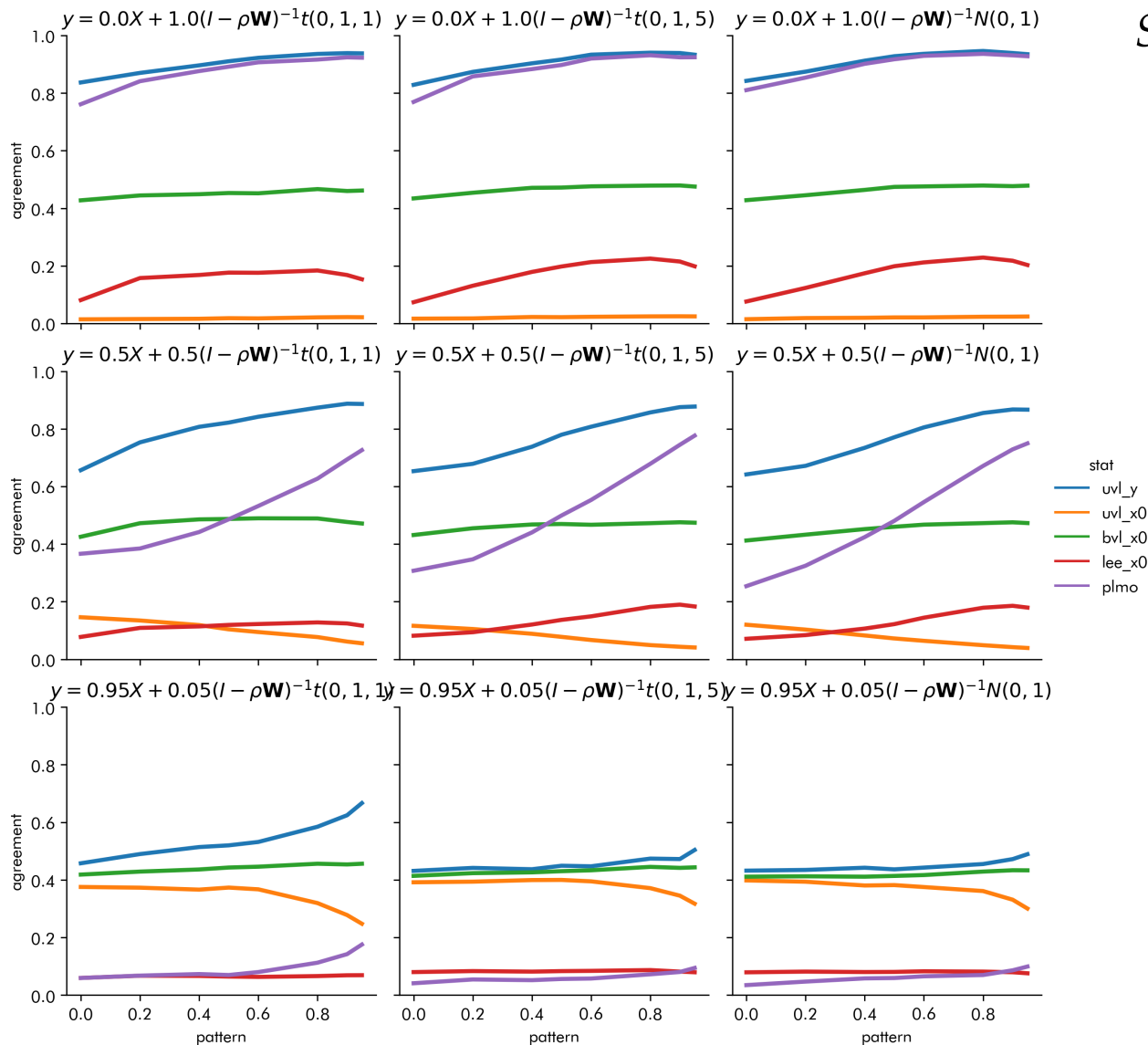


SAR t -distributed random effects process (SARTRE)

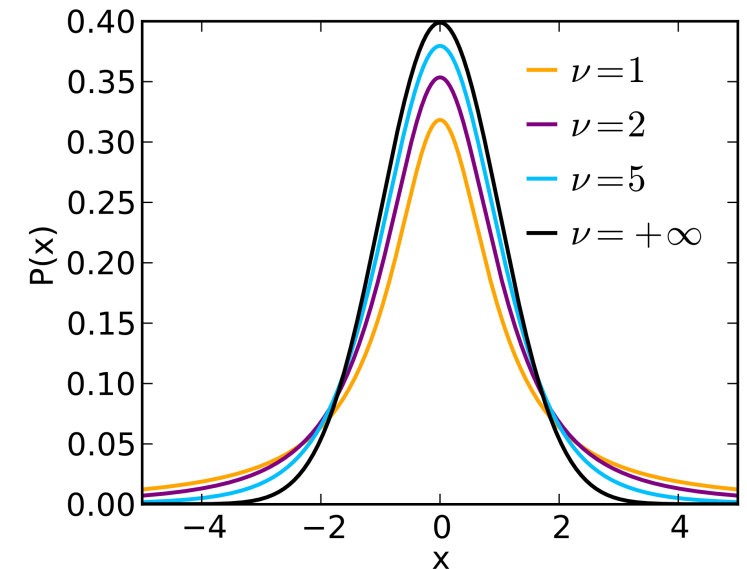
$$y = \alpha x + (1 - \alpha)(I - \lambda\mathbf{W})^{-1}t(\mu, \tau^2, \nu)$$

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SURPRISE! SIMULATION RESULTS FIRST!



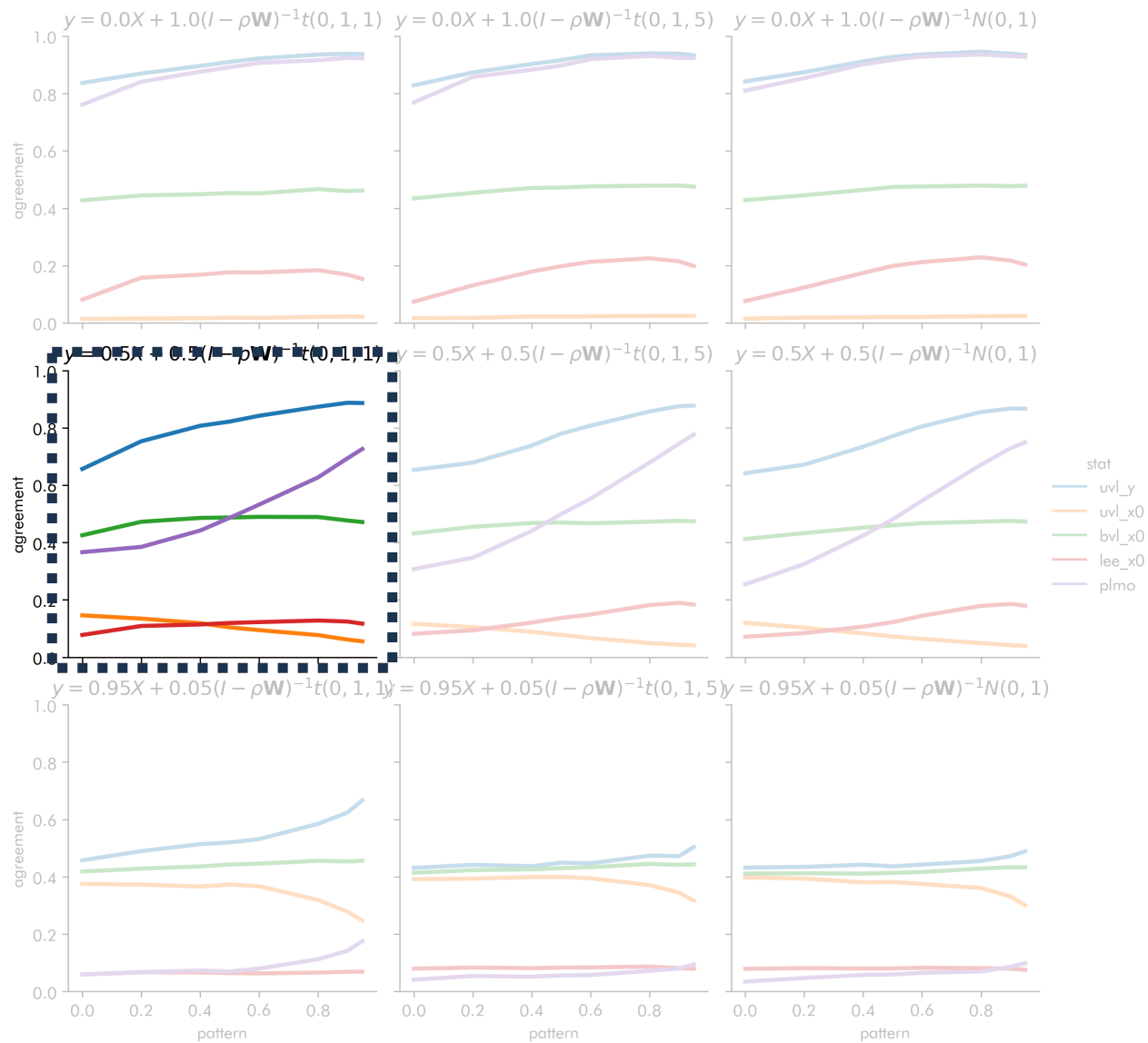
SAR t -distributed random effects process (SARTRE)



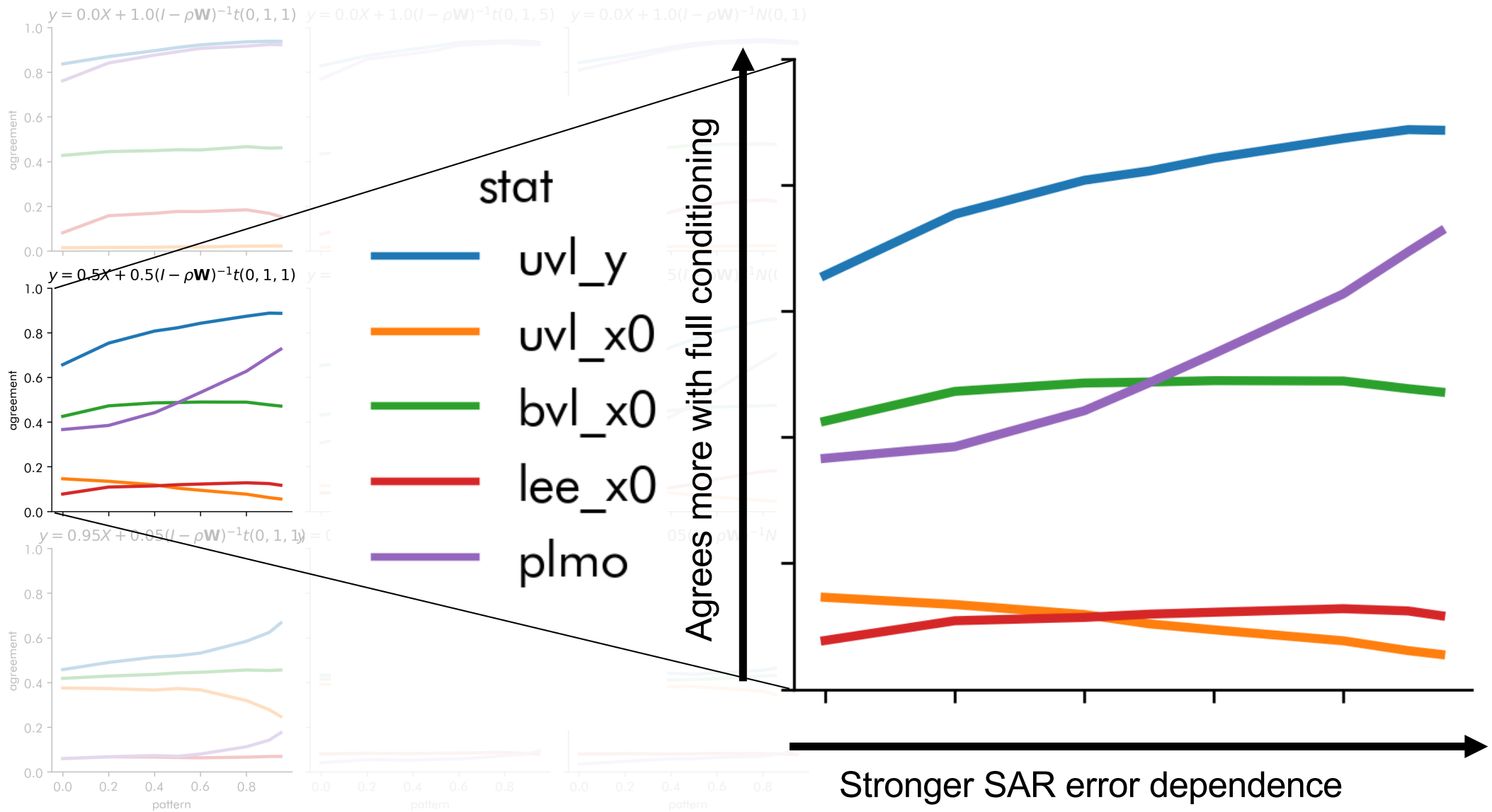
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$$\nu \rightarrow \infty$$

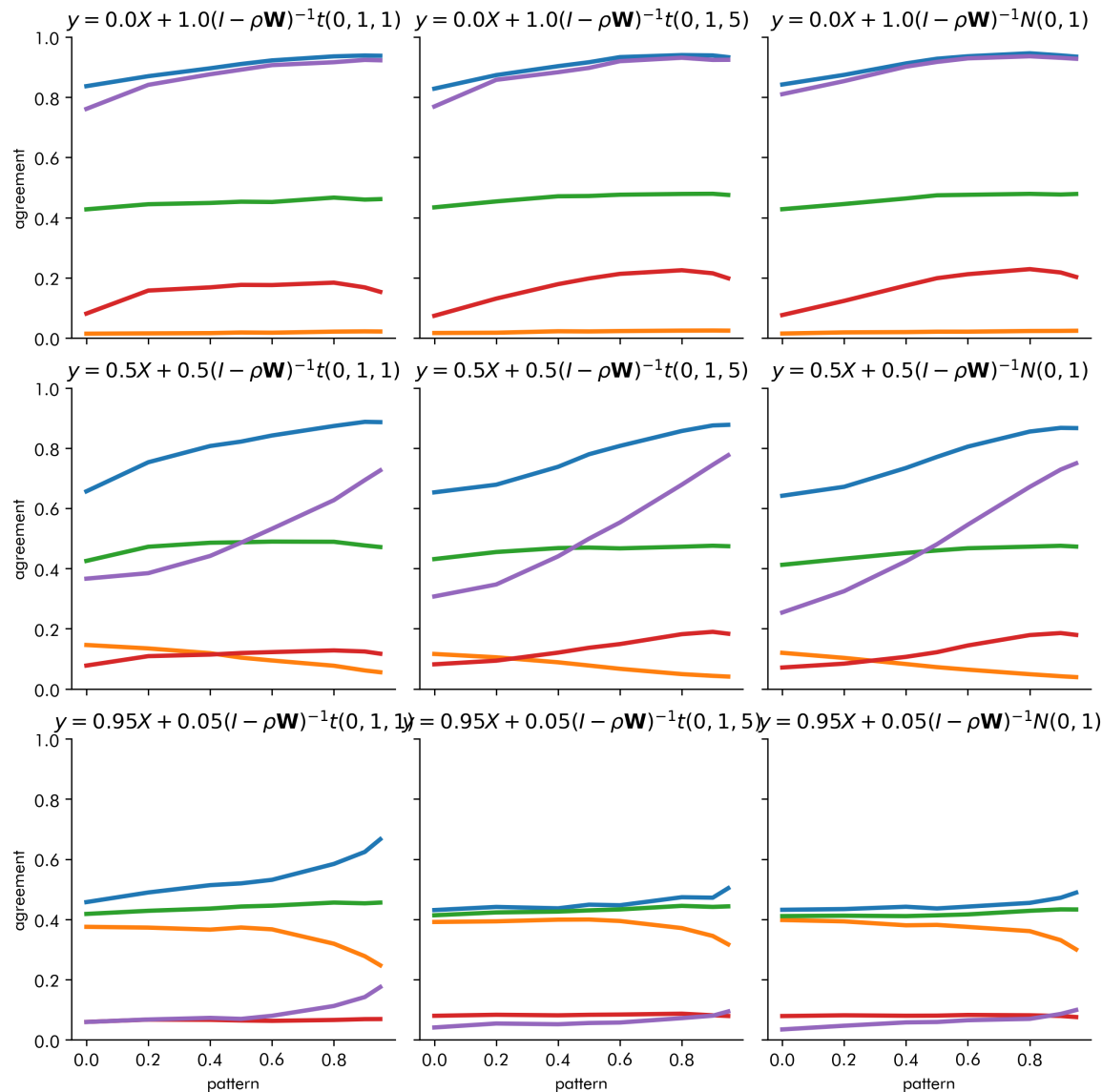
SURPRISE! SIMULATION RESULTS FIRST! $t(\mu, \tau^2, \nu) \rightarrow \mathcal{N}(\mu, \tau^2)$



SURPRISE! SIMULATION RESULTS FIRST!

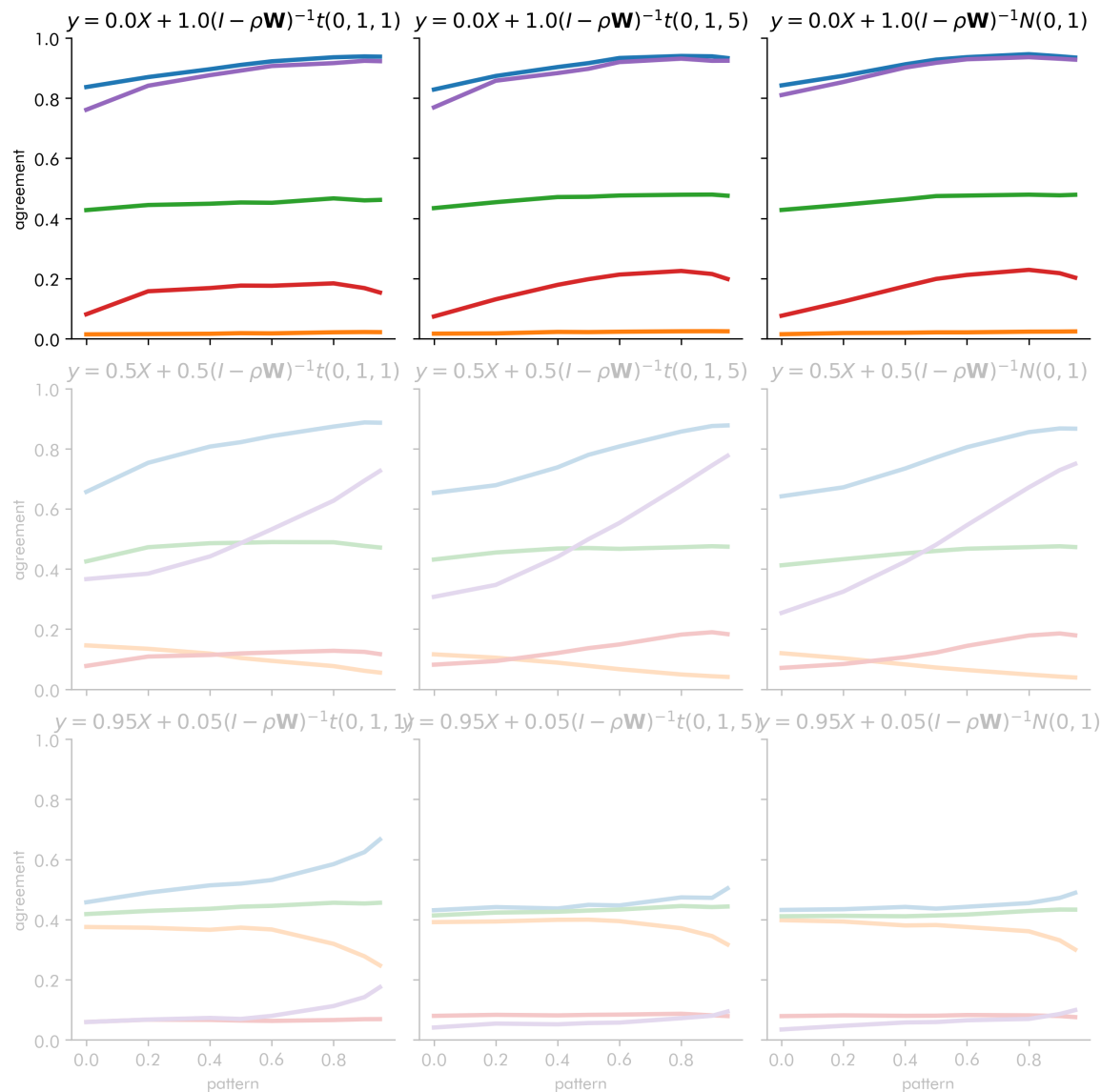


SURPRISE! SIMULATION RESULTS FIRST!



- Increasing SD right within facets
- Increasing “agreement” w/ aux. reg. moving up within facets
- Thinner tails moving right over facets
- Increasing confounding moving down over facets

SURPRISE! SIMULATION RESULTS FIRST!



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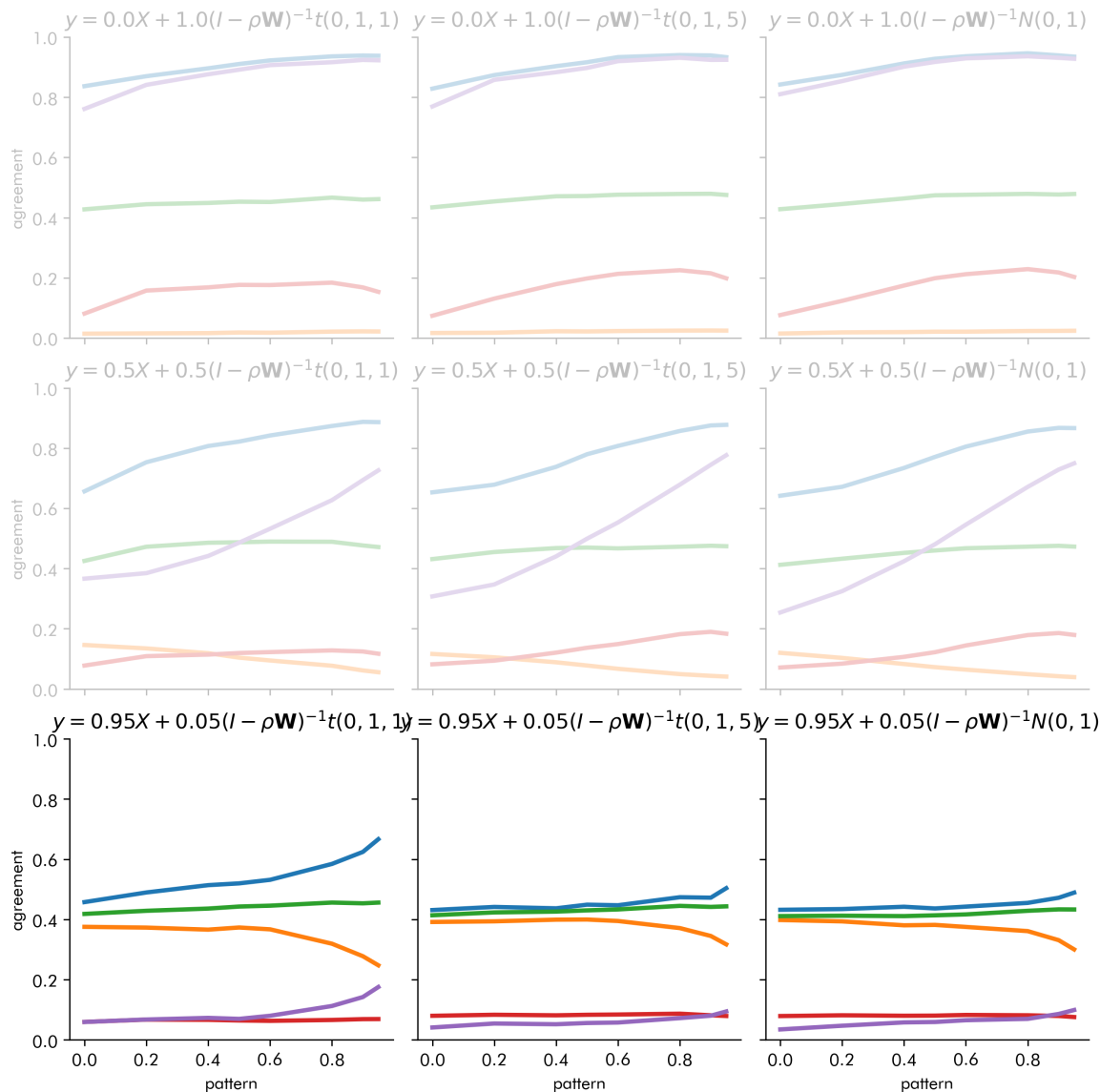
WHEN x IS IRRELEVANT,

(P)LMO both correctly ignore x

Pairwise stats incorrectly sensitive to x

plmo

SURPRISE! SIMULATION RESULTS FIRST!



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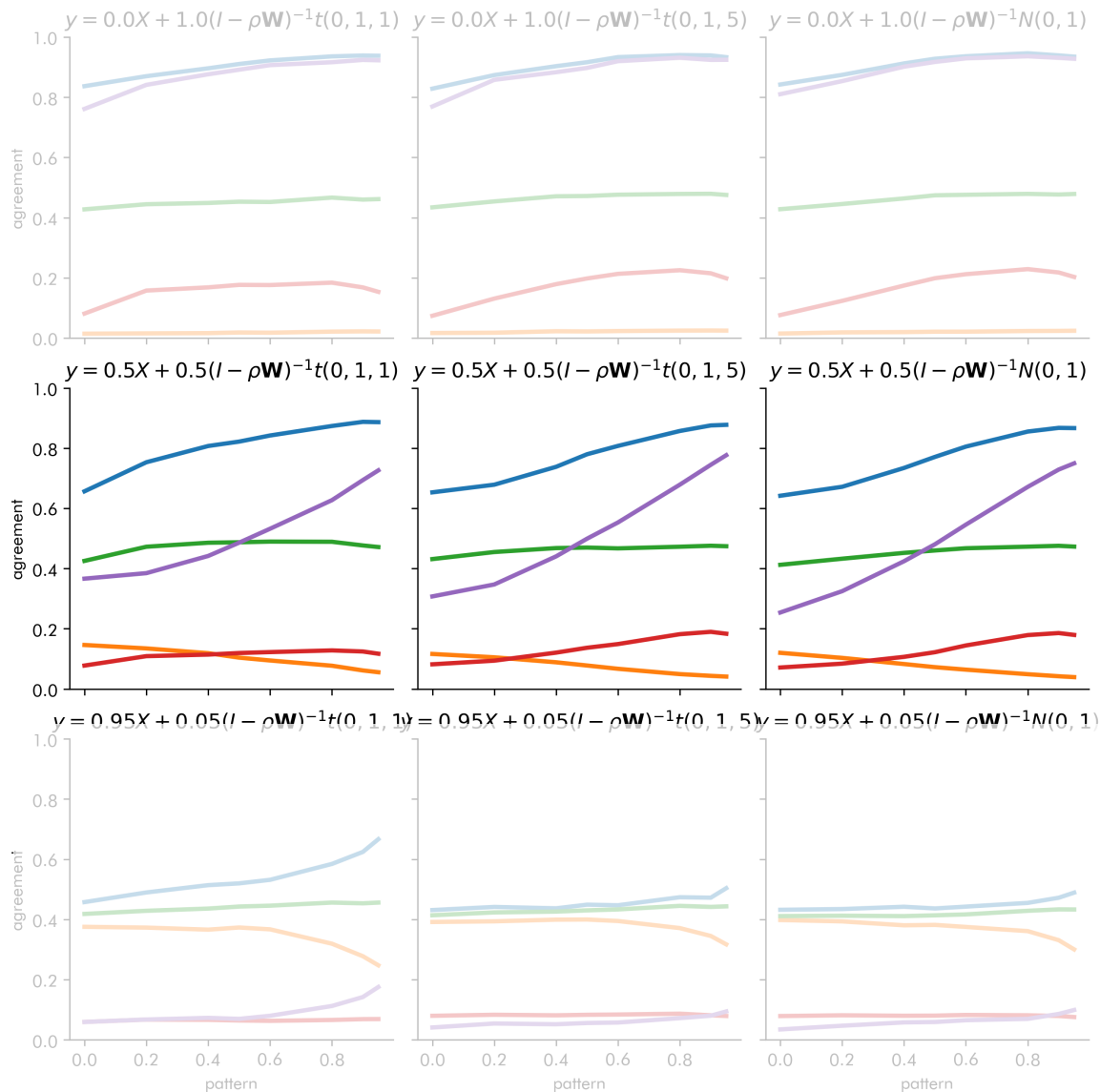
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WHEN MAINLY x MATTERS,

Lee & PLMo disagree with full conditioning
Univariate approaches broadly agree

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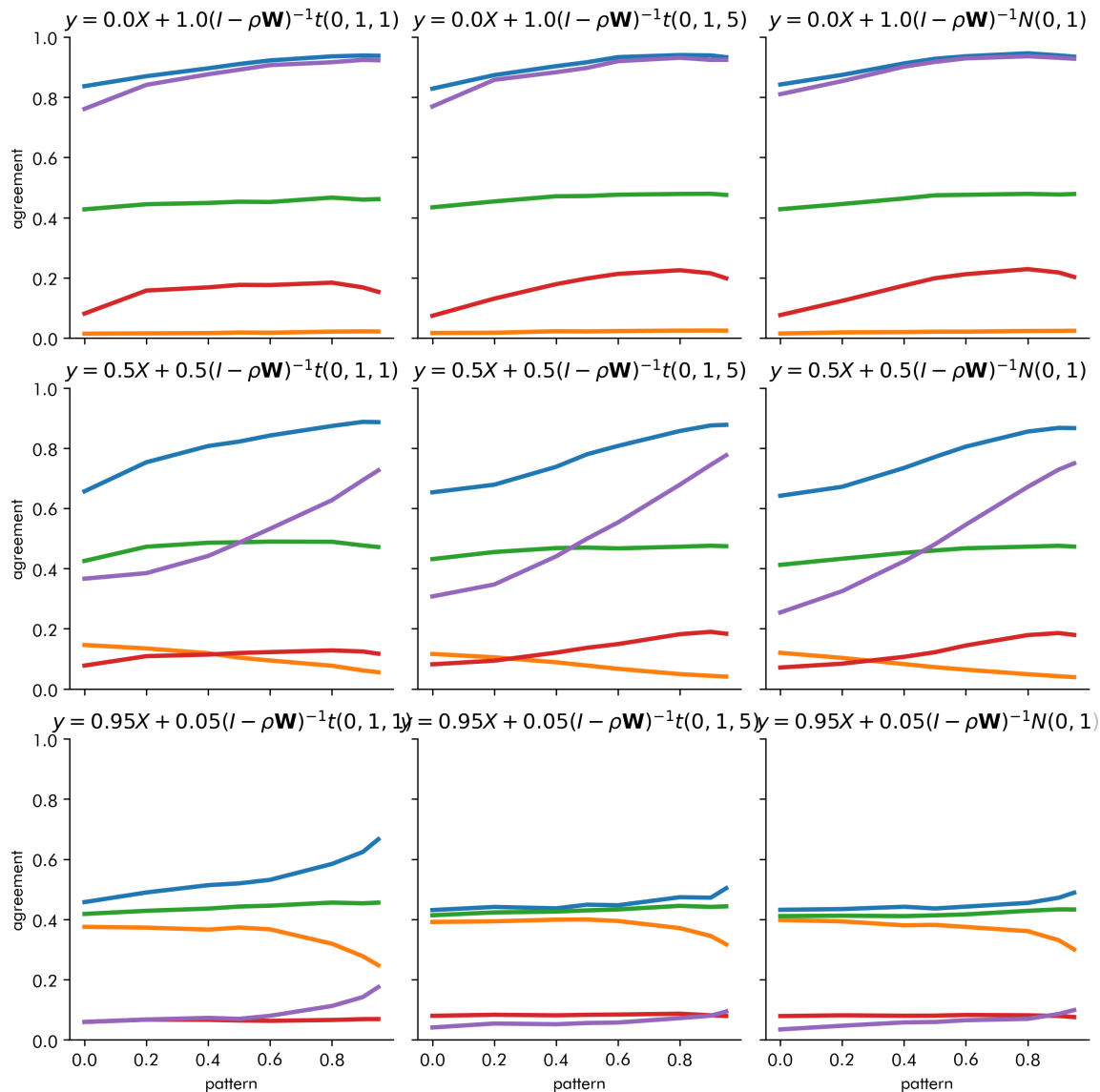
WHEN BOTH MATTER

LMo_x and Lee both problematic

BVL_{Mo} fails to recognize SD+

(P)LMo become more similar as SD+

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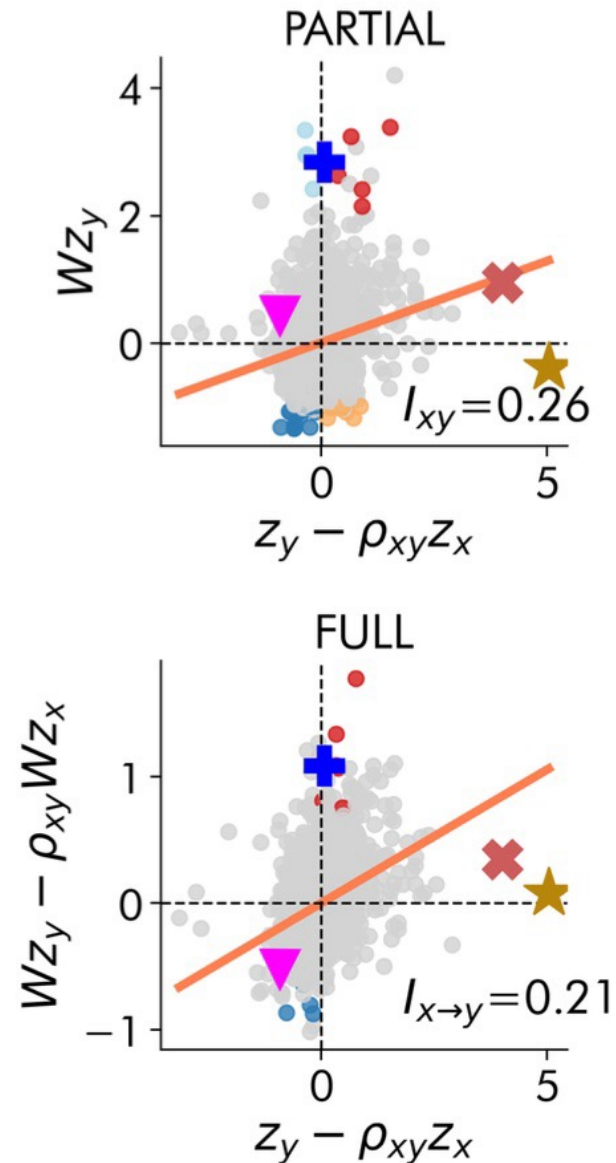
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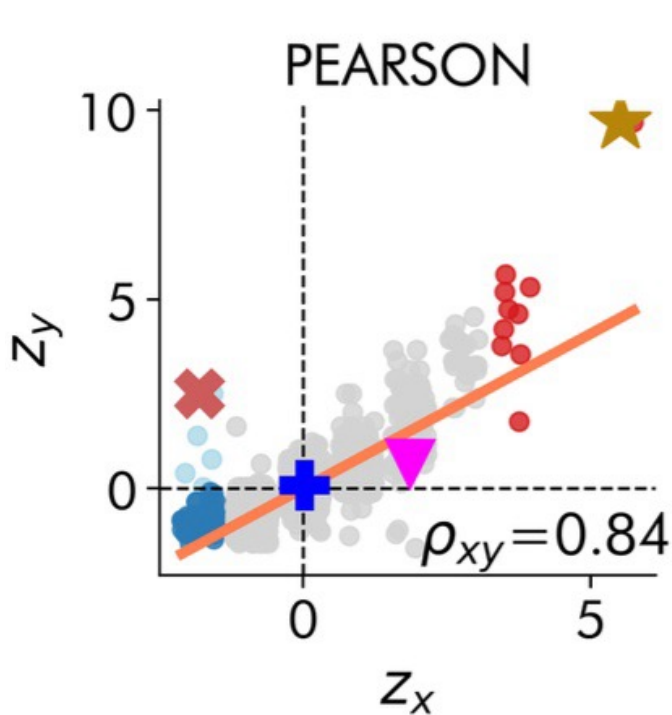
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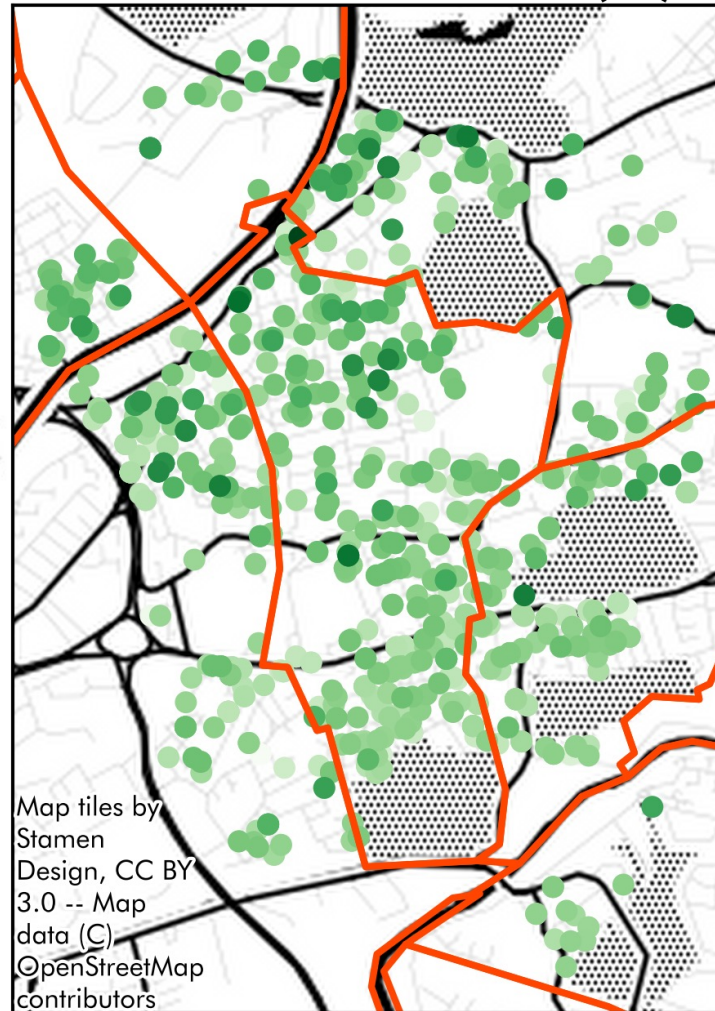
SURPRISE! SIMULATION RESULTS FIRST!



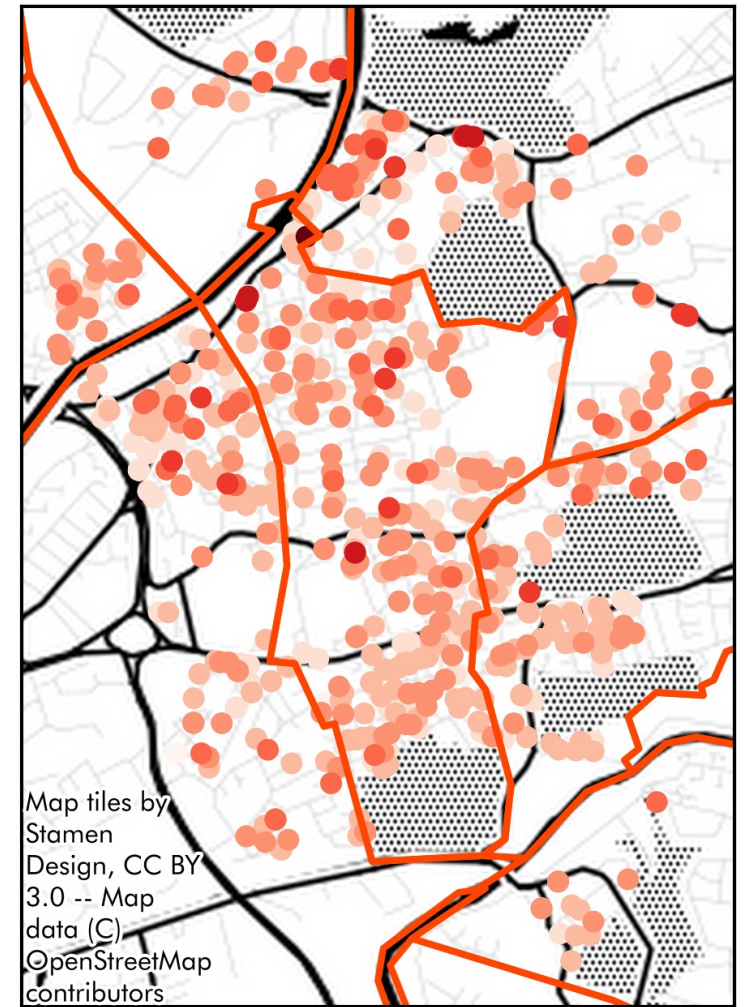
GOOD DEALS MAY NOT HAVE STRONG RELATIVE PREMIA



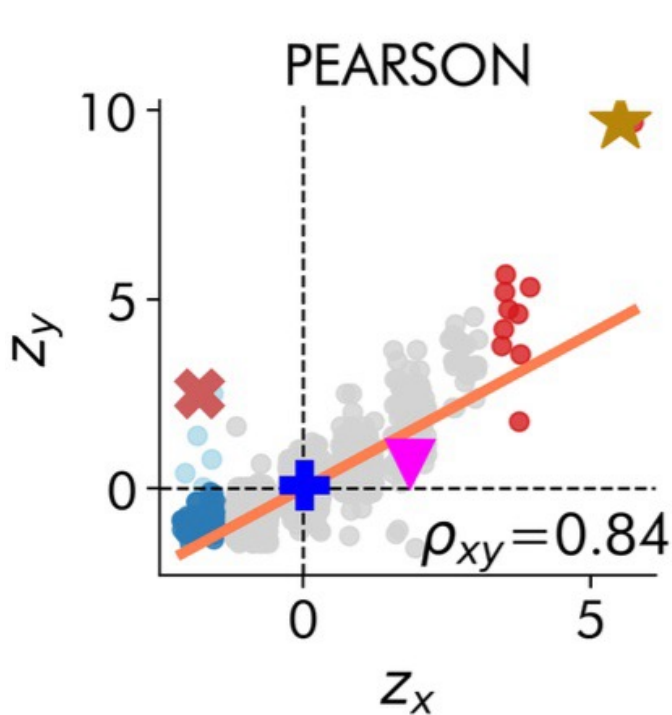
WEEKLY RENT (£)



BEDROOMS

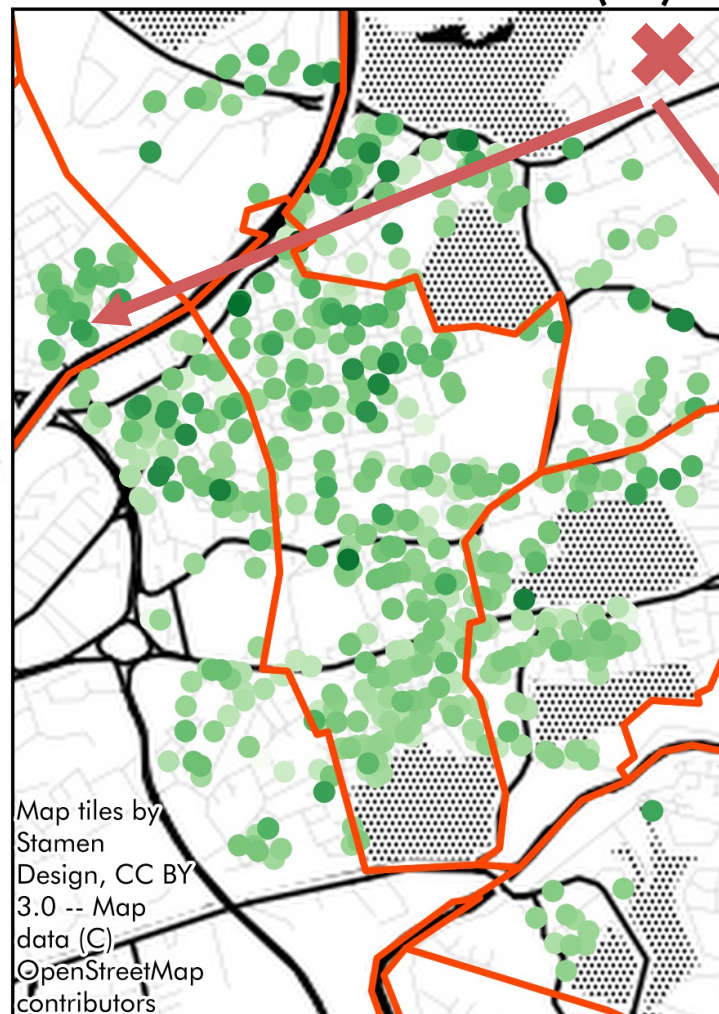


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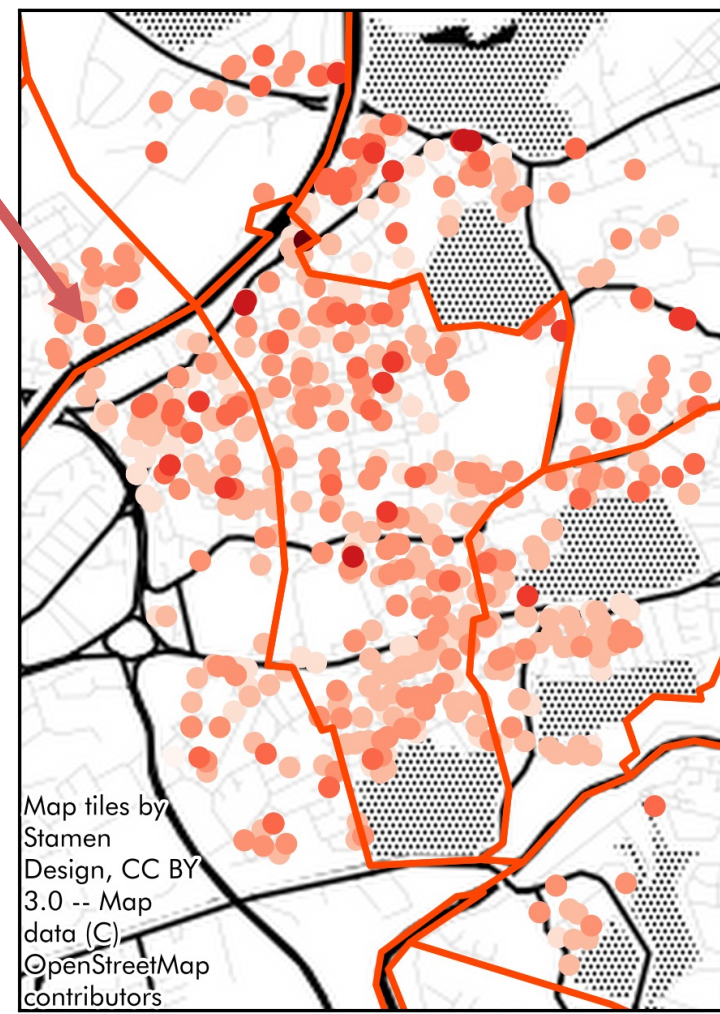


✗ is a very expensive studio

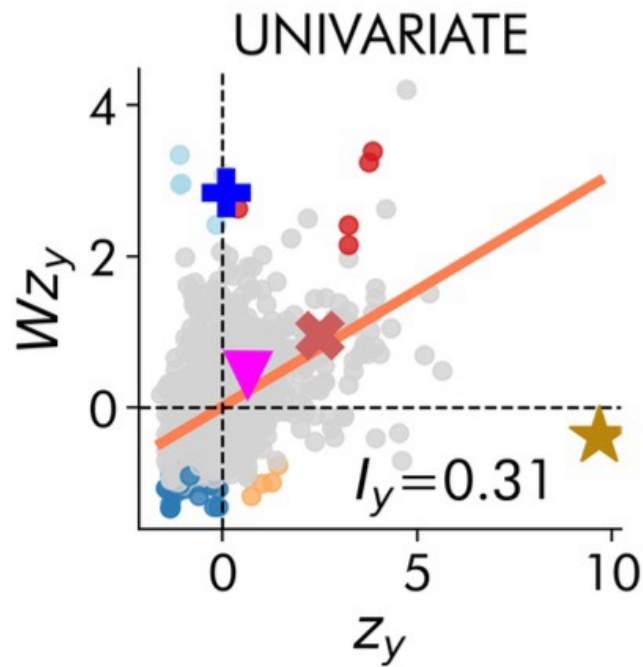
WEEKLY RENT (£)



BEDROOMS

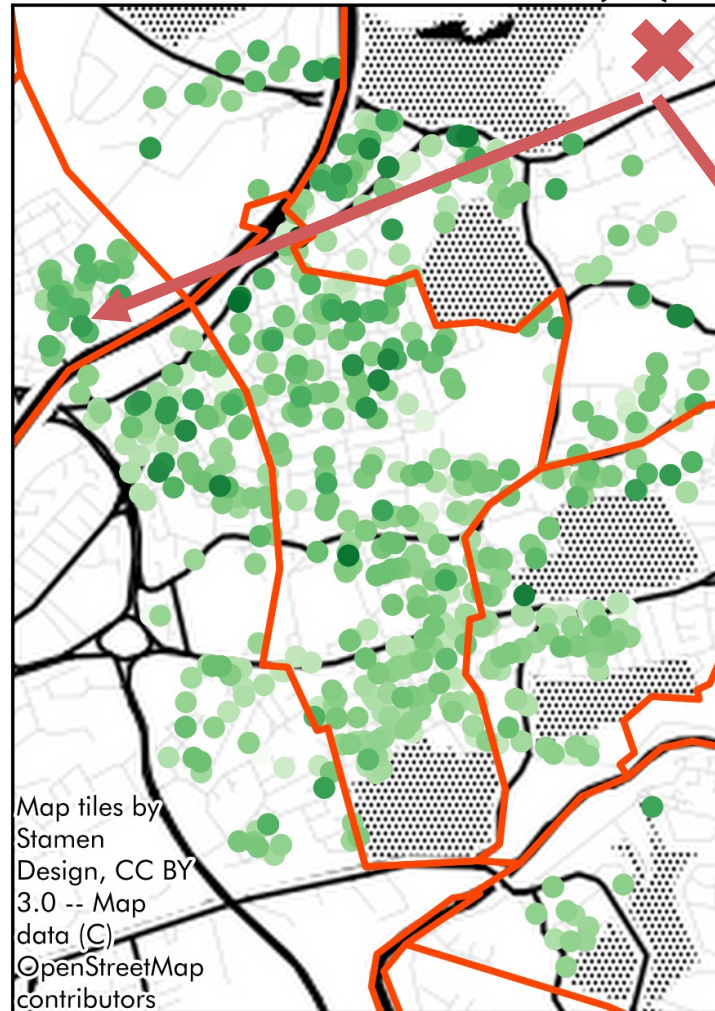


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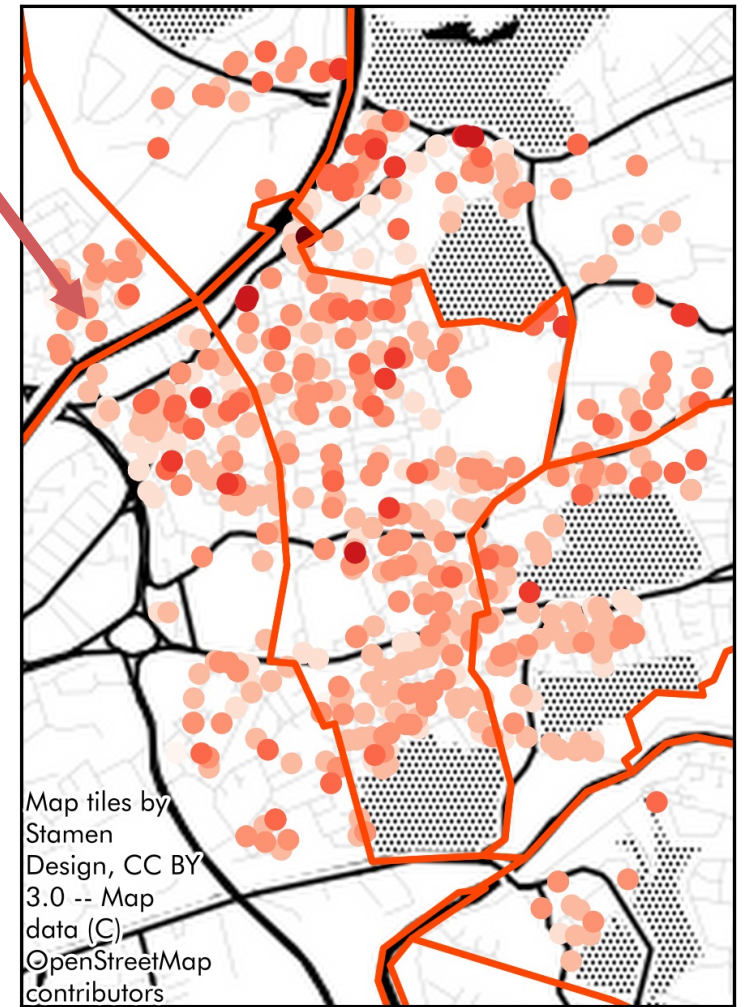


✗ is a very expensive studio, but it's not a rent outlier.

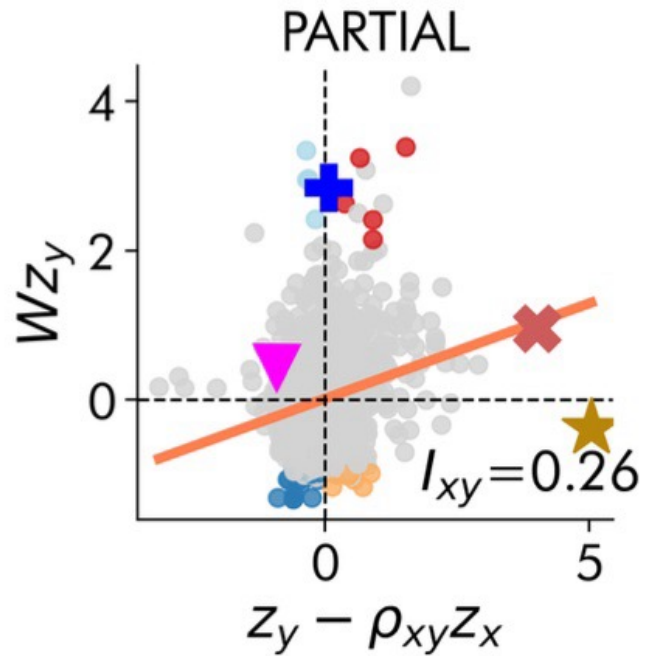
WEEKLY RENT (£)



BEDROOMS

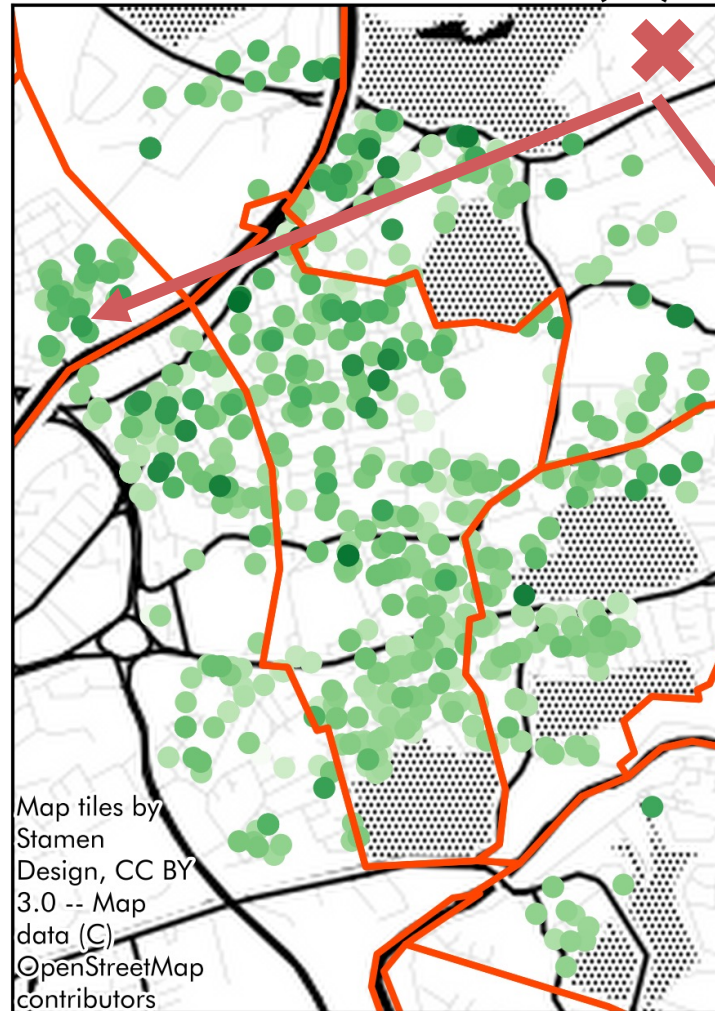


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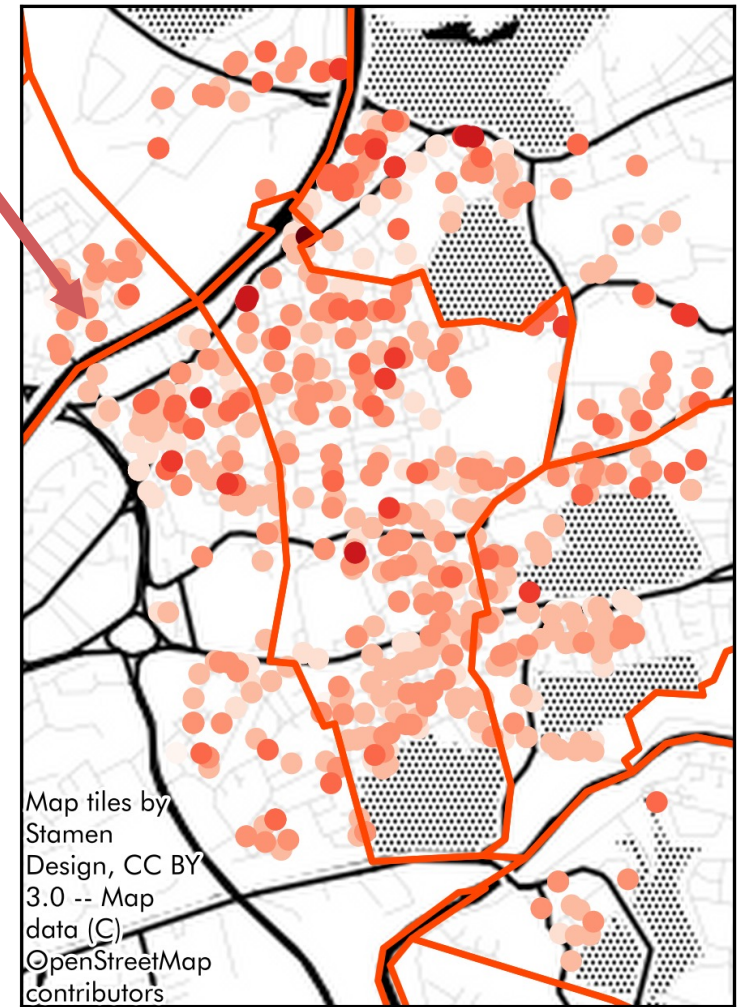


✗ is expensive for a studio, surrounded by expensive properties.

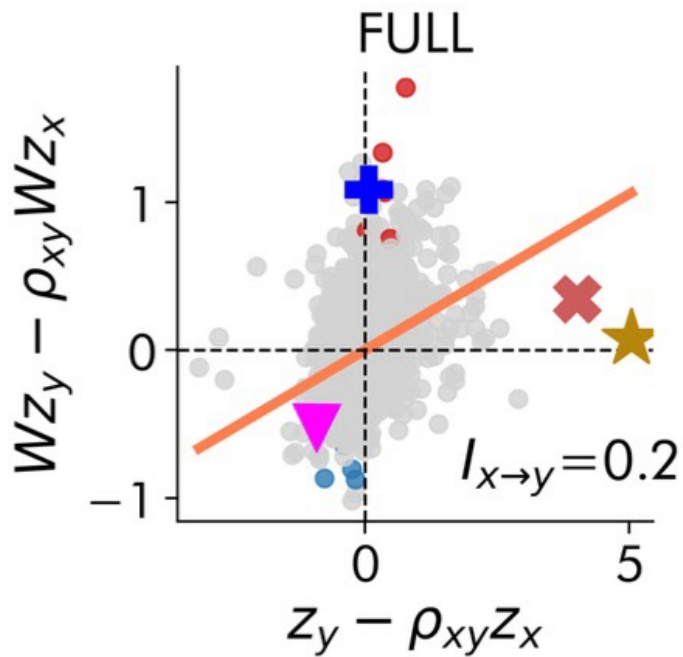
WEEKLY RENT (£)



BEDROOMS

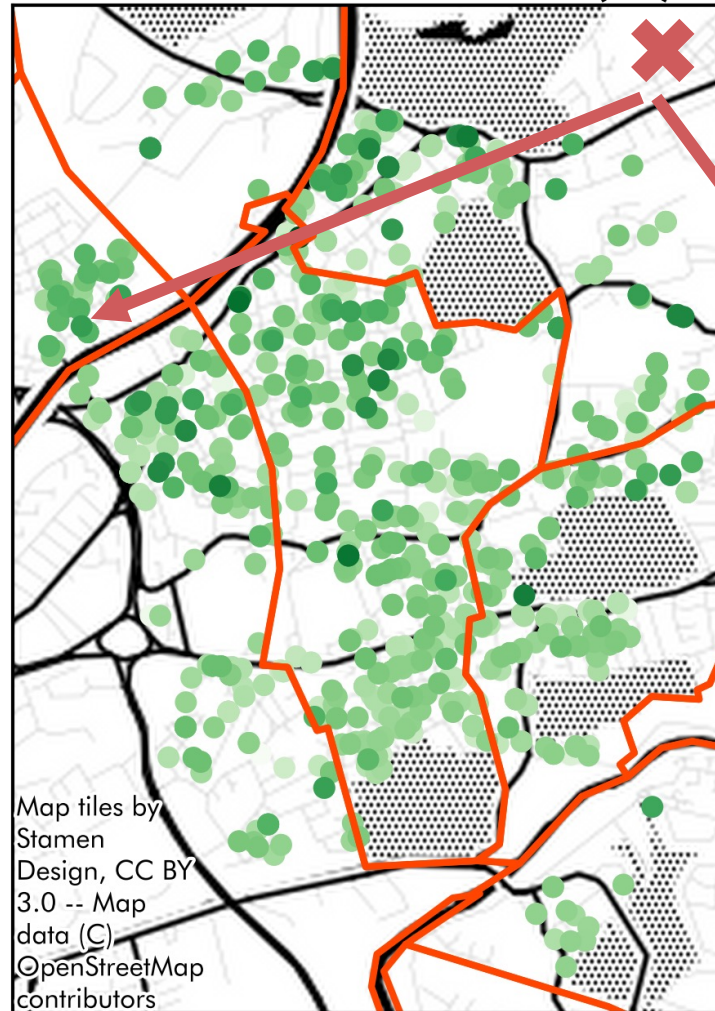


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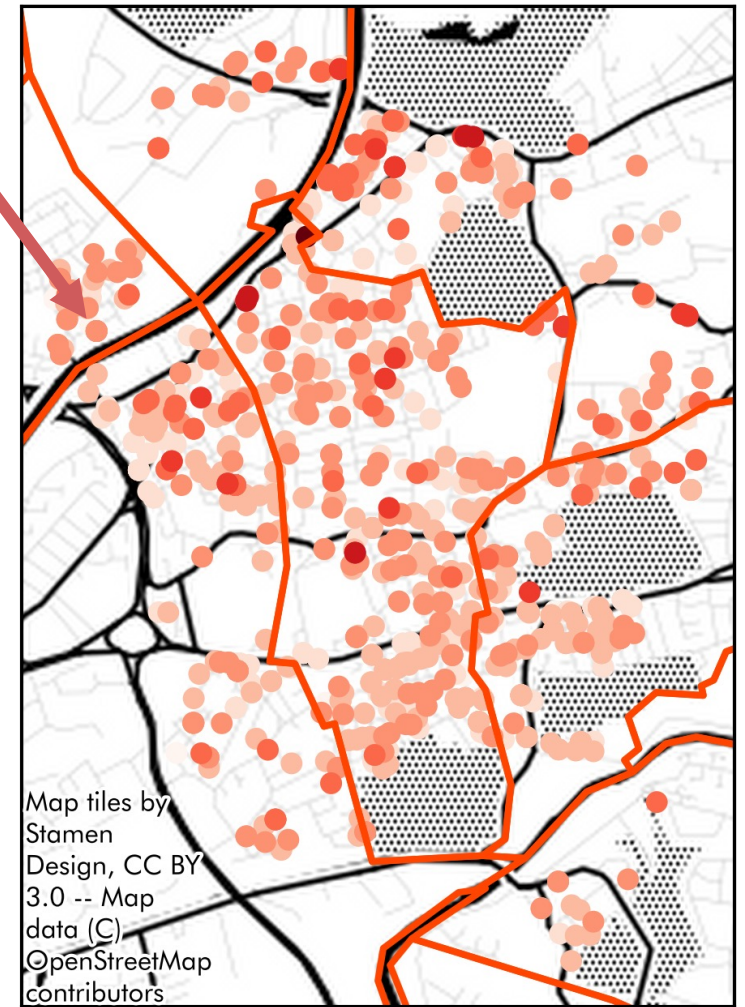


✖ is expensive for a studio, surrounded by other properties that are expensive for their size

WEEKLY RENT (£)



BEDROOMS



GOOD DEALS MAY NOT HAVE STRONG RELATIVE PREMIA

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HANDLING LOCAL CONFOUNDING

LEVI JOHN WOLF

UNIVERSITY OF BRISTOL

`levi.john.wolf@bristol.ac.uk`

